PETW CFD Research Work Summary: AutoCFD and CM1-CFD

Chapter 1 Scope, Date, Evidence Policy and Executive Synthesis

1.1 Report identity and date

This study is dated 2026-08-12 and is written against a frozen, read-only evidence snapshot. The snapshot is described by `FROZEN_SNAPSHOT_MANIFEST.json` with `schema_version` `1.0`, `created_utc` `2026-08-12T20:22:17.416080+00:00`, `file_count` `92`, and `total_bytes` `1533051`. The parent fact ledger used to bind numerical values is `PARENT_FACT_LEDGER.json` with `schema_version` `1.0` and `created_utc` `2026-08-12T20:36:36.135368+00:00`. The bilingual structure, table identifiers, figure identifiers and status vocabulary are fixed by `BILINGUAL_REPORT_CONTRACT.json`, schema `petw.cfd.bilingual-report-contract.v1`.

The manifest declares four operating boundaries that hold for the whole of this study: no remote write, no HPC submission, no scientific claim promotion, and reviewer output that is advisory and parent-verified. This study neither requested, simulated, nor performed any write, remote access, scheduler action, or solver execution.

The 92 snapshot files consist of 70 Python source files and 22 Markdown or JSON evidence files. The Python files divide as 19 in the cylinder product line, 16 in the generic control plane, 16 in Case011Up, 13 in the D420 empty-domain line, 5 in the D420 M1 URANS line, and 1 figure-generation script. Every program mechanism stated in this study is bound to one of the 70 source files, and every numerical result is bound to one of the 22 evidence files or to the parent fact ledger read from them.

The external anchor `f278ed027895a8e1d3eb76e36b9acd9559c3fbdbd7acd3bbc8f73f49a0b49419` supplied by the caller does not appear in any field inside the manifest. This review was read only and had no hashing tool, so the anchor is recorded as caller-supplied and not recomputed in this session, that is, `UNVERIFIED`. The 92 per-file `sha256` values in the manifest were likewise referenced as declared and not recomputed.

1.2 Evidence policy

The evidence policy of this study is the policy stated in the parent fact ledger: values are parent-read from local frozen source files, and interpretation limits travel with each value. Four operating rules follow from it.

First, a value is reported together with the artifact that produced it and with the boundary sentence attached to that artifact in its own source file. A number does not become more authoritative by being repeated in a summary.

Second, status terms are drawn from one closed ten-term vocabulary (Table T01). No synonym is introduced, and no term is upgraded by narrative context.

Third, eight promotions are prohibited outright, and this study applies them as hard constraints: a directory named final, accepted or ready does not become `SCIENCE_ACCEPTED`; a scheduler `submitted` or `COMPLETED` state does not become scientific acceptance; a solver `End` marker does not become physical validation; postprocessing success does not become a controlled comparison; a ready package does not become a submitted job or a solver run; unit-density forces do not become structural design loads; coarse patch completeness does not become medium contract closure; and historical D420 M0 or M1 output does not become a matched-drive comparison.

Fourth, historical test counts are treated as historical frozen-file evidence and are not replayed in packaging. Where this study cites 195 control-plane tests passed, 12 benchmark tests passed, 15 analysis and verifier tests passed, execution and retrieval checks passing 35 of 35, parent verification passing 25 of 25, or CFD and CM1 semantic validation passing 16 of 16, those counts are quoted from the frozen evidence files that recorded them at the time of the corresponding round. No test suite was executed in this reporting environment, and this study therefore makes no claim of a current full-suite pass.

1.3 Status vocabulary

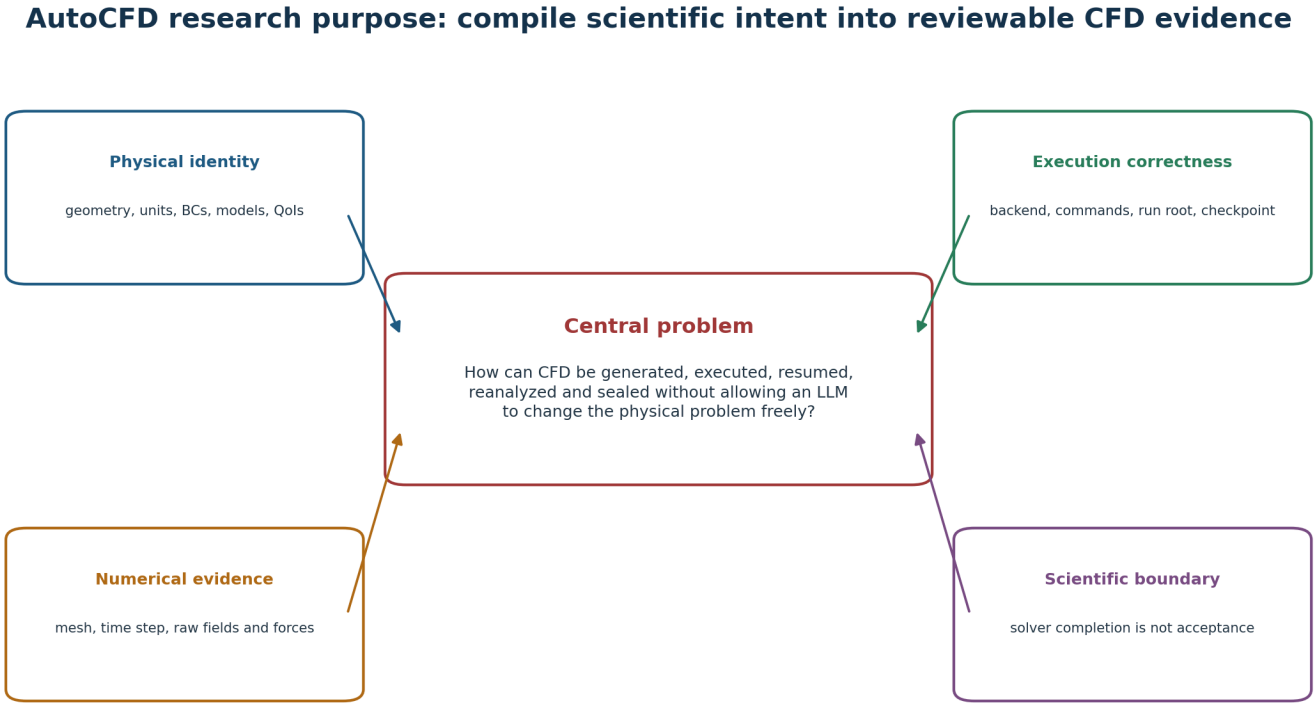


Figure F01. Method diagram. The four framing concerns of AutoCFD, physical identity, execution correctness, numerical evidence and scientific boundary, converge on one central problem statement: how CFD can be generated, executed, resumed, reanalyzed and sealed without allowing a language model to change the physical problem freely. This is a statement of research intent and method scope, not a validation result.

Table T01. Evidence status vocabulary

ID	Status term	Definition used throughout this study	What it explicitly does not mean	Representative instance in the frozen snapshot
V1	`IMPLEMENTED`	Program object exists in the snapshot and is statically verifiable by reading its own frozen bytes	Does not mean the object has been executed, nor that its physics is correct	70 Python source files; 24 `route_b_cli` subcommands; `task6_candidate.py` deterministic candidate
V2	`PACKAGE_READY`	A static case package is frozen with verified internal identity and complete file ledger	Does not mean submitted, meshed on the target machine, or solved	D420 three-arm package, 5,293 manifest-listed files, `STATIC_THREE_ARM_PACKAGE_OK`
V3	`SUBMITTED`	A scheduler accepted the job and consumed the single approved submission	Does not mean the application started, produced output, or produced science	Job `3391286`, submitted exactly once after live checks and `sbatch --test-only`
V4	`SOLVER_RAN`	The solver executed and terminated normally with an `End` marker and no fatal marker	Does not mean converged, mesh independent, physically validated, or accepted	Cylinder five cases; Condor32 cluster `1542.0`; jobs `3382272`, `3396039`, `3384063`, `3386652`
V5	`FAILED`	The scheduler terminal or the application terminal is a failure state, preserved rather than retried	Does not attribute the cause to physics, solver, or platform without separate evidence	Job `3391286`, scheduler `FAILED 2:0`, elapsed `00:00:57`, zero scientific output
V6	`POSTPROCESSED`	Raw output was recomputed from unchanged raw files by a pinned analyzer	Does not mean the recomputed quantities were accepted as physics	Cylinder `METRICS_EXTRACTED`; Case02Up 50,877 samples; job `3393320` four-time reconstruction
V7	`TERMINAL_CLASSIFIED`	Scheduler, application, wrapper, artifact, usability and claim terminals are recorded separately	Does not mean the terminals agree, and success in one category never substitutes for another	`TERMINAL_CLASSIFICATION_3396039.json`, six coexisting terminals
V8	`SCIENCE_ACCEPTED`	has explicitly approved a scientific interpretation of a model output	Cannot be produced by any automated path, directory name, or status field	No instance exists anywhere in this snapshot
V9	`BLOCKED`	A claim is explicitly blocked by the evidence file that owns it	Does not mean the underlying engineering artifact failed	`CLAIM_BLOCKED_FOR_ACCEPTANCE` for job `3396039`; `NOT_CONSISTENT_CONTROLLED_COMPARISON` for job `3393320`
V10	`UNVERIFIED`	A value appears in a report but cannot be rebound to a source file inside this snapshot	Does not assert the value is wrong, only that it is unsupported here	V12 display values `33.7 m/s` and `34.1 m/s`; caller-supplied anchor `f278ed02...`

1.4 Headline terminals

The single most consequential structural fact of this snapshot is that terminals are layered and none implies another. `TERMINAL_CLASSIFICATION_3396039.json` records six terminals at once: scheduler `TERMINAL_COMPLETED`, application `TERMINAL_COMPLETED`, wrapper `TERMINAL_EVIDENCE_COMPLETE`, artifact `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE`, usability `PARTIAL_MODEL_OUTPUT_USABLE`, and claim `CLAIM_BLOCKED_FOR_ACCEPTANCE`. Any CFD automation system that reports only the first two categories will systematically overstate its own maturity.

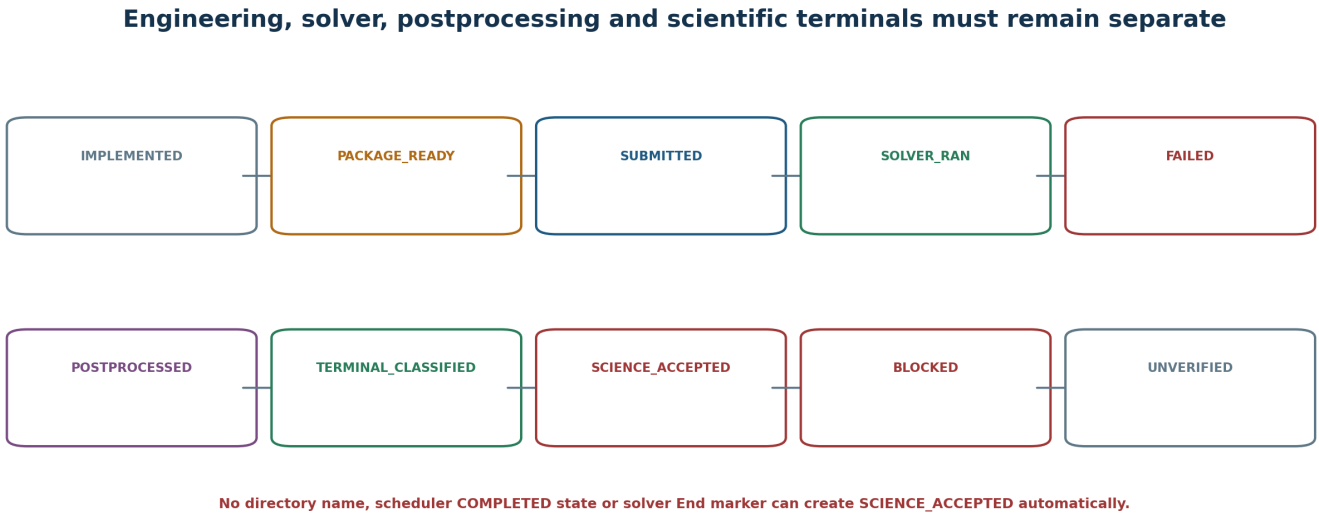


Figure F08. Method diagram. The ten-term closed status vocabulary arranged as separate terminals, with the governing rule that no directory name, scheduler `COMPLETED` state or solver `End` marker can create `SCIENCE_ACCEPTED` automatically. This is the evidence-classification policy of the program, not a validation result.

Table T02. Headline terminal matrix

ID	Line or artifact	Engineering terminal	Scientific terminal	Governing evidence
H01	Cylinder five-case campaign, DFG 2D-2 `Re=100`, local WSL, OpenCFD v2506 serial	`SOLVER_RAN` for all five cases, then `POSTPROCESSED` and packet sealed at `PACKET_SEALED`	`EW_NOT_RELEASED`; grid convergence, time-step convergence, physical validation and automatic benchmark pass all prohibited	`ROUND5_FINAL_DISPOSITION.md`, `SCIENTIFIC_DIAGNOSTIC.md`, `CAMPAIGN_SUMMARY.json`, `CAMPAIGN_COMPARISON.json`
H02	Task 6 `fine_dt0025` cylinder candidate	`IMPLEMENTED` and authorized to mesh, dispatch stops at `MESH_QUALIFIED`	No scientific terminal; no solver authorization exists	`task6_candidate.py`, `task6_execution_lock.py`, `task6_staged_dispatch.py`, `task6_mesh_runner.py`
H03	Case011Up Condor32 production, HTCondor cluster `1542.0`	`SOLVER_RAN`, continued to final logged time `10.999827824854936` against target `11`, exit code `0`	Reference-normalized `Cd_ref = 0.76965972` is an arithmetic output, not an approved physical coefficient for nonstationary inflow	`CONDOR32_PRODUCTION_COMPLETION_EVIDENCE.md`
H04	Case011Up coarse empty ABL, jobs `3382272`, `3383251`, `3383356`	`SOLVER_RAN` then `PATCH_EXTRACTION_EXACT_41_COMPLETE`	; recommended path `B_ENGINEERING_SMOKE_ONLY_PENDING_RESOLVED_GATE`	`terminal_receipt_3382272.json`, `CASE011UP_V046_EMPTY_ABL_FULL_EVIDENCE_PARENT_VERIFIED.md`
H05	Case011Up medium empty ABL, job `3396039`, 680,000 cells	`SOLVER_RAN` and `POSTPROCESSED`, wrapper `PASS`	`PROFILE_CONTRACT_INCOMPLETE` and `CLAIM_BLOCKED_FOR_ACCEPTANCE`; `TERMINAL_CLASSIFIED` across six categories	`MEDIUM_EMPTY_ABL_SCIENCE_REPORT_3396039.md`, `TERMINAL_CLASSIFICATION_3396039.json`, `READ_ONLY_PROFILE_GRID_REANALYSIS_3396039.md`
H06	Case02Up LayerForce50, attempt 14, 50 layers to `11 s`	`POSTPROCESSED`, 50,877 unique strictly increasing samples, 601 columns per row	Unit-density solver forces at `rhoInf=1`; not approved physical or design loads	`CASE02UP_LAYERFORCE50_RESULT_REVIEW.md`
H07	D420 historical M0, job `3384063`	`SOLVER_RAN`, 1,160,000 cells, 48 ranks, final time `260 s`, maximum Courant `0.228`	No matched-domain scientific comparison exists	`D420_M1_VS_M0_SCIENTIFIC_ACCEPTANCE_PROTOCOL.md`
H08	D420 historical M1, job `3386652`	`SOLVER_RAN`, final time `80 s`, output at `50, 60, 70, 80 s` on all 48 ranks	Same as H07; execution and output availability only	`D420_M1_VS_M0_SCIENTIFIC_ACCEPTANCE_PROTOCOL.md`
H09	D420 rebound postprocess, job `3391286`	`SUBMITTED` then `FAILED`, scheduler `FAILED 2:0`, elapsed 57 s, scientific output count `0`	`D420_REBOUND_POSTPROCESS_FAILED_MISSING_PROCESSOR_MESH_NO_SCIENTIFIC_OUTPUT`	`AU-20260810-CM1-CFD-Job-3391286-failure.md`
H10	D420 meshfix v7 postprocess, job `3393320`	`POSTPROCESSED`, four metrics JSON and four figure NPZ files, `new_solver_run false`	and independently `NOT_CONSISTENT_CONTROLLED_COMPARISON` with eight listed conflicts	`D420_MESHFIX_V7_TERMINAL_3393320.json`, `D420_MESHFIX_V7_BOUNDARY_CONSISTENCY_AUDIT_20260811.json`
H11	D420 matched-drive three-arm package	`PACKAGE_READY`, 5,293 files, `submitted false`, `solver_ran false`	Proves input identity and static case completeness only; no mesh quality, solver completion, numerical validity or mechanism claim	`20260811_D420_three_arm_production_package_ready.md`
H12	V12 CM1-to-CFD display archive	34 package files, 33 SHA entries, ZIP `testzip=PASS`, SHA-256 `5d0ca61b...`, `1096344` bytes	`ACCEPTED_BOUNDED_DISPLAY_ARCHIVE` under decision record `D-20260812-001`; a display and provenance archive, not a controlled scientific CFD result	`AU-20260812-005`, `AU-20260812-007`

1.5 Executive synthesis

AutoCFD answers three questions in sequence, and the snapshot answers them unevenly.

The first question is whether a fixed two-dimensional benchmark can be generated, meshed, initialized, solved, continued, analyzed and sealed by one public workflow. The answer here is a complete affirmative on the engineering axis. The cylinder product line owns a six-mode public CLI, a ten-stage state machine, per-stage receipts, checkpoint and resume semantics with completeness criteria, an analyzer loaded only from locked bytes, and a packet verifier that rebuilds comparison documents semantically rather than trusting stored hashes. Five real cases ran to `30 s` with `blockMesh` and `checkMesh` return code `0`, `Mesh OK=true`, `pimpleFoam` return code `0`, an `End` marker present, no `FOAM FATAL` marker, and periodic lift detected.

The second question is whether building CFD can produce complete force, pressure and height-resolved output usable for a structural handoff. The answer is a partial affirmative with explicit open items. Case011Up produced a real building time history on HTCondor and real ABL profiles on TACC, and Case02Up produced a fully resolved 50-layer synchronized force history. What is missing is unification: the building line executes through `fake-exec` plus external Condor and Slurm, so terminals are reassembled after the fact and there are no checkpoint semantics or per-stage receipts for those runs.

The third question is whether TACC OpenCFD v2412 can complete a medium empty-ABL case on real HPC and return profile evidence sufficient to decide the next Building RANS step. The answer is that the solver completed while the sampling contract did not. Job `3396039` reached time `1000` with 6000 residual records and 1000 continuity records on a 680,000-cell mesh reporting `Mesh OK`, and simultaneously returned `39/33/41/31` profile rows against a required 41, so the artifact terminal is `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE` and the claim terminal is `CLAIM_BLOCKED_FOR_ACCEPTANCE`.

The engineering conclusion supported by this snapshot is controlled automation under a fixed scientific contract. The scientific conclusion supported by this snapshot is narrower: model outputs exist, are reproducible from raw data, and are classified honestly, and none has been promoted to physical validation. There is no `SCIENCE_ACCEPTED` instance anywhere in the 92 files.

Chapter 2 AutoCFD Research Purpose and the Manual-CFD Problem

2.1 The stated purpose

The purpose recorded in the parent fact ledger is to compile fixed scientific intent into repeatable, fail-closed CFD execution and evidence without granting automatic scientific acceptance. The two evidence anchors for that statement are `petwcfd_cylinder/science_contract.py` and `control_plane/petwcfd/workflow_plan.py`, and both encode the second half of the sentence as executable code rather than as prose. The contract loader refuses to load a contract whose `claim_boundary.scientifically_validated` is not `false`, whose `automatic_pass_tolerance_approved` is not `false`, or whose `release_status` is not `NOT_RELEASED`. The workflow planner hard-codes `execution_authorized` to `false` on every stage it emits.

The purpose is explicitly not to substitute a language model for CFD judgment. It addresses a concrete research-engineering problem. In manual CFD work, geometry, units, boundary conditions, turbulence models, reference quantities and quantity-of-interest definitions drift across iterations, and after drift the work is neither reproducible nor attributable. PETW needs a way to compile one fixed scientific question into one computational object that is repeatably executable, pre-checkable, resumable, recomputable and auditable.

2.2 The task boundary that framed the work

`STATE.md`, updated 2026-06-15, profile [path], coordinator [path], writes this purpose as a task boundary. The mission is to automate PETW CFD setup, execution packaging, postprocessing and validation interfaces for thunderstorm-wind structural-response studies. The scope is CFD geometry and mesh preparation, inflow boundary-condition generation, solver setup, run manifests, postprocessing scripts, and wind-tunnel validation comparison packets. The handoff boundary receives wind-field feature packets from Windfield and validation data from WindTunnel, and hands pressure, force and flow-field evidence to Structural and DesignLoad; long runs and remote or HPC submissions go through ComputeOps and require PI approval when high impact. The human-in-the-loop rule states that the agent produces evidence, artifacts, uncertainty notes and decision prompts while PI remains the scientific decision maker, and that the agent must distinguish evidence from assumptions and flag claims requiring PI review. The safety rule states that remote PETW files remain read-only unless PI explicitly authorizes writes, that long remote or HPC jobs require PI approval, and that secrets and tokens must not be written to chat or vault files.

The same file records the PI update of 2026-06-15 22:14 EDT: the CFD retest has been sent, and no CFD result may be treated as evidence before returned outputs, logs or validation notes arrive. That rule persists unchanged through every later stage of the program and is the direct ancestor of the `UNVERIFIED` terminal in Table T01.

2.3 The three-level question

AutoCFD decomposes the purpose into three questions asked in order of increasing physical difficulty and decreasing evidence completeness.

Level one is a fixed two-dimensional benchmark with a published authority curve, chosen so that the automation loop can be closed end to end and any defect is attributable to the automation rather than to the physics. This is the DFG 2D-2 time-periodic channel-cylinder case at `Re=100`.

Level two is a finite-height building at model scale with an atmospheric-boundary-layer inlet, chosen because it is the first case whose outputs are structurally meaningful, that is, forces, pressures and height-resolved sectional loads. This is Case011Up and its structural-interface sibling Case02Up.

Level three is a nonstationary, multiscale coupled drive in which the inflow itself is a time-varying field produced by a separate meteorological model. This is the CM1 D420 downburst line, which stresses the AutoCFD method with conservative input mapping, a matched clock, and a comparison design that must be controlled rather than merely executed.

2.4 The manual-CFD problem, itemized

The failure modes below are not hypothetical. Each row of Table T03 is present in the snapshot either as a defect that occurred and was corrected, or as a checker written specifically to make that defect impossible. This is what distinguishes the AutoCFD response from generic software hygiene: the gates were installed against observed failures.

Table T03. Manual workflow failure modes and automation response

ID	Manual failure mode	Research consequence	AutoCFD response	Implementing object
M01	Scientific intent restated informally each iteration, with fields added or dropped	The same nominal case serializes differently; identity cannot be computed	Intent reduced to exactly `{schema_version, case_id, profile, design}`; any extra or missing field rejected	`control_plane/petwcfd/intent.py`
M02	Numbers entered as floats and reformatted by tooling	Byte-level identity of intent is unstable, so `intent_sha256` is meaningless	Numbers arrive as decimal strings through `canonical_decimal_string`, at most 64 characters, adjusted exponent magnitude at most 100, trailing zeros stripped, negative zero normalized to `0`	`intent.py`
M03	Units carried in variable names or in the analyst's head	Silent unit errors survive review	Units inferred deterministically from path suffix: `_kg_m3` to `kg/m3`, `_m2_s` to `m2/s`, `_m_s` to `m/s`, `_s` to `s`, `_m` to `m`, everything else to `1`	`control_plane/petwcfd/resolver.py`
M04	Derived quantities computed ad hoc in a spreadsheet	The path from one design input to solver inputs cannot be reconstructed	Every resolved leaf records `source`, `value`, `units`, `formula_id`, `dependencies`; five frozen formula identifiers; 128-digit decimal context	`resolver.py`
M05	Physical consistency assumed rather than checked	Inconsistent `Umax`, `Re`, `Aref` combinations reach the solver	Contract loader enforces `Umax = 1.5 Umean` to `1.0e-12`, `Re = Umean D / nu`, `Uref_basis = Umean` at `abs_tol=0.0`, `Aref = D * thickness`	`petwcfd_cylinder/science_contract.py`
M06	Analysis window chosen after seeing the result	Window shopping inflates apparent agreement	Exactly three windows frozen (`20_to_25_s`, `22_5_to_27_5_s`, `25_to_30_s`) with exactly one primary window	`science_contract.py`
M07	Solver binary or environment changes between runs	Results are not attributable to a version	Six pre-execution checks including `WM_PROJECT_VERSION v2506`, `WM_PROJECT_DIR`, `WM_OPTIONS`, per-executable SHA-256, symlink rejection, `LD_LIBRARY_PATH` allowlist	`petwcfd_cylinder/runner.py`, `__main__.py`
M08	Arbitrary commands typed at a prompt	The command set is not part of the evidence	Commands come only from `COMMAND_ARGV`; `result.args` must equal canonical argv or the runner raises "executor actual argv differs from canonical executable"	`runner.py`
M09	Executable code injected through solver dictionaries	Case setup becomes an arbitrary program	Only the `#eval` directive permitted; `dynamicCodeDict`, `redirectType`, `code*`, `coded*` rejected; `libs` only in `system/controlDict`, exactly twice, in `rawForces` and `forceCoeffs`, value `(forces)`	`control_plane/petwcfd/preflight.py`
M10	Reusing or overwriting a run directory	Prior state contaminates the new result and cannot be separated	New root required; `atomic_write_relative_no_replace` creates if absent and fails if present; "new campaign root must not already exist"; independently reproduced on D420 as "package root already exists (no overwrite)"	`safe_io.py`, `task8_campaign.py`
M11	Manual restart from a directory that merely looks like a time	Continuation starts from an incomplete or aliased state	Checkpoint requires `U`, `p` and `phi` present and non-empty; canonical decimal directory names; "numeric checkpoint alias rejected"; "incomplete checkpoint"	`petwcfd_cylinder/restart.py`
M12	Metrics copied from a previous report into the next one	An early parsing defect propagates silently forever	Every stage recomputes from unchanged raw files through an analyzer pinned by digest and loaded only from locked bytes	`science_analysis.py`, `replay.py`
M13	Coefficient files parsed by fixed column index	Format change silently swaps quantities; `Cd(r)` was read as `Cl`	Named-column parsing from the actual header, exactly one `Time` header line, no duplicate names, row width equal to header width, `Time` strictly increasing	`science_analysis.py`, `parse_named_table`
M14	Directory named final or accepted treated as approval	Engineering completion silently becomes scientific acceptance	Claim boundary booleans hard-coded false in receipts and packets; ten-term status vocabulary; six separate terminals	`receipts.py`, `TERMINAL_CLASSIFICATION_3396039.json`
M15	Input inventory assumed complete because the run directory looks complete	`reconstructPar` failed after submission with zero scientific output	Processor-level mesh payload promoted to an explicit inventory item: `points`, `faces`, `owner`, `neighbour`, `boundary` for every processor view	`AU-20260810-CM1-CFD-Job-3391286-failure.md` and the meshfix successor

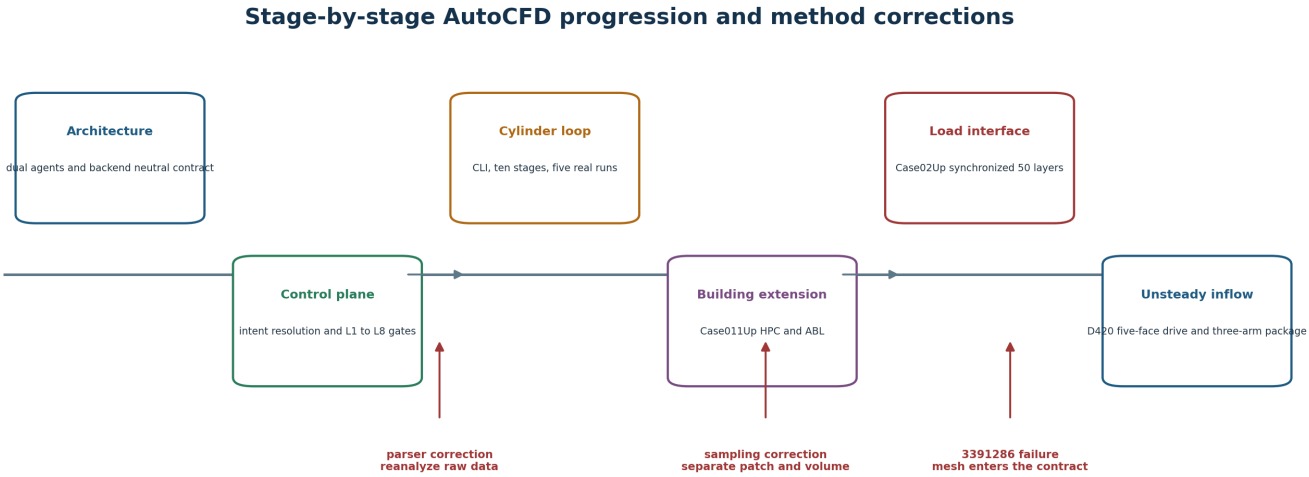


Figure F02. Method diagram. The six-stage AutoCFD progression, architecture, control plane, cylinder loop, building extension, load interface and unsteady inflow, with the three method corrections that entered from below: the parser correction and raw-data reanalysis, the sampling correction separating patch and volume operators, and the job `3391286` failure that moved the processor mesh into the contract. This diagram records program history and method structure, not a validation result.

Chapter 3 Problem Decomposition and Method Formation

3.1 The dual-agent, backend-neutral framing

`20260620_PETW_CFD_Dual_Agent_Fluent_OpenFOAM_Architecture_Note.md`, dated 2026-06-20, records a clarification that set the scope of everything after it. The intended PETW architecture is not a deterministic Fluent front end followed by an agentic OpenFOAM back end. Both sides are intended to be agent driven. The Fluent side operates through MCP or PyFluent for geometry cleanup, watertight or fault-tolerant workflows, face-zone naming, boundary-layer meshing, mesh-quality diagnostics, visual inspection, setup checking and postprocessing. The OpenFOAM side starts by selecting and extending an existing agent-CFD platform rather than by building a new generic CFD agent.

The note defines four layers: a Fluent-side agent layer, a manifest and evidence layer, an OpenFOAM-side agent layer, and a human-in-the-loop gate. The second layer is the technical pivot for all later work. It is a backend-neutral schema recording geometry version, units, face-zone to patch mapping, mesh metrics, near-wall targets, reference quantities, force and moment definitions, validation gates, and human review status.

The note states its own status as PI framing and candidate application strategy, not a validated workflow, and explicitly requires that no manuscript or proposal imply that PETW has implemented or validated the full dual-agent architecture. The snapshot contains no Fluent-side implementation file, so on the evidence the dual-agent architecture is currently implemented on the OpenFOAM side only. This is an important scope statement for the whole of this study: the architecture is `IMPLEMENTED` on one side and remains a design intent on the other.

Four-layer AutoCFD architecture: authority, execution and scientific judgment remain separate

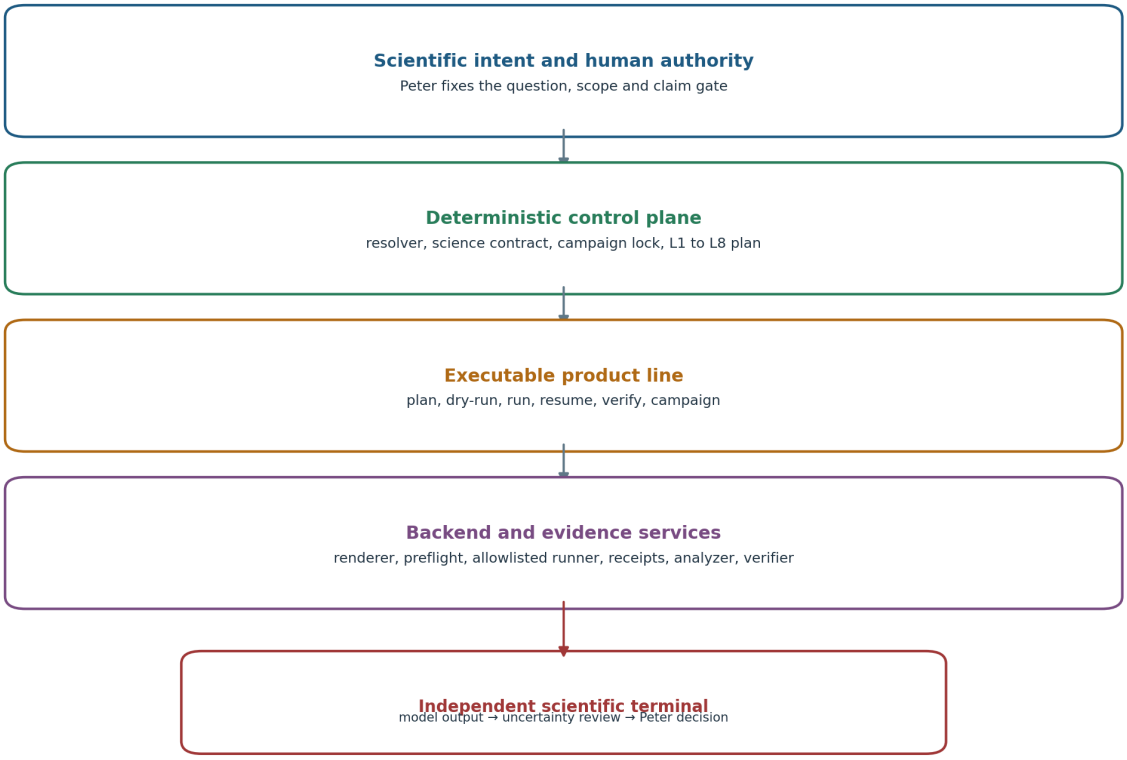


Figure F03. Method diagram. The four architectural layers defined in the 2026-06-20 note, with the manifest and evidence layer as the backend-neutral pivot and the human-in-the-loop gate as the terminal authority. This is the intended architecture and its scope statement, not a validation result.

3.2 Minimizing and determinizing scientific intent

The backend-neutral schema was realized in two places: the intent, profile and resolver triple of the generic control plane, and the case-specific science contracts.

`control_plane/petwcfd/intent.py` reduces user input to the smallest possible object. The top-level field set must equal exactly `{schema_version, case_id, profile, design}`, and any extra or missing field is rejected. `schema_version` must be `petwcfd.intent.v1`. `profile.id` currently accepts only `cylinder2d_re100_similarity_v1`. `design` carries only `diameter_m`. Numbers arrive as decimal strings rather than floats and pass through `canonical_decimal_string`. This removes the possibility that the same intent serializes to different bytes twice, which is exactly what makes `intent_sha256` usable as an identity rather than as decoration.

`profiles.py` freezes the profile registry. The file digest of `cylinder2d_re100_similarity_v1` is `b7b722f9f2d2ef9ad9f5d217279cfcbbcecc360c5d6320e003f8ed2e18051ac0` and its canonical digest is `4dff313f82bfd5ad307578807b9d879278442ee78f4a4332833b400883392b7a`. `resolver.py` rejects any profile path containing a symlink component and requires the file digest to equal both the registered digest and the digest declared in the intent.

3.3 The semantic resolver: units, provenance and derivation formulas

The core of `resolver.py` is not numerical computation but provenance. Every resolved leaf field records a five-tuple of `source`, `value`, `units`, `formula_id` and `dependencies`. Units are inferred deterministically from path suffixes. Derivations are carried by five frozen formula identifiers: `GEOM_CYLINDER_SIMILARITY_V1` for span, domain lengths and cylinder center; `FLOW_FIXED_NU_TARGET_RE_V1` for `U_ref` from diameter, target Reynolds number and fixed `nu`; `FORCE_CYLINDER_PROJECTED_AREA_V1` for `A_ref`, `L_ref` and center of rotation; `MESH_FROZEN_MEDIUM_V1`; and `TIME_LEGACY_MESH_ONLY_V1`. Resolution runs in a 128-digit decimal context and emits a `resolved_payload` plus an eleven-field receipt whose `solver_eligible` and `qualification_status` feed every later gate.

The scientific value of this step is that the path from one user-supplied diameter to the full set of solver inputs can be independently recomputed and explained without trusting any natural-language description of what was done.

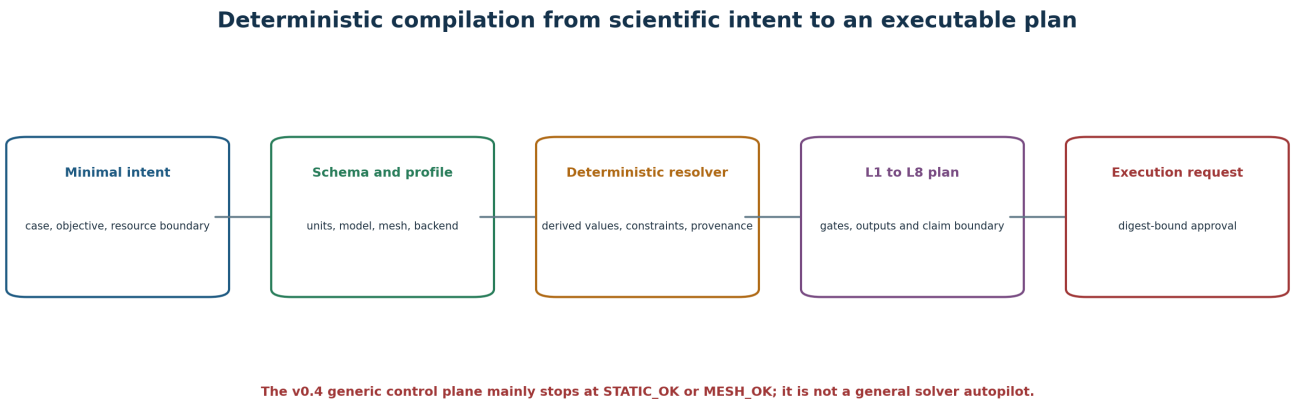


Figure F04. Method diagram. The deterministic path from a four-field intent through profile digest verification, unit inference and formula-identified derivation to an eleven-field receipt and a gated plan whose `execution_authorized` is hard-coded false. This is the method of compilation, not a validation result.

3.4 Science contracts that encode physical consistency

The cylinder `science_contract.py` validates physics, not only schema. `Umax` must equal `1.5 * Umean` to relative tolerance `1.0e-12`. The declared `Re` must equal `Umean * D / nu`. The coefficient `Uref_basis` must be `Umean` with exact numerical equality at `abs_tol=0.0`, `Lref_basis` must be `cylinder_diameter`, and `Aref_basis` must be

`cylinder_diameter_times_thickness` with `Aref` equal to `D * thickness`. The analysis windows are frozen to exactly three, `20_to_25_s`, `22_5_to_27_5_s` and `25_to_30_s`, with exactly one primary window, `25_to_30_s`. Finally, `claim_boundary.scientifically_validated` must be `false`, `automatic_pass_tolerance_approved` must be `false`, and `release_status` must be `NOT_RELEASED`, or contract loading fails outright. The rule that scientific validation cannot be granted automatically is therefore part of the loader rather than a sentence in a document.

The building-side `autocfd_case011up/contract.py` does the same for building geometry and boundaries. `case_id` must be `Case011Up`; geometry type must be `finite_height_square_prism`; span must be `(0.06, 0.06, 0.48001)` m; the domain must be `((-2.5, 7.5), (-1.7, 1.7), (0.0, 2.5))` m; the active STL digest must equal `d3d99859602b7a4f7162a1f76c409b12c0ae935eca3152a8b4ba464f21d75af1`; and the six boundary types are fixed as `inlet=abl_log_law`, `outlet=pressure_outlet`, `ground=rough_wall`, `building=no_slip_wall`, `top=symmetryPlane`, `lateral=symmetryPlane`. Force outputs require `n_bins=20`, `cumulative=false`, `direction=[0,0,1]`, `CofR_m=[0,0,0]` and `closure_required=true`; flow-profile fields must be exactly `U`, `k`, `epsilon` and `nut`. The contract is staged: `pressure_dimensions` may remain `inspect_runtime_field` at the `static` stage but must be frozen at any solver stage, and `p_ref` must be numeric rather than `[label]` at any solver stage.

The D420 `contract.py` freezes the fullest identity in the snapshot: `case_id D420EmptyDomain`; production packet `178929854` bytes with SHA-256 `b0dc479d5a48d649fe29a485c681992e227c3ed500dda1459f46c39a83471069`, scientific-content SHA-256 `7cc2aa62f475cf239fb4bdc582dbc52e5f9020acc37fd380bfd22b630758a50a` and fixture SHA-256 `b2140991a3f7ce452a4f787846dde6959a69c77d1fea1b720c2e8c1932cf8481`; a clock contract of 261 records with CFD relative time `[0.0, 260.0] s`, source time `[2120.0, 2380.0] s`, write index `[115, 375]`, cadence `1.0 s` and `no_extrapolation true`; M0 topology `[200, 200, 29]` with 1,160,000 cells and 1,212,030 points; C0 topology `[100, 100, 29]` with 290,000 cells and scientific use `NONE_ENGINEERING_MECHANICS_ONLY`; boundary condition types `d420TimeVaryingMappedInletOutletVelocity` and `d420ComplementaryPressure`; and a velocity policy of `preserve_raw_source_vectors_exactly_no_correction`. Of the seven D6B decision keys, `initial_projection_policy` must decide exactly `PROJECTION_FORBIDDEN_WITHOUT_SEPARATE_APPROVAL`; `[label]` is permitted at the `static` and `c0_smoke` stages, while the `m0_package` stage lists and rejects every pending field.

3.5 Mesh profiles, backends and target registries

`autocfd_case011up/mesh_profiles.py` requires the registered profiles to be exactly `coarse`, `medium` and `fine`; `fine` must remain `enabled=false`; the medium background cell total must exceed coarse; the medium surface refinement maximum level must not be coarser than coarse; and the canonical digest of the whole payload must equal `76adabd68fd65378752104c8594ca73804023f04fa364e5d2962209c2a174f17`.

`openfoam_backends.py` allows exactly one active backend, `tacc_openfoam_com_v2412`, host `TACC_Stampede3`, `exact_version v2412`, with the activation string fixed to `module purge; module load intel/24.0 impi/21.11 openfoam/v2412`, required executables exactly `foamDictionary`, `blockMesh`, `surfaceFeatureExtract`, `snappyHexMesh`, `checkMesh`, `simpleFoam` and `pimpleFoam`, and `qualification` restricted to `STATIC_ONLY` or `UNQUALIFIED`.

`stampede3_targets.py` freezes four execution targets: `empty_abl` (coarse, `skx-dev`, 1 node, 4 ranks, `00:30:00`, six required gates), `empty_abl_medium` (medium, 4 ranks, `00:45:00`), `building_rans` (medium, 48 ranks, `02:00:00`, eight required gates of which the first is `empty_abl_acceptance`), and `building_urans` (`blocked_pending_mesh_derived_time_controls`, eleven required gates). The registry requires `execution_authorized`, `remote_write_authorized` and `submission_authorized` to be `false`, `allocation` to be `null`, and `evidence_status` to be `DATED_COMPUTEOPS_ADVICE_REQUIRES_FRESH_RECHECK`.

This registry is the clearest example in the project of a scientific gate written into code: `building_rans` cannot legally start before `empty_abl_acceptance`, and that acceptance has not been granted to date.

The limitation of the contract layer is that it can prove internal consistency and non-tampering but cannot prove that the contract is physically correct. Round 5 records one real gap at this layer: the generated source manifests record an earlier contract digest, but those historical contract bytes were not preserved, and `round5_benchmark/CONTRACT_TRANSITION_LIMITATION.json` records that evidence limitation, so byte-complete contract continuity across generation and continuation is not claimed.

3.6 What is invariant and what is case specific

Table T04. Invariant versus case-specific decisions

ID	Decision	Class	Where frozen	Procedure to change it
D01	Intent field set and canonical decimal representation	Invariant across all cases	`intent.py`	Code change plus new schema version `petwcfd.intent.vN`
D02	Unit inference from path suffix	Invariant	`resolver.py`	Code change; affects every derived quantity
D03	Five derivation formula identifiers	Invariant registry, case-selected members	`resolver.py`	New `formula_id` added to the frozen registry
D04	Claim-boundary booleans false and `release_status NOT_RELEASED`	Invariant and non-overridable	`science_contract.py`, `receipts.py`, `workflow.py`	Cannot be changed by configuration; would require code change and separate approval
D05	Ten-stage lifecycle names and order	Invariant for the cylinder line	`receipts.py`, `workflow.py`	Code change; `plan.stages` is validated field by field
D06	Canonical executable map and per-executable SHA-256	Invariant per backend build	`runner.py`	New backend registration with new digests
D07	Cylinder geometry, `Umean`, `Umax`, `Re`, `nu`, `Aref` basis	Case specific, frozen per case	`science_contract.py`	New contract file and new campaign lock
D08	Three analysis windows and one primary window	Case specific, frozen	`science_contract.py`	New contract; windows cannot be selected after seeing results
D09	Case011Up geometry, domain, STL digest, six boundary types	Case specific, frozen	`autocfd_case011up/contract.py`	New contract; STL digest must be re-registered
D10	Force output `n_bins=20`, `direction=[0,0,1]`, `CofR_m=[0,0,0]`	Case specific	`autocfd_case011up/contract.py`	Case02Up uses 50 layers, which is why it is a separate product line
D11	Mesh profile registry `coarse`, `medium`, `fine` with `fine` disabled	Case specific with invariant ordering constraints	`mesh_profiles.py`	Payload canonical digest must be updated deliberately
D12	Execution target gates, in particular `empty_abl_acceptance` before `building_rans`	Case specific gate, invariant ordering	`stampede3_targets.py`	grants the named gate; code does not grant it
D13	D420 clock contract, 261 records, `no_extrapolation true`	Case specific, frozen	`autocfd_d420_empty_domain/contract.py`	New packet identity and new contract
D14	D420 velocity policy `preserve_raw_source_vectors_exactly_no_correction`	Invariant policy for this line	`contract.py`, `mask_reference.py`	Would require separate approval; no flux correction is computed anywhere downstream
D15	M1 URANS parameters `U_REF_M_S=15.0`, `TURBULENCE_INTENSITY=0.10`, `TURBULENCE_LENGTH_M=500.0`, `CMU=0.09`	Case specific	`render_m1.py`	New render parameters; `k` and `omega` follow by frozen formula

3.7 Division of authority between agent, code and PI

The snapshot contains no code that calls a language-model API. Language-model participation appears in the evidence as three boundary rules rather than as an execution path.

First, `workflow_plan.py` sets `automatic_correction_authorized` permanently to `false`, alongside `mesh_execution_authorized`, `solver_execution_authorized`, `remote_execution_authorized` and `hpc_submission_authorized`, all also permanently `false`.

Second, candidate construction is deterministic rather than generative. `task6_candidate.py` defines `fine_dt0025` as the fine mesh controls combined with the medium `dt=0.0025` `controlDict`, and hard-codes the SHA-256 of thirteen generated files, the mother repository `https://github.com/AndreWeiner/ml-cfd-lecture.git` at commit `7bcd52a3722795c5e038354421d843495c5ef326`, path `test_cases/cylinder2D`, and the mother manifest digest `a779e3e8d1ecd95c4c73910a6552b36c48e153d1a99c13521536e20bd0f4f4de`.

Third, any correction that touches execution requires a new approval artifact. `task6_execution_lock.py` carries its own lock and authorization digests, `2cc9b6c5...` and `229a08f7...`, under policy identifier `petwcfd.task6-fine-dt0025-bounded-local-serial.v1`. `task6_staged_dispatch.py` permits only four stages, `CONTRACT_LOCKED`, `GENERATED`, `MESH_BUILT` and `MESH_QUALIFIED`, explicitly forbids `setExprBoundaryFields`, `potentialFoam` and `pimpleFoam`, and pins the argv, path, SHA-256 and byte size of `blockMesh` and `checkMesh`. `task6_mesh_runner.py` declares that only a fresh root may execute and that execution stops at `MESH_QUALIFIED`.

The `architecture_references` field of the workflow plan lists six external systems, `AutoCFD_NL2FOAM`, `ChatCFD`, `CFD_copilot`, `Foam_Agent_2_0`, `OpenFOAMGPT_2_0` and `PhyNiKCE`, with `knowledge_status` set to `SOURCE_CLAIMED_PATTERNS_LOCAL_CONTRACT`. That is an honest label: what is borrowed is the role decomposition these systems claim, the local contract is authoritative, and none of their code was reused or performance verified.

Table T05. Authority division: agent or LLM, deterministic code, and PI

ID	Decision class	Agent or LLM authority	Deterministic code authority	[redacted] authority	Evidence
A01	Choice of scientific question and case definition	None	None	Sole; frozen into a contract before any code runs	`science_contract.py`, `autocfd_case011up/contract.py`
A02	Orchestration order of stages	Proposal only	Sole enforcement; `plan.stages` validated field by field against frozen definitions	Approves the plan digest	`workflow.py`, `runner.py`
A03	Which OpenFOAM commands may execute	None	Sole; `COMMAND_ARGV` plus `CANONICAL_EXECUTABLES` plus per-executable SHA-256	Approves the backend registration	`runner.py`, `__main__.py`
A04	Environment and library path	None	Sole; six pre-execution checks, nine-variable child environment, four approved library directories	None routinely	`__main__.py` `_subprocess_executor`
A05	Candidate case construction for a new numerical point	None in this snapshot	Sole; deterministic recombination with thirteen pinned file digests	Approves execution scope separately	`task6_candidate.py`
A06	Automatic correction of a failing run	Forbidden	Forbidden; `automatic_correction_authorized` permanently false	Sole; a new approval artifact is required	`workflow_plan.py`, `task6_execution_lock.py`
A07	Mesh execution authorization	None	Enforces; refuses when [redacted] is not `true`	Grants, bound to `case_id`, `profile_sha256`, `resolved_payload_sha256` and `allowed_until`	`execution_request.py`
A08	Solver execution authorization	None	Enforces; [redacted] must be `false` in the generic control plane, so it cannot run a solver at all	Grants separately in the cylinder line through the execution binding	`execution_request.py`, `runner.py`
A09	Remote submission	None	Enforces; `authorizes_submission`, `remote_write_authorized`, `submission_authorized` all false; `fresh_presubmit_required` true	Sole; one submission scope at a time	`route_b_request.py`, `stampede3_targets.py`
A10	Interpretation of a model output as physics	Advisory text only, flagged for review	Refuses; claim-boundary booleans false, `NOT_RELEASED`	Sole; no automated path exists	`science_contract.py`, `TERMINAL_CLASSIFICATION_3396039.json`
A11	Granting `empty_abl_acceptance`	None	Enforces the ordering constraint into `building_rans`	Sole; not granted to date	`stampede3_targets.py`
A12	Tolerance changes for tool comparison	Proposal with recorded rationale	Applies only the documented value	Notified; the change must not alter scientific or remote scope	2 mm `zhval` tolerance change in the D420 G2 readback
A13	Selection of a physical anchor or crop origin	None	Refuses to select; records candidates and their exact coordinates	Sole; no anchor selected to date	`2026-08-05_CM1_D420_G2_readback_and_CFD_pilot_contract_review.md`

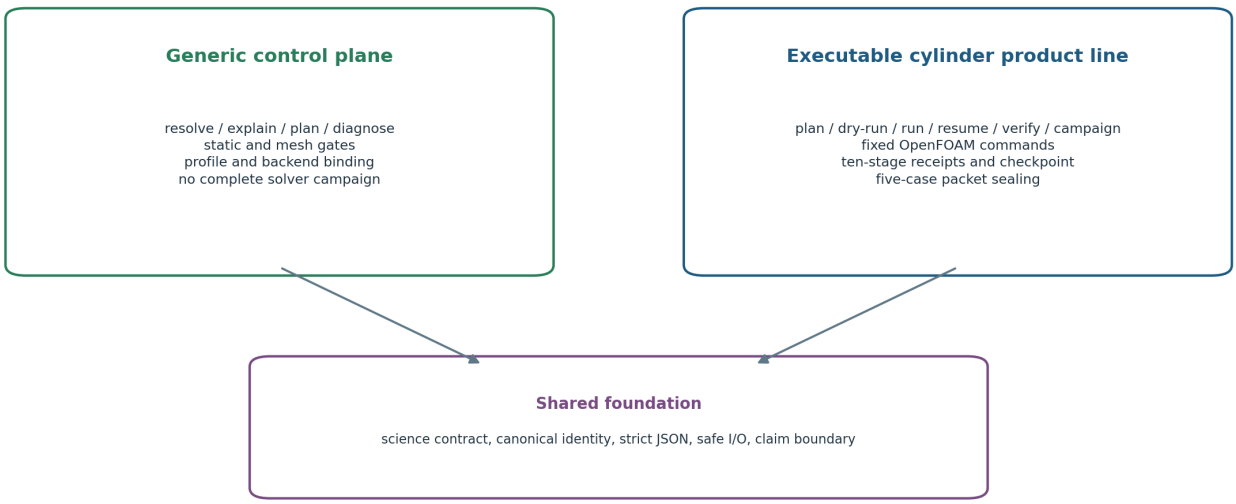


Figure F05. Method diagram. The separation between the generic L1 to L8 control plane, whose maximum permitted gate is `MESH_OK` and whose solver authorization is permanently false, and the case-specific executable product lines that own a full state machine under an explicit execution binding. This is a division of program authority, not a validation result.

Chapter 4 Executable Automation Mechanism

4.1 Two public command-line surfaces and one case-specific surface

The snapshot contains three distinct public command-line surfaces, and confusing them is the most common way to overstate what AutoCFD can do.

The generic control plane CLI, `control_plane/petwcfd/__main__.py`, provides `resolve`, `explain`, `plan`, `diagnose` and a deliberately restricted `run`. The `--until` argument accepts only `STATIC_OK` or `MESH_OK`. If the caller passes `--manifest`, the program raises "direct manifest execution is disabled in PETWCFD v0.4 development" and requires `--intent` together with a digest-bound `--approval` and a `--profile-dir`. `execution_request.py` binds that approval three ways: `case_id` must equal the resolved `case_id`; `profile_sha256` and `resolved_payload_sha256` must equal the corresponding receipt digests; and `allowed_until` must equal the requested gate. It also requires [label] to be `true` and [label] to be `false`. By construction this control plane cannot be used to run a solver.

The cylinder product-line CLI, `petwcfd_cylinder/__main__.py`, provides exactly six subcommands and is the only surface in the snapshot that completes a full solver loop. Its module docstring states the constraint plainly: OpenFOAM commands are fixed in runner code and require the authorized, preloaded local OpenFOAM.com v2506 serial environment, and no user-supplied command is accepted.

The Case011Up case-specific CLI, `autocfd_case011up/route_b_cli.py`, provides twenty-four subcommands across three families: the base smoke family `render`, `preflight`, `plan`, `request`, `fake-exec`, `ingest`, `analyze`, `replay`; the v0.6 postprocess family with the `pp-` prefix; and the time-5 qualification-profile family with the `pp5-` prefix. Every one of the twenty-four is a local static action. The decisive detail is that the execution verb is `fake-exec` and not `run`: the CLI can write a synthetic acceptance bundle to exercise the ingest, analyze and replay chain, but real execution must be carried out externally by Condor, Slurm or ComputeOps.

Table T06. Public CLI semantics

ID	Surface	Subcommand	Required arguments	Action	Side-effect boundary	Failure gate
C01	Cylinder	`plan`	`--science-contract`, `--lock`, `--case-id`, `--root`	Loads contract and lock, generates and writes `PLAN.json`	No OpenFOAM invocation	`ScienceContractError`, `CampaignLockError`, `WorkflowError`, exit 1
C02	Cylinder	`dry-run`	`--science-contract`, `--lock`, `--case-id`, `--root`	Same loaders as live execution, prints an `openfoam_would_execute` list	`openfoam_executed` permanently `false`	Same as C01
C03	Cylinder	`run`	`--science-contract`, `--lock`, `--approval`, `--source-case`, `--root`, optional `--until` from the ten frozen stages, default `PACKET_SEALED`	Loads the verified execution binding, then executes the fixed workflow in a new run root	New root only; no overwrite; nine-variable child environment; 7200 s timeout	`RunnerError` exit 1; `subprocess.TimeoutExpired` exit 2
C04	Cylinder	`resume`	Same arguments as `run`	Validates existing receipts, checkpoints and control dictionaries, then advances only incomplete stages	Never re-executes a completed stage; restores `controlDict` bytes in a `finally` block	`RECEIPT_VALIDATION_ERROR` written to `failures/`; `TERMINAL_CHECKPOINT_WITHOUT_RECEIPT`; `CHECKPOINT_AFTER_TERMINAL_TIME`
C05	Cylinder	`verify`	`--lock`	Replays the campaign lock and evidence bindings, prints the replay receipt	Produces no new solver output	`CampaignLockError` exit 1
C06	Cylinder	`campaign`	`--science-contract`, `--lock`, `--approval`, `--decision`, `--source-root`, `--root`, mutually exclusive `--dry-run` or `--resume`	Processes the five cases in code-frozen order and seals a unified packet	Non-resume mode requires a non-existent root; resume mode resumes only existing child roots and creates only missing ones	"new campaign root must not already exist"; per-case status mismatch during sealing
C07	Control plane	`resolve`, `explain`, `plan`, `diagnose`	`--intent`, `--profile-dir`	Resolution, provenance explanation, eight-layer plan, diagnostics	Static only	Profile digest mismatch; symlink component in profile path
C08	Control plane	`run`	`--intent`, `--approval`, `--profile-dir`, `--until` restricted to `STATIC_OK` or `MESH_OK`	Static and mesh gates only	Cannot reach a solver stage under any argument combination	"direct manifest execution is disabled in PETWCFD v0.4 development"; approval triple-binding failure
C09	Case011 Up	`render`, `preflight`, `plan`, `request`, `fake-exec`, `ingest`, `analyze`, `replay`	Package and bundle paths	Deterministic render, fail-closed static verification, exact plan, governed ComputeOps request, synthetic acceptance bundle, terminal ingest, deterministic analysis, replay without dispatch	All local and static; `request` authorizes nothing	Unresolved `{{` or `}}` token; scheduler or command token present; "request does not match deterministic rebuild from the package"
C10	Case011 Up	`pp-` family, eight subcommands	Package and bundle paths	Same eight-step chain for the v0.6 postprocess qualification package	Local static only	Same gates as C09
C11	Case011 Up	`pp5-` family, eight subcommands	Package and bundle paths	Same eight-step chain for the time-5 qualification-profile package	Local static only	Same gates as C09

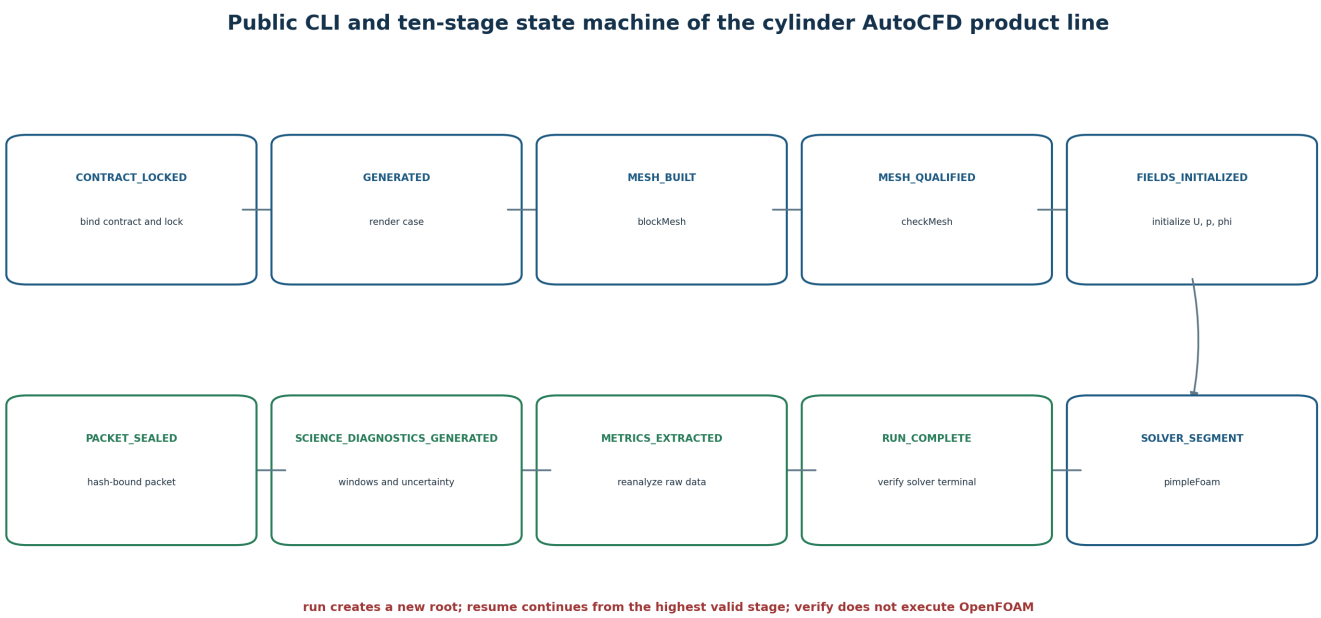


Figure F06. Method diagram. The six cylinder CLI modes and the ten frozen lifecycle stages they can reach, showing that `plan`, `dry-run` and `verify` produce no solver output while `run`, `resume` and `campaign` are the only paths to `SOLVER_SEGMENT` and beyond. This diagram documents command semantics, not a validation result.

4.2 The ten-stage lifecycle

`receipts.py` freezes the ten stage names under receipt schema `petwcfd.stage-receipt.v2`. `workflow.py` binds each stage to a description and an OpenFOAM command list under plan schema `petwcfd.workflow-plan.v1`, and validates that `plan.stages` equals the frozen definition field by field, that `plan.openfoam_stages` equals `[2, 3, 4, 5]`, that the three boolean fields in `plan.claim_boundary` are JSON `false`, and that `release_status` is `NOT_RELEASED`. The plan key set must equal exactly `{schema_version, case_id, science_contract_sha256, campaign_lock_sha256, run_root, stages, total_stages, openfoam_stages, claim_boundary}`, and each stage record must carry exactly `{index, name, description, openfoam_commands, requires_openfoam}`.

Table T07. Ten-stage lifecycle: inputs, actions, outputs and failure gates

#	Stage	Inputs	Action and canonical command	Outputs	Failure gate
0	`CONTRACT_LOCKED`	Canonical science contract, campaign lock, execution authorization, approved plan, analysis implementation	Loads, validates and byte-fingerprints all five artifacts; no OpenFOAM execution	Stage receipt with claim boundary	Canonical JSON re-serialization mismatch; `trust_anchors.canonical_science_contract` field mismatch; loader-private binding token failure
1	`GENERATED`	Locked contract, verified source manifest	Generates `blockMeshDict`, `controlDict`, `transportProperties`, `fvSchemes`, `fvSolution` and boundary conditions from the locked contract; no OpenFOAM execution	Case tree written into a new root by absent-or-identical writes	"source changed while snapshotting"; manifest `execution_performed` not `false`; source file set not equal to manifest keys plus `petwcfd_case_manifest.json`
2	`MESH_BUILT`	Generated case tree	Executes `blockMesh`	Mesh, log, stage receipt	Executable digest mismatch `11bfeeb39cbfbf37e6846e4712b027025b7954a3985eafc2f2ee19a038ad780b`; non-zero return code; argv mismatch
3	`MESH_QUALIFIED`	Built mesh	Executes `checkMesh`	`Mesh OK` assessment, log, stage receipt	Executable digest mismatch `95efe7503bf03c8d0b7a761264f4b5d26b071ab5fb62b5a0420a0e45893d059b`; quality assessment failure
4	`FIELDS_INITIALIZED`	Qualified mesh, boundary expressions	Executes `setExprBoundaryFields` then `potentialFoam -initialiseUBCs -writephi`	Initial `U`, `p`, `phi`	Executable digests `c7b6c9e4...` and `c253b638...`; argv must equal the canonical tuple exactly
5	`SOLVER_SEGMENT`	Initialized fields, locked time controls	Executes `pimpleFoam` over the locked segment; during continuation temporarily rewrites `startFrom` to `latestTime` and restores original bytes in a `finally` block	Time directories, force and probe raw files, log	Executable digest `76523e34...`; "physics dictionaries changed during continuation"; incomplete checkpoint
6	`RUN_COMPLETE`	Solver logs and time directories	Verifies the solver end state, `endTime` reached and residual behaviour; no OpenFOAM execution	End-state verification record	Missing `End` marker; `FOAM FATAL` marker present
7	`METRICS_EXTRACTED`	Unchanged raw coefficient, probe, force and moment files	Runs the analyzer pinned by the campaign lock through `_locked_reanalyzer`; no OpenFOAM execution	`METRICS.json` with status `MODEL_OUTPUT_REANALYSIS`	Header line count not exactly one; duplicate column names; row width mismatch; `Time` not strictly increasing; analyzer `__module__` mismatch
8	`SCIENCE_DIAGNOSTICS_GENERATED`	Extracted metrics, frozen windows, bound authority values	Generates window analysis, contrast and authority comparison in Python; no OpenFOAM execution	`DIAGNOSTICS.json` with status `MODEL_OUTPUT_DIAGNOSTICS_NOT_APPROVAL`	Window set not equal to the three frozen windows; primary window not unique
9	`PACKET_SEALED`	All case artifacts	Seals the output packet and byte-fingerprints it; no OpenFOAM execution	`RUN_RECEIPT.json` with `stage=PACKET_SEALED` and `status=COMPLETE`, sealed packet	Exact 21-path file set mismatch; `SHA256SUMS.txt` byte mismatch; semantic rebuild mismatch of `CAMPAIGN_COMPARISON.json`, `NUMERICAL_UNCERTAINTY.json` or `SCIENCE_REVIEW_REQUEST.md`

4.3 Runner mechanics: what happens before any command

`_subprocess_executor` performs six checks before any command is launched. First, `WM_PROJECT_VERSION` must be `v2506`, `WM_PROJECT_DIR` must be `/usr/lib/openfoam/openfoam2506`, and `WM_OPTIONS` must be `linux64GccDPInt32Opt`. Second, `LD_PRELOAD` and `LD_AUDIT` must be empty. Third, `argv[0]` must be in the `CANONICAL_EXECUTABLES` map. Fourth, the target executable must be a real file and not a symlink. Fifth, its SHA-256 must equal the value registered in code. Sixth, every `LD_LIBRARY_PATH` entry must be one of four approved directories, namely the platform `lib`, `lib/sys-openmpi`, `lib/dummy`, and `/usr/lib/x86_64-linux-gnu/openmpi/lib`; two known bashrc directories are explicitly skipped by returning `None`; empty entries are rejected with "empty LD_LIBRARY_PATH entries are forbidden"; and any entry that is not an exact approved directory raises. The child process receives only nine environment variables, `PATH`, `LANG`, `LC_ALL`, `HOME`, `USER`, `TMPDIR`, `WM_PROJECT_VERSION`, `WM_PROJECT_DIR` and `WM_OPTIONS`, plus the filtered `LD_LIBRARY_PATH`, with a timeout of `7200` seconds.

Commands do not come from configuration; they come from `COMMAND_ARGV`. `MESH_BUILT` maps to `("blockMesh",)`, `MESH_QUALIFIED` to `("checkMesh",)`, `FIELDS_INITIALIZED` to `("setExprBoundaryFields",)` and `("potentialFoam", "-initialiseUBCs", "-writephi")`, and `SOLVER_SEGMENT` to `("pimpleFoam",)`. After execution, `result.args` must equal the canonical argv exactly, or the runner raises "executor actual argv differs from canonical executable".

`load_verified_execution_binding` reads and binds five artifacts before any stage: the science contract with SHA-256 `ea82f44c7d0e94de0970a6bcb56a4e36544a15993db8ab9cc40d9ee3daee69b2`, the campaign lock `713a64359d71ecc8244fe280c07caab741987954e38990634d28cd5a86308e28`, the execution authorization

`79df24ba1d4f1c04d6600ad2ecb19fb65ba1422f8fe862437260da02ba2bf2f0`, the approved plan `03ee24dff81221a6ccdd44f8708f5fa75d9572c401d37ed4d4c1c624406d4df6`, and the analysis implementation `41b7a579bae29c798618f0037a8f593f27e286a9a0b391ea68f4d19eed3a9452` at 54,308 bytes. Contract and lock must additionally be canonical JSON, meaning re-serialization reproduces the same bytes. The authorization `plan_sha256_at_authorization` must equal the actual plan digest. The lock `trust_anchors.canonical_science_contract` must equal `{path, sha256, size_bytes}` field by field, and the `analysis_implementation` anchor must agree with the file on disk in all three fields.

`VerifiedExecutionBinding` is loader-private: its constructor checks `token is _BINDING_TOKEN` and otherwise raises `TypeError`, and `_validate_execution_binding` rejects any plain mapping and re-reads all five files, recomputing size and digest, before each execution. A caller cannot fabricate a binding dictionary that merely looks correct.

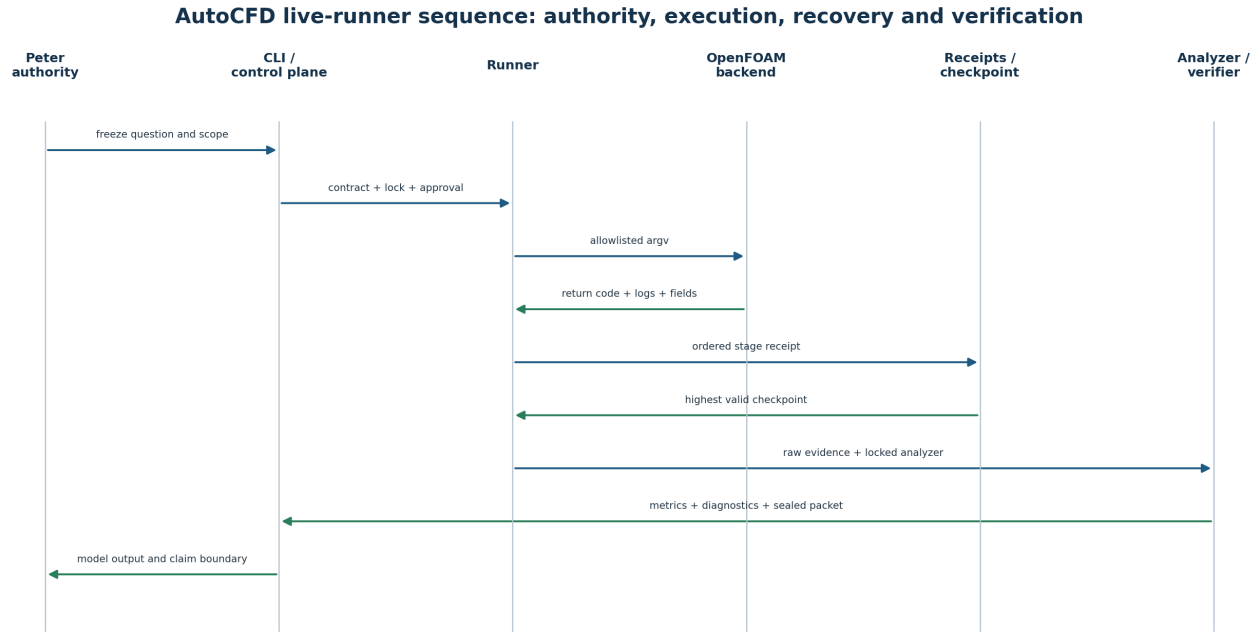


Figure F07. Method diagram. The per-stage runner sequence: load and revalidate the five-artifact binding, select argv from the code-owned map, run the six pre-execution environment and fingerprint checks, execute, compare actual argv against canonical argv, re-hash the frozen sources and the four physics dictionaries, then write the receipt. This documents the execution mechanism, not a validation result.

4.4 New roots, absent-or-identical writes and no overwrite

`copy_verified_source` copies only files listed in the manifest and verifies each digest, re-checks during copying that the source has not changed with "source changed while snapshotting", rejects a source manifest whose `execution_performed` is not `false`, and requires the actual source file set to equal the manifest key set plus `petwcfd_case_manifest.json`. Writes go through `safe_io.atomic_write_relative_no_replace`, which creates if absent and fails if present, without overwriting. `safe_io` itself requires the platform to provide `O_DIRECTORY`, `O_NOFOLLOW` and `O_NONBLOCK` plus `dir_fd` support for `open`, `link`, `unlink` and `mkdir`, and refuses to operate otherwise; all path traversal opens directories level by level through descriptors, which forecloses symlink substitution.

`run_campaign` calls `create_directory(root)` in non-resume mode and reports "new campaign root must not already exist" if the directory is present. The same principle is independently reproduced on the D420 side: one background build, process `proc_461cad454fee`, reused an existing temporary package root and was refused by the renderer with "package root already exists (no overwrite)", while the clean-root build, process `proc_30cb98283905`, completed normally in `111.92` seconds.

4.5 Checkpoint and resume semantics

`highest_complete_checkpoint` in `restart.py` accepts nothing that merely looks like a time directory. Directory names must match canonical decimal syntax; two different names resolving to the same numeric value raise "numeric checkpoint alias rejected"; negative values and non-directories are rejected; and the directory must contain `U`, `p` and `phi` all present and non-empty, with any absence reported as "incomplete checkpoint". The partial-checkpoint policy is fail-closed.

The resume path validates that the root is safe and free of symlinks, reloads the source and runtime manifests where runtime mode permits `0/U` and `0/p` to differ from source, recognizes and repairs one known crash residue, namely that if the root `controlDict` equals the version with `startFrom startTime` rewritten to `latestTime` the original is restored, ensures the eight runtime directories exist, and takes the highest complete checkpoint as `continuation_start`. During continuation, `SOLVER_SEGMENT` temporarily rewrites `startFrom` to `latestTime` and unconditionally restores the original bytes in a `finally` block.

If `continuation_start >= 30.0` without a valid `SOLVER_SEGMENT` receipt, the runner writes `TERMINAL_CHECKPOINT_WITHOUT_RECEIPT` at exactly `30.0` or `CHECKPOINT_AFTER_TERMINAL_TIME` above `30.0` and stops. Those are precisely the two situations in which manual continuation most often goes wrong, and the program refuses rather than guessing.

The execution loop in `run_workflow` does one of two things per stage. If a receipt exists, it calls `_replay_completed_stage` to revalidate that receipt, and a failure is written to `failures/` with classification `RECEIPT_VALIDATION_ERROR`. If no receipt exists, it runs the stage commands in order, writes logs, validates output, produces artifacts, and then writes the receipt. After each stage completes, the frozen source files and the four physics dictionaries `constant/transportProperties`, `constant/turbulenceProperties`, `system/fvSchemes` and `system/fvSolution` are re-hashed against the baseline, and any difference raises "physics dictionaries changed during continuation".

4.6 Raw reanalysis rather than copied numbers

`METRICS_EXTRACTED` calls the analyzer pinned by the lock. `parse_named_table` in `science_analysis.py` requires exactly one comment header line beginning with `Time`, reporting zero as missing and two or more as duplicate or conflicting; rejects duplicate column names; requires each row width to equal the header width; and requires `Time` to be strictly increasing. Fields are located by name rather than by fixed column index. Reference quantities are fixed in the module: `D_REF_M=0.1`, `U_REF_M_S=1.0`, `RHO_REF_KG_M3=1.0`, `A_REF_M2=0.001`, with pressure probes at `(0.15, 0.2, 0.005)` and `(0.25, 0.2, 0.005)`.

`_locked_reanalyzer` in `replay.py` compiles the analyzer module only from already verified locked bytes and requires `reanalyze_campaign` to be callable with a `__module__` equal to the module name created from those bytes, so a same-named function imported from elsewhere is rejected.

4.7 Evidence services, packet sealing and the claim gate

`task8_campaign.py` first verifies the PI Task 7 decision artifact with SHA-256 `d9db5929b8f24cce4682a94c2cb3e201295fa9fb6b46ca3c99e7dbd97a90c98b`, requiring `decision_maker` PI, `direct_instruction` `"4"`, `selected_option` `4`, `branch` `NO_ADDITIONAL_RUN`, `task8_transition.authorized` `true`, and a `scope` string equal to "canonical five-case DFG 2D-2 Re=100 campaign, local WSL, OpenFOAM.com v2506, serial only".

Case order is frozen in code as `CASE_ORDER = tuple(CANONICAL_CASE_SPECS)`, and every child case source manifest digest is preflighted before dispatch. In resume mode only an already existing child root is resumed and only a missing one is created. During sealing, `_read_case_artifacts` requires exact statuses per case: `RUN_RECEIPT.json` must have `stage=PACKET_SEALED` and `status=COMPLETE`, `METRICS.json` must be `MODEL_OUTPUT_REANALYSIS`, and `DIAGNOSTICS.json` must be `MODEL_OUTPUT_DIAGNOSTICS_NOT_APPROVAL`.

The difference between final verification and trusting a lifecycle status field is clearest in `verify_campaign_packet`. It re-enumerates the actual file set of the packet and requires it to equal exactly 21 fixed paths, six fixed files plus three artifacts for each of five cases; recomputes `SHA256SUMS.txt` and compares byte for byte; re-checks the lock and contract digests; and then rebuilds `CAMPAIGN_COMPARISON.json`, `NUMERICAL_UNCERTAINTY.json` and

`SCIENCE_REVIEW_REQUEST.md` semantically from the raw artifacts through `_campaign_documents` and compares them against the stored files. Any mismatch fails. A false closure achieved by editing upstream semantics and recomputing all hashes is therefore caught by semantic rebuilding.

Four boolean fields in `NUMERICAL_UNCERTAINTY.json` are hard-coded to `false`: `formal_order_of_accuracy_estimated`, `richardson_extrapolation_performed`, `gci_performed` and `uncertainty_bound_approved`. This is a code-level honesty statement that the five-case contrasts expose trends only and establish no order of accuracy, mesh independence or approved uncertainty interval.

On the Case011Up side, the static evidence services are different in kind but identical in intent. `render_case.py` is a deterministic OpenCFD v2412 candidate renderer that produces static artifacts only and never invokes OpenFOAM, Slurm, SSH or an arbitrary command; any residual `{{` or `}}` template token in rendered text raises an error. It freezes fifteen v2412 `meshQualityControls` entries, `maxNonOrtho 65`, `maxBoundarySkewness 20`, `maxInternalSkewness 4`, `maxConcave 80`, `minFlatness 0.5`, `minVol -1e30`, `minTetQuality 1e-30`, `minArea -1`, `minTwist 0.02`, `minDeterminant 0.001`, `minFaceWeight 0.02`, `minVolRatio 0.01`, `minTriangleTwist -1`, `nSmoothScale 4` and `errorReduction 0.75`, and preflight requires exactly those fifteen with no missing, unexpected or duplicated key.

The static gates in `autocfd_case011up/preflight.py` and `route_b_plan.py` include a two-way file ledger, symlink rejection, scheduler token rejection covering `#SBATCH`, `--partition`, `--nodes`, `--ntasks`, `--account` and `TG-CIV`, command token rejection covering `sbatch`, `srun`, `ssh`, `scp`, `rsync`, `mpirun`, `blockMesh`, `snappyHexMesh`, `checkMesh`, `simpleFoam`, `pimpleFoam` and `foamDictionary`, unresolved template token rejection, execution flags forced false, and an `Allrun.source-disabled` whose content must equal `#!/bin/sh\necho 'SOURCE_DISABLED: package never executes locally' >&2\nexit 77\n` byte for byte.

`source_manifest.py` freezes six source roles with their evidence grades: `active_geometry_stl` verified, `excluded_alternate_geometry_stl` not adopted, `historical_blockmesh_dictionary` partially verified, `historical_snappy_dictionary` partially verified, `sectional_force_metadata` partially verified, and `uncertainty_record` partially verified. Writing "partially verified" into the manifest rather than into a footnote is the key design decision at this layer.

The ComputeOps request built by `route_b_request.py` sets `authorizes_submission`, `remote_write_authorized` and `submission_authorized` to `false`, sets `fresh_presubmit_required` to `true`, requires `run_root.create_only` to be `true` and `resource_binding.status` to be `REQUESTED_FRESH_COMPUTEOPS_ADVICE`, and requires the request to rebuild deterministically from the package, with "request does not match deterministic rebuild from the package" as a real failure message. The request also enumerates ten terminal classes: the success class `BUILDING_RANS_SMOKE_COMPLETE`, nine failure classes `SOURCE_HASH_MISMATCH`, `RUN_ROOT_PREEXISTS`, `V2412_IDENTITY_MISMATCH`, `MESH_FAILED`, `BUILDING_PATCH_MISSING`, `PRESSURE_SOLVE_INACTIVE`, `REQUIRED_OUTPUT_MISSING`, `NONFINITE_EVIDENCE`, `MPI_ABNORMAL_TERMINATION` and `RECEIPT_PACKAGE_IDENTITY_MISMATCH`, and the indeterminate class `EVIDENCE_INDETERMINATE`.

Table T08. Component responsibilities: renderer, preflight, runner, resume, postprocess, evidence and claim gate

ID	Component	Owns	Must not do	Characteristic failure gate	Product lines that have it
P01	Renderer	Deterministic generation of case dictionaries and package trees from a locked contract	Never invokes OpenFOAM, Slurm, SSH or an arbitrary command; never overwrites an existing root	Residual `{{` or `}}` token; "package root already exists (no overwrite)"	Cylinder (`GENERATED`), Case011Up `render_case.py` , D420 `render_m0.py` , `render_c0.py` , `render_m1.py`
P02	Static preflight	Semantic verification of a package before anything runs	Never repairs; never relaxes a gate to let a package pass	Section digest mismatch in `blockMeshDict` `vertices` , `blocks` , `edges` , `boundary` ; boundaryField digest mismatch in `0/U` and `0/p` ; forbidden dictionary directive; scheduler or command token found; fifteen `meshQualityControls` keys not exactly matched	Control plane `preflight.py` , Case011Up `preflight.py` , D420 `preflight.py`
P03	Runner	Selecting argv from code, verifying environment and executable identity, executing, writing logs and receipts	Never accepts a user-supplied command; never runs with `LD_PRELOAD` or `LD_AUDIT` set; never reuses a root	Executable digest change; argv mismatch; unapproved `LD_LIBRARY_PATH` entry; 7200 s timeout exit 2	Cylinder only
P04	Resume	Receipt revalidation, checkpoint discovery with completeness criteria, safe `controlDict` mutation and restoration	Never advances a completed stage; never accepts a partial or aliased checkpoint; never leaves a mutated `controlDict` behind	"incomplete checkpoint"; "numeric checkpoint alias rejected"; `TERMINAL_CHECKPOINT_WITHOUT_RECEIPT` ; `CHECKPOINT_AFTER_TERMINAL_TIME` ; "physics dictionaries changed during continuation"	Cylinder only
P05	Raw reanalysis	Recomputation of every metric from unchanged raw files by a digest-pinned analyzer loaded from locked bytes	Never cites a number from a previous report; never resolves columns by fixed index	Header count not one; duplicate column names; row width mismatch; non-monotone `Time` ; analyzer `__module__` mismatch	Cylinder full; Case02Up full for layer forces; Case011Up and D420 partial
P06	Evidence service	Stage receipts, terminal receipts, terminal classification, source manifests with evidence grades, package identity reports	Never upgrades a grade; never converts a partially verified role into a verified one	`RECEIPT_VALIDATION_ERROR` ; identity mismatch between manifest and disk	All lines, at differing depth
P07	Packet sealing and final verification	Exact 21-path enumeration, byte-for-byte `SHA256SUMS.txt` , semantic rebuild of comparison and uncertainty documents	Never trusts a stored hash alone; never trusts a lifecycle status field	Any file-set, hash or semantic-rebuild mismatch	Cylinder full; others partial or package-level
P08	Claim gate	Enforcing that engineering completion cannot become scientific acceptance	Never grants `SCIENCE_ACCEPTED` ; never approves a tolerance automatically	Contract load failure when claim-boundary booleans are not false; `CLAIM_BLOCKED_FOR_ACCEPTANCE` ; `NOT_CONSISTENT_CONTROLLED_COMPARISON`	All lines

Figure F08, introduced in Chapter 1, is the governing diagram for P08 and is the reference for every terminal statement in Chapters 5 and 6.

Chapter 5 Stage-by-Stage AutoCFD Development

This chapter follows the development order shown in Figure F02. Each stage is presented as the problem that forced it, the method introduced, the software objects implemented, the evidence obtained, what failed, and how the failure changed the next stage. Numerical model-output tables belong to Chapter 7; this chapter states the terminals and the methodological consequences.

5.1 Stage S1: architecture, dual agents and a backend-neutral contract

The problem was that an architecture drawn as a diagram cannot be checked. The 2026-06-20 note fixed the scope of automation, agents hold orchestration authority and not scientific judgment, and defined the four layers described in Section 3.1. It produced no executable object and did not define the fields of the contract. It therefore forced the work to move from drawing an architecture to writing the manifest and evidence layer as verifiable code. Its own declared status was PI framing and candidate application strategy, not a validated workflow, with an explicit prohibition on implying that the full dual-agent architecture is implemented or validated. Terminal: `IMPLEMENTED` as a scope document only.

5.2 Stage S2: the control plane, intent resolution and L1 to L8 gates

The problem was that the backend-neutral schema had to become bytes. Stage S2 delivered the intent, profile and resolver triple, the case science contracts, the mesh profile registry, the backend registry, and the execution target registry described in Chapter 3, together with `workflow_plan.py`.

`workflow_plan.py` compiles a resolution into an eight-layer plan under `schema_version` `petwcfd.workflow_plan.v1`, in which each stage carries `status`, `authority` and `evidence_requirements`, and every stage has `execution_authorized` hard-coded to `false`. The controlling boolean is `qualified`, defined as `receipt["qualification_status"] == "MESH_PROFILE_QUALIFIED_ANCHOR"`. When false, L3 through L6 become `BLOCKED_UNQUALIFIED`, `selected_backend` becomes `null`, `max_permitted_gate` drops from `MESH_OK` to `RESOLUTION_ONLY`, and `blocking_reasons` records `mesh_profile_not_qualified`. Whether true or false, the five capability flags `mesh_execution_authorized`, `solver_execution_authorized`, `remote_execution_authorized`, `hpc_submission_authorized` and `automatic_correction_authorized` are all `false`. L7 postprocessing and L8 validation loop are always `DISABLED_UNQUALIFIED`, with authority [label] and `future_scientific_review` respectively.

`backend.py` gives backend identity an operational meaning: local OpenCFD v2506, environment script `/usr/lib/openfoam/openfoam2506/etc/bashrc`, expected build `_d6aa836d-20260127`, expected patch `260127`, `WM_OPTIONS` `linux64GccDPInt32Opt`, `WM_COMPILER` `Gcc`, `WM_MPLIB` `SYSTEMOPENMPI`. `probe_environment` runs fifteen checks including banner version, build, patch, whether the executable tree resolves under `project_dir`, and a `file_fingerprint` of the environment script and three executables covering `sha256`, `device`, `inode`, `uid`, `gid`, `mode` and `size`. Only all fifteen passing returns `ENV_OK`.

The stage ended by exposing its own limit. The generic control plane stops at static and mesh gates because it lacks three necessary things: stage receipts, checkpoint semantics, and an analyzer binding that recomputes metrics from raw data. A solver campaign does not need more gates; it needs a resumable execution state machine. That directly produced Stage S3. Terminal: `IMPLEMENTED`, maximum permitted gate `MESH_OK`.

5.3 Stage S3: the cylinder loop, one CLI, ten stages, five real runs

[path] is the only product line in this project that completes a full solver loop. It is not the same code as the generic control plane and is not an extension of it; the two share design principles and are frozen independently. Its mechanism is the subject of Chapter 4.

`ROUND5_FINAL_DISPOSITION.md`, dated 2026-08-03, records the first complete loop. PI restricted the round to two items: minimal control-plane technical-debt closure and a strictly two-dimensional cylinder benchmark. Case011Up, Case02Up, building sectional forces, remote and HPC work, and all later-round tasks were not executed in that round.

The benchmark is the DFG 2D-2 time-periodic channel-cylinder case at `Re=100` based on `Umean=1.0 m/s` and `D=0.1 m`, with a `Umax=1.5 m/s` parabolic inlet, front and back `empty`, on OpenFOAM.com v2506 build `_d6aa836d-20260127` patch `260127`, local serial execution only. All five cases satisfied `blockMesh` return code `0`, `checkMesh` return code `0`, `Mesh OK=true`, solver final time `30 s`, `pimpleFoam` return code `0`, `End` marker true, `FOAM FATAL` marker false, and periodic lift detected. The analysis window is `25` to `30 s`. Terminal: `SOLVER_RAN` for all five, then `POSTPROCESSED`, then `PACKET_SEALED`, with the scientific terminal [label].

The parser correction is the most important methodological event of this stage. The first parser used a fixed old-format index and read `Cd(r)` as `Cl` in the v2506 coefficient file. An independent CFD review identified the defect; self-checking did not. The corrected parser resolves the uniquely named `Time`, `Cd` and `Cl` columns from the actual header, uses a real v2506-format regression fixture, recomputes every result from unchanged raw files, hashes coefficient, probe, raw force and raw moment evidence, passes the authority positive-path test, and rejects metric tampering in final campaign verification. A consequence was also recorded: an unnecessary low-Courant diagnostic launched under the false near-steady interpretation was stopped and recorded once the defect was identified.

This correction established a core AutoCFD principle: a local tool defect is repaired from frozen raw data and recomputed rather than by rerunning CFD. It also explains why every later stage forces recomputation from raw files instead of citing numbers from a previous report.

Round 5 records its test evidence as 195 control-plane tests passed, 12 benchmark tests passed, 15 analysis and verifier tests passed, Ruff checks passed, a security pattern scan with 0 matches, final campaign verifier `VERIFIED_MODEL_OUTPUT_PACKET`, and final verification SHA-256 `479ff703dfb1b0f4421933945163e3a5cd50c45460432bfed77841a9b20cd4b0`. The control-plane freeze diff shows 5 changed files relative to the Round 4 candidate, 0 added, 0 removed except the excluded old candidate ledger, and the Round 4 candidate unmodified. These counts are historical frozen-file evidence recorded by that round and were not replayed in packaging or in this reporting environment.

The disposition lists the unapproved conclusions verbatim: automatic benchmark pass, mesh independence, time-step independence, scientific validation, design guidance and code recommendation.

Stage S3 also produced the closest prepared state to new data in the snapshot. Task 6 `fine_dt0025` is deterministically constructed, authorized to mesh, and not authorized to solve, stopping at `MESH_QUALIFIED`.

5.4 Stage S4: the building extension, Case011Up on HPC and in the ABL

Case011Up is the only building case with a static package, real HPC solving, postprocessing and terminal classification together, and its automation is not owned by the cylinder public CLI. The first methodological fork appears here: the Case011Up public CLI has `fake-exec` and not `run`. It can produce a synthetic acceptance bundle to exercise the ingest, analyze and replay chain, but real execution must be carried out externally by Condor, Slurm or ComputeOps. That is the structural difference from the cylinder product line, and it is the origin of the maturity split in Chapter 6.

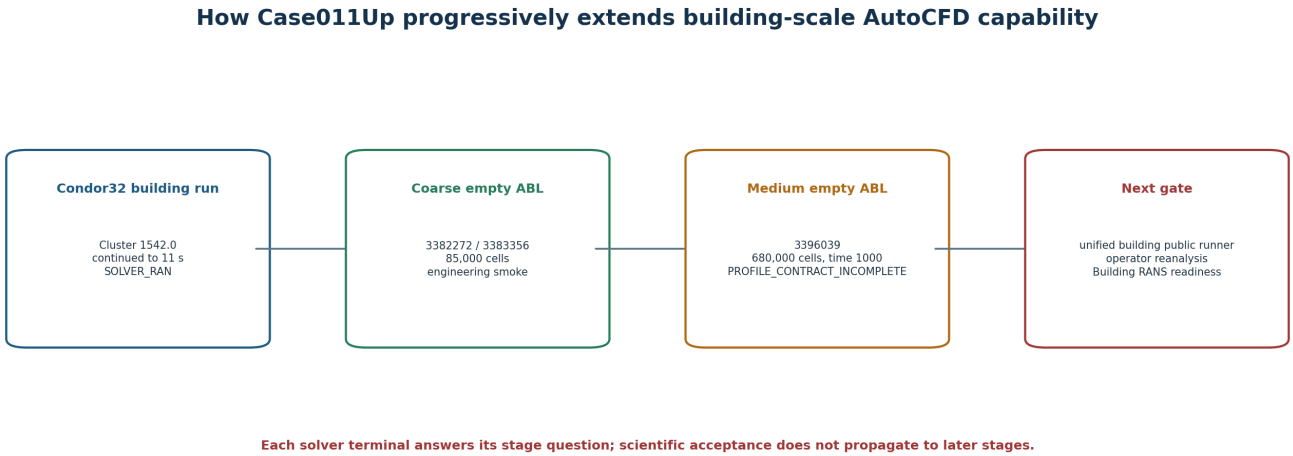


Figure F13. Program evidence. The Case011Up sequence, Condor32 cluster `1542.0` continued to `11 s` with terminal `SOLVER_RAN`, coarse empty ABL jobs `3382272` and `3383356` at 85,000 cells used for engineering smoke, medium empty ABL job `3396039` at 680,000 cells to time `1000` with terminal `PROFILE_CONTRACT_INCOMPLETE`, and the next gate covering a unified building public runner, operator reanalysis and Building RANS readiness. Each solver terminal answers its own stage question and scientific acceptance does not propagate to later stages. This is program execution history, not a validation result.

Condor32 production. `CONDOR32_PRODUCTION_COMPLETION_EVIDENCE.md`, recorded 2026-07-11 13:01 EDT, reports completion evidence for HTCondor cluster `1542.0`. The allocation was 32 CPUs and 16 GB requested memory; the remote host recorded at submission was [path]; and the isolated run root was [path]. The source restart state was the reconstructed written time `0.00909599` from the preserved Condor64 source run, with target end time `11`. The final logged physical time before OpenFOAM termination was `10.999827824854936`, followed by the terminal step and completion markers, giving a continuation interval of approximately `10.99090401` simulation-time units relative to the restart state. Final cumulative timing was `ExecutionTime = 59874.78 s`, that is `16:37:54.78`, and `ClockTime = 61095 s`, that is `16:58:15`; the local read-only monitor process `proc_19a9abfa94bd` left the queue normally with exit code `0`; and the fetched remote log tail contained both `End` and `Finalising parallel run`.

The terminal solver block reported cumulative continuity error `-1.5812877e-08` and executed five function objects, `baseForces`, `storyForces`, `forceCoeffs`, `fieldMinMax` and `fieldAverage`. The stage established that a building case can run to a target time on a real scheduler and leave a readable terminal record. It did not resolve reference quantities, units, a scientifically selected time-dependent or fixed reference velocity, complete time histories, restart continuity, probe caveats, or physical validation. The evidence file also states that the monitor fetched bounded tails of scheduler, reconstruction, decomposition and solver logs and that this is not a full-log warning audit.

Coarse empty ABL and the sampling-operator correction. `terminal_receipt_3382272.json` gives the machine-readable terminal for the formal empty-ABL source job under schema `petw-case011up-v046-two-phase-terminal-v1`: `status PASS`, `completion_class END_TIME_1000_REACHED`, host [path], account `TG-CIV260040`, partition `skx-dev`, `ntasks 4`, a pressure smoke of 5 iterations to time 5 with return code `0`, production of 1000 iterations to time 1000 with return code `0`, `profile_postprocess_used false`, archive SHA-256 `7adb6c1263bf21a4261beecf5a2435163f0e8df57d9e4508b44c9ca52d73a074`, package SHA-256 `dd9fc73b0307e30456b5b6c8337bb3573c59734d4db7db2ebc7027c5c063ff6f`, and terminal time `2026-08-07T19:36:12.093067+00:00`.

The operator problem and its correction are recorded in `CASE011UP_V046_EMPTY_ABL_FULL_EVIDENCE_PARENT_VERIFIED.md`, dated 2026-08-08. The original boundary-plane volume sampler returned inlet, middle and outlet counts of `22/41/24` in parallel. A v2 serial same-operator diagnostic job `3383251`, scheduler `COMPLETED 0:0`, proved the serial result was still `22/41/24`, so parallel decomposition was not the cause. A v3 patch-aware job `3383356`, scheduler `COMPLETED 0:0`, elapsed 35 s, returned `41/41/41` with all values finite, terminal class `PATCH_EXTRACTION_EXACT_41_COMPLETE`, and maximum returned coordinate errors of inlet `3.388e-18 m`, middle `8.217e-32 m` and outlet `3.388e-18 m`, far below the `1e-6 m` gate. Execution, retrieval and source-immutability checks were recorded as passing 35 of 35, which is historical frozen-file evidence and was not replayed here.

On numerical consistency, patchCloud and the original volume sampler agree at every height the original operator could return: inlet overlaps at 22 heights, outlet at 23 frozen grid heights with the extra non-frozen `z=0.2 m` point in the original file excluded, and middle at 41 heights, with a maximum absolute difference of `1e-6` corresponding to output precision in the last digit; v3 and v2 serial outputs are value-identical at overlapping points. The conclusion is that patchCloud is a numerically consistent method for completing boundary patch evidence but that patch results must not be renamed volume-sampler evidence. That distinction is carried in labels: inlet and outlet are labelled `PATCH_EXTRACTION`, and middle and upstream are labelled `VOLUME_EXTRACTION`.

The mesh-resolution evidence is equally important for interpreting the stage. The frozen domain height is `2.5 m`, the coarse background mesh is `100x34x25`, and the formal empty-domain mesh of 85,000 cells is identical to the pure background mesh. Therefore `dz = 0.1 m`, the first cell centre is `z=0.05 m` and the first cell top is `z=0.1 m`; of the 41 ABL points, the first 2 lie below the first cell centre and the first 4 below the first cell top, with the fifth point at `z=0.105002 m` just above the first cell top. A candidate `y+` computed at the first cell centre with `u*=0.5739 m/s` and `nu=1.5e-5 m2/s` is approximately 1913; the file also states that the Wiki [A04] range `30 <= y+ <= 500` is a smooth-wall wall-function guideline rather than the frozen gate for this rough-wall case and can serve only as a near-wall compatibility risk signal.

A parent-level factual correction is recorded: the original Fable5 decision package later described resolved-height `epsilon` as preserved within a few percent, which is inconsistent with its own tables and is not adopted. This is the second recorded instance of an external review correcting an internal narrative, matching the pattern of the Round 5 parser event. The candidate turbulence-intensity diagnostic `Iu = sqrt(2k/3)/|Ux|` is forced to the label `CANDIDATE_INTERPRETATION_NOT_CONTRACT` because the contract freezes neither the formula nor a tolerance, and the Wiki mean `U` and turbulence-intensity 10% clause is recorded as a `Normative draft` and not a PI Case011Up gate.

The final recommendation is B: allow coarse building-RANS engineering and pipeline smoke while deferring scientific production. The verified status is `FABLE5_FINAL_SCIENCE_VERIFIED_WITH_PARENT_CORRECTIONS`, the scientific status is [label], and the recommended path is `B_ENGINEERING_SMOKE_ONLY_PENDING_RESOLVED_GATE`.

Medium empty ABL: solver completion with an incomplete sampling contract. `MEDIUM_EMPTY_ABL_SCIENCE_REPORT_3396039.md` and `TERMINAL_CLASSIFICATION_3396039.json` together provide the most instructive terminal separation in this snapshot. The scheduler terminal is Slurm job `3396039`, `COMPLETED`, exit code `0:0`, elapsed `00:12:08`, node `c455-011`, with no queue record after completion. The application terminal is OpenCFD v2412 `simpleFoam` reaching time `1000`, ending normally, finalizing the parallel run, with 6000 residual records, 1000 continuity records, no nonfinite solver-log lines, and a production mesh of 680,000 cells reporting `Mesh OK`. The wrapper evidence terminal is `PASS` with package SHA-256 `09b175d52bb45469ac5e8abd6c76b2e431f0c4329b161c82df4856d444abef12`, input identity SHA-256 `921e263f8f9d6ed9bf78507f077f6ee7195a335e74256783f93c53329ba2c044`, and remote run root [path].

The artifact terminal is `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE`. The contract declares six flow profiles, ten output times from 100 to 1000, and 41 points per profile. Observed: the five interior or wake profiles returned 41 rows at every time, while `inlet_profile` returned 39 rows at every time; the additional ABL diagnostic profiles returned 33 rows for `abl_inlet`, 41 for `abl_middle` and 31 for `abl_outlet`; and all observed rows had nine finite numeric columns. Column mapping is set by the runtime serialized suffix `epsilon_k_nut_U`, parsed as `x y z epsilon k nut Ux Uy Uz`; the `controlDict` field declaration `(U k epsilon nut)` is retained as provenance and does not determine the mapping. The analysis status is therefore `PROFILE_CONTRACT_INCOMPLETE` rather than pass.

`READ_ONLY_PROFILE_GRID_REANALYSIS_3396039.md` locates the missing coordinates exactly through local read-only reanalysis. At time 1000, `upstream_profile` and `abl_middle` are complete at 41 rows; `inlet_profile` has 39 rows and is missing `z = 0.0096002 m` and `0.569051855 m`; `abl_inlet` has 33 rows and is missing `0.0100000`, `0.0337505`, `0.6750140`, `0.6987645`, `0.7225150`, `0.7462655`, `0.7700160` and `0.7937665 m`; and `abl_outlet` has 31 rows and is missing `0.0100000`, `0.0337505`, `0.0575010`, `0.0812515`, `0.3187565`, `0.4612595`, `0.6037625`, `0.6275130`, `0.8650180` and `0.8887685 m`. The row-count pattern is invariant across times 100 through 1000, and each retained profile is finite and monotone in z. The reanalysis applied no interpolation, padding or reconstruction, and ran no solver, postProcess, scheduler or remote write. It confirms a deterministic sampling-cardinality and coordinate-lattice deficiency but does not identify whether the cause is boundary-plane and cell-intersection behavior, the production post-processing condition, or their interaction.

PI accepts the wall-function RANS treatment for this medium empty-ABL baseline, so the absence of wall-resolved prism layers and a runtime `y+` report is not a blocker for this baseline. That decision does not waive the remaining evidence limits, and the next scientific issue is profile completeness and ABL interpretation rather than conversion of

this baseline to a wall-resolved mesh.

The claim terminal is `CLAIM_BLOCKED_FOR_ACCEPTANCE`. The single approved claim is that a real medium empty-ABL model output was generated and analyzed with an incomplete profile lattice. Blocked claims include that medium empty-ABL is accepted, that ABL is preserved or validated, that the result is wind-tunnel validated, and that the result supports Building RANS. The stop condition states that one scheduler submission reached terminal state and that no retry, resubmission, cancel, cleanup, root reuse or successor execution is authorized from this iteration.

5.5 Stage S5: the load interface, Case02Up and synchronized 50-layer forces

`CASE02UP_LAYERFORCE50_RESULT_REVIEW.md`, recorded 2026-07-17, represents a different product line: real solver output already exists, and the automation is concentrated in postprocessing and the structural handoff.

The analysis reads all base-name and OpenFOAM rollover `forceBin*.dat` files from the completed attempt14 run and obtains 50,877 unique, strictly increasing samples spanning `0.0000600024 s` through `11.0 s`, with 601 columns per row, that is, time plus 50 non-cumulative layers with 12 values per layer.

A semantic correction underlies the whole analysis: OpenFOAM v2506 source code shows that the header `z co-ords` are bin end points rather than centers. The corrected 50 layer centers span `z=0.0265350` to `2.4737147`, the printed end points span `0.0515062` to `2.4986859`, and the nominal spacing is `0.0499424`. This is the third instance of a semantic correction obtained by reading source code or through independent review, matching the parser and `epsilon` cases.

The initialization transient occupies the first four samples, with `Fx` values `-455.7531`, `284.3468`, `-10.2555` and `-1.7519` at `t = 0.0000600024`, `0.0001320090`, `0.0002184160` and `0.0003221050 s`. After `0.001 s` the maximum observed `|Fx|` before 1 s is only `0.1748`. The impulse contaminates full-record extrema and the `0` to `1 s` RMS but does not affect the main `t >= 1 s` results unless the initialization procedure itself is under study.

A consistent contract-side observation is that `autocfd_case011up/contract.py` fixes `force_outputs.n_bins` to 20 while this case uses 50 layers, which confirms that the layer-force postprocessing belongs to a separate product line rather than to the Case011Up contract. The scientific boundary is that the active output uses `rhoInf=1`, so every value in this line is a unit-density solver force and not an approved physical design load. No wind-tunnel comparison, grid-sensitivity study, time-step-sensitivity study, density conversion, reference-area normalization or structural-response transfer has been approved. Terminal: `POSTPROCESSED`.

5.6 Stage S6: unsteady inflow, the D420 five-face drive and the three-arm package

Stage S6 pushes the AutoCFD method into time-varying, multiscale, conservative input mapping and matched-drive comparison. It is the only line that simultaneously exposed an input-contract defect, coupled confounders, and the gap between package readiness and execution. Its mechanism and evidence are the subject of Chapters 9 and 10; the development-stage facts are these.

The interface chain was built and its conservation semantics were frozen: `packet_reader.py` fixes production schema `petw-d420-d4b-production-packet-v1`, face order `west, east, south, north, top`, face normals and point orders, a flux replay relative tolerance of `1.0e-9` and a cadence of `1.0 s`; `mask_reference.py` is the Python golden reference for the inflow-outflow switching rule, with `dot < 0` inflow, `dot >= 0` outflow, the exact-zero case frozen to outflow, interpolation permitted only between two adjacent 1 s records with weight inside `[0, 1]`, and no flux correction computed anywhere; `initial_field_mapper.py` is a pure reorder that refuses to run unless the projection policy is frozen to `PROJECTION_FORBIDDEN_WITHOUT_SEPARATE_APPROVAL` and refuses to overwrite an existing `0/U`; and `boundarydata_exporter.py` writes five-face `constant/boundaryData` with float64 `%.17g` points and float32 `%.9g` velocities that round-trip exactly, into an owned root, atomically, never overwriting.

Two engineering failures in the G2 readback were preserved rather than hidden. The first module attempt failed before NetCDF open because NetCDF required an explicit compiler module. A complete first read then found that a 0.1 mm threshold on `zhval` versus `1000*zh` was too strict for independent float32 unit representations, with an observed

maximum difference of `0.954 mm`; the comparison tolerance was changed to a documented 2 mm without changing scientific or remote scope, and the final full rerun passed. This is a clear example of a tool tolerance that is not a scientific tolerance.

The decisive failure of the stage is job `3391286`. The successor package was submitted exactly once after live account, partition, source and output checks and a successful `sbatch --test-only`. The job started on `c454-101` in `skx-dev` with account `TG-CIV260040` and then failed during the first M0 `reconstructPar` call with `FOAM FATAL ERROR: (openfoam-2412) Cannot find file "points" in directory "polyMesh" in times "0" down to constant`. A read-only remote inventory established the root cause: the root-level clean M0 source view has a readable `constant/polyMesh/points` at `62,279,881` bytes, but each of its 48 processor directories contains only the selected time directories `50,60,70,80`, and both M0 and M1 have `processorN/constant/polyMesh/points` present at `0/48`. The failure is therefore not caused by the root symlink target or by file permissions: `reconstructPar` requires the processor-level decomposed mesh payload, and the clean-view QA checked selected processor time fields without checking this required processor mesh inventory. The terminal is `D420_REBOUND_POSTPROCESS_FAILED_MISSING_PROCESSOR_MESH_NO_SCIENTIFIC_OUTPUT` with zero scientific output.

The repair boundary was written as explicitly as the diagnosis: copying the root global mesh into every processor directory is not an acceptable repair; a safe repair requires either the original processor-level mesh payloads from the source of truth or a separately reviewed exact decomposition procedure whose generated processor mesh and field ordering are independently proven compatible; and running `decomposePar`, regenerating processor meshes, modifying source views or submitting a repaired successor requires a new package, new output and control roots, and a new explicit PI approval. The one approved successor submission was consumed by this job, no retry was performed, and the failed job remains preserved.

The successor, job `3393320`, completed postprocessing and produced four metrics JSON files and four figure-input NPZ files with `automatic_acceptance` `false` and `new_solver_run` `false`. An independent audit classified the same job as `NOT_CONSISTENT_CONTROLLED_COMPARISON` and listed eight direct conflicts between M0 and M1 case setups. The response was not to argue about the metrics but to redesign the comparison: the matched-drive three-arm package binds `D420_MC100_LAMINAR`, `D420_MF010_LAMINAR` and `D420_MF010_KOMEGASST_URANS` to the same five-face time-varying boundaryData manifest, the same `0` to `260 s` clock, the same `noSlip` ground treatment, no restart and the same CM1-derived initial velocity mapping, with `k`, `omega` and `nut` present in the URANS arm alone. All 5,293 manifest-listed files matched paths and sizes, giving `STATIC_THREE_ARM_PACKAGE_OK`. Terminal: `PACKAGE_READY`, `submitted` `false`, `solver_ran` `false`.

5.7 Stage table and correction table

Table T09. Six development stages

ID	Stage	Date anchor	Problem that forced it	Method introduced	Principal software objects	Evidence obtained	What it did not settle
S1	Architecture: dual agents and a backend-neutral contract	2026-06-20	An architecture drawn as a diagram cannot be checked	Four layers with a manifest and evidence layer as the pivot; agents hold orchestration and not scientific judgment	Architecture note only; no executable object	Scope statement with an explicit prohibition on claiming implementation or validation	Contract fields; any Fluent-side implementation, which is absent from the snapshot
S2	Control plane: intent resolution and L1 to L8 gates	Through 2026-08-03 freeze	The schema had to become verifiable bytes	Minimal canonical intent, digest-pinned profiles, provenance-carrying resolution, staged contracts, gate registries	`intent.py` , `profiles.py` , `resolver.py` , `workflow_plan.py` , `backend.py` , `preflight.py` , `execution_request.py` , `science_contract.py` , `mesh_profiles.py` , `openfoam_backends.py` , `stampede3_targets.py`	`IMPLEMENTED` with `max_permitted_gate` `MESH_OK` ; all five capability flags false	Solver execution; it lacks receipts, checkpoint semantics and an analyzer binding
S3	Cylinder loop: one CLI, ten stages, five real runs	Round 5, 2026-08-03	Gates alone cannot run a campaign; a resumable execution state machine is needed	Six-mode CLI, ten-stage receipts, five-artifact execution binding, checkpoint completeness, locked reanalysis, semantic packet rebuild	`__main__.py` , `runner.py` , `workflow.py` , `receipts.py` , `restart.py` , `replay.py` , `science_analysis.py` , `task8_campaign.py` , `task6_*`	Five cases `SOLVER_RAN` to `30 s` , then `POSTPROCESSED` and `PACKET_SEALED` ; verifier `VERIFIED_MODEL_OUTPUT_PACKET`	Mesh independence, time-step independence, physical validation; the `6.31%` medium-to-fine lift-amplitude change remains open
S4	Building extension: Case011Upon HPC and in the ABL	2026-07-11 through 2026-08-11	Structural relevance requires a building case, real HPC and ABL inflow	Deterministic renderer, fail-closed static preflight, governed request with ten terminal classes, `fake-exec` instead of `run` , external Condor and Slurm execution, terminal recovery through receipts	`render_case.py` , `preflight.py` , `route_b_plan.py` , `route_b_request.py` , `route_b_cli.py` , `source_manifest.py` , `stampede3_targets.py`	Condor32 `SOLVER_RAN` to `11 s` ; coarse ABL `PATCH_EXTRACTION_EXACT_41_COMPLETE` ; medium ABL `SOLVER_RAN` plus `PROFILE_CONTRACT_INCOMPLETE` and `CLAIM_BLOCKED_FOR_ACCEPTANCE`	`empty_abl_acceptance` , which gates `building_rans` ; checkpoint semantics and per-stage receipts for building runs
S5	Load interface: Case02Up synchronized 50 layers	2026-07-17	Structural handoff needs height-resolved, time-synchronized forces, not aggregate coefficients	Full raw ingestion across rollover files, layer-centre semantic correction from solver source, descriptive interval statistics for nonstationary inflow	`CASE02UP_LAYERFORCE50_RESULT_REVIEW.md` analysis chain	`POSTPROCESSED` : 50,877 unique strictly increasing samples, 601 columns, 50 layer centers `0.0265350` to `2.4737147 m`	Physical density conversion, reference-area normalization, structural transfer; all values remain unit-density at `rhoInf=1`
S6	Unsteady inflow: D420 five-face drive and the three-arm package	2026-08-05 through 2026-08-12	A nonstationary multiscale drive stresses input mapping, clock matching and comparison control	Frozen packet schema and face order, golden-reference inflow mask, pure-reorder initial mapping, exactly round-tripping boundaryData export, matched-drive three-arm design	`packet_reader.py` , `mask_reference.py` , `initial_field_mapper.py` , `boundarydata_exporter.py` , `contract.py` , `window.py` , `subwindow_extractor.py` , `render_m1.py`	Historical M0 `3384063` and M1 `3386652` `SOLVER_RAN` ; `3391286` `FAILED` with zero output; `3393320` `POSTPROCESSED` but `NOT_CONSISTENT_CONTROLLED_COMPARISON` ; three-arm `PACKAGE_READY`	Any matched-drive solver result; `Vhat_3s` ; design pressure; structural design load

Table T10. Failures, corrections and methodological consequences

ID	Failure or defect	How it was found	Correction applied	Methodological consequence	Resulting terminal
E01	Coefficient parser read `Cd(r)` as `Cl` using a fixed old-format column index	Independent CFD review, not self-checking	Named-column parsing from the actual v2506 header, real-format regression fixture, full recomputation from unchanged raw files	Tool defects are repaired from frozen raw data and recomputed, never by rerunning CFD; every later stage recomputes rather than citing	Five raw runs reanalyzed; campaign `POSTPROCESSED` and re-sealed
E02	An unnecessary low-Courant diagnostic launched under the false near-steady interpretation caused by E01	Follow-on review of the parser defect	The diagnostic was stopped and the fact recorded rather than deleted	Actions taken under a defective interpretation are recorded as part of the evidence trail	Recorded and closed
E03	Boundary-plane volume sampler returned `22/41/24` instead of `41/41/41`	Serial same-operator diagnostic job `3383251` ruled out parallel decomposition	Patch-aware operator in job `3383356` returned `41/41/41` with coordinate errors far below the `1e-6 m` gate	Patch extraction and volume extraction are separate labelled operators; patch results must never be renamed volume-sampler evidence	`PATCH_EXTRACTION_EXACT_41_COMPLETE`
E04	Decision package described resolved-height `epsilon` as preserved within a few percent, inconsistent with its own tables	Parent verification	The description was not adopted; the tabulated deviations stand	External review carries semantic misreadings that deterministic checkers cannot catch	Parent-corrected; `FABLE5_FINAL_SCIENCE_VERIFIED_WITH_PARENT_CORRECTIONS`
E05	`forceBin` header `z co-ords` misread as bin centers	Reading OpenFOAM v2506 solver source code	Corrected 50 centers `0.0265350` to `2.4737147 m` against printed end points `0.0515062` to `2.4986859 m`	Output semantics are established from solver source, not from field-name intuition	Layer analysis `POSTPROCESSED` on corrected centers
E06	`reconstructPar` failed because processor-level `polyMesh` payload was absent at `0/48` while the root view was complete	Read-only remote inventory after the job failed	Processor mesh promoted to an explicit inventory item: `points`, `faces`, `owner`, `neighbour`, `boundary` per processor view; shortcut repairs explicitly forbidden	The input contract matters as much as the output contract; the correct repair adds a required item rather than relaxing a check	`3391286` `FAILED`, zero scientific output; successor `3393320` completed postprocessing
E07	Identical directory labels treated as identity between M0 and M1 `boundaryData`	Byte-level consistency audit	Sizes, point counts and per-file hashes compared directly; `same_path_hashes_differ` recorded as `true`	Structural sameness of a directory tree is not evidence of identical content	`NOT_CONSISTENT_CONTROLLED_COMPARISON` with eight listed conflicts
E08	Medium empty-ABL profiles returned `39/33/41/31` rows against a required 41, invariant across ten output times	Read-only local grid reanalysis	Missing heights enumerated exactly; no interpolation, padding or reconstruction applied	A blocked scientific artifact can coexist with a fully successful scheduler and application terminal; terminals must be layered	`PROFILE_CONTRACT_INCOMPLETE` and `CLAIM_BLOCKED_FOR_ACCEPTANCE`
E09	Package root reuse attempted by a background build	Renderer refusal during the build	"package root already exists (no overwrite)"; the clean-root build completed in `111.92 s`	No-overwrite and new-root discipline is enforced identically across independent product lines	Clean build succeeded; reuse attempt refused
E10	A 0.1 mm `zhval` comparison threshold was too strict for independent float32 unit representations, observed difference `0.954 mm`	Complete first read of the G2 readback	Tolerance changed to a documented 2 mm with no change to scientific or remote scope	A tool tolerance is not a scientific tolerance and must be labelled as such	Full rerun passed
E11	Display values `33.7 m/s` and `34.1 m/s` cited in the existing work summary	Attempted rebinding against the 92 snapshot files	Marked as appearing only in the existing report and not rebound here; not treated as verified	Values that cannot be rebound to a source file are given an explicit terminal rather than being repeated	`UNVERIFIED`

How failures and corrections changed the AutoCFD method

Issue	Observed failure	Method correction	Result
Coefficient parser	fixed index read Cd(r) as Cl	parse fields from the v2506 header	reanalyzed five raw runs
ABL sampler	boundary volume sampler missed rows	separate patch and volume operators	coarse 41/41/41
Job 3391286	processor mesh absent from source view	contract points/faces/owner/neighbour/boundary	3393320 postprocessing completed
Medium profiles	39/33/41/31 rows repeated	read-only grid analysis, no padding	PROFILE_CONTRACT_INCOMPLETE remains

Figure F21. Program evidence. Four issue rows recorded in the frozen files, the coefficient parser reading `Cd(r)` as `Cl`, the boundary volume sampler missing rows, job `3391286` with processor mesh absent from the source view, and the repeated `39/33/41/31` medium profile rows, each with its observed failure, the method correction adopted, and the resulting state. This records program history and method change, not a validation result.

Chapter 6 Program Product Lines and Capability Maturity

6.1 Maturity is a property of the product line, not of the project

Placing the stages side by side exposes a structural fact: AutoCFD automation maturity is a property of the product line, not of the project. The reason can be read directly from source rather than inferred.

The cylinder product line owns five things no other line owns. `run_workflow` is the state machine that actually executes OpenFOAM. `restart.py` is a checkpoint discoverer with completeness criteria. `receipts.py` is a per-stage receipt template with a non-overridable claim boundary. `replay.py` is a replayer that loads the analyzer only from locked bytes. `task8_campaign.py` is a packet verifier with semantic rebuilding.

Case011Up has a renderer, preflight, plan, request, ingest, analyze and replay, but its execution verb is `fake-exec` rather than `run`. Real execution happens outside the product line under Condor and Slurm, and terminals are recovered afterward through `terminal_receipt_*.json` and `TERMINAL_CLASSIFICATION_*.json`.

D420 has a renderer, exporter, mapper, preflight and package identity, but `analyze_m0_raw` in `post_analysis.py` is an explicit fail-closed outline: it checks the run root, `postProcessing/targetProbes`, logs and the target-points hash, and even when all of those pass it still raises "analyze_m0_raw is an outline: full raw analysis is gated on M0 terminal PASS after the PI D7 execution chain".

Case02Up is the inverse of D420: no generation or execution product-line code appears in the snapshot, only a highly mature postprocessing and structural-interface result review.

The statement that AutoCFD is automated therefore has to be split four ways on the evidence. The end-to-end loop for the fixed two-dimensional benchmark is owned by one unified public product line. The building case has mature static and request-side automation while execution and recovery are not owned by the same state machine. Layer-force postprocessing and structural-interface semantics are highly automated while their generation and execution are not in the snapshot. The CM1-CFD input mapping and static package are mature while the raw-data analyzer remains a gated outline.

Figure F05, introduced in Chapter 3, is the reference diagram for this division: the generic control plane carries authority only up to `MESH_OK`, while the executable product lines carry a full state machine under an explicit execution binding.

6.2 Module and product inventory

Table T11. Module and product inventory

ID	Product line	Snapshot module count	Representative modules	Public entry point	Execution verb	Line terminal
L01	Generic control plane, `petwcfd_round5_science_20260803/control_plane/petwcfd/`	16 Python files	`intent.py`, `profiles.py`, `resolver.py`, `workflow_plan.py`, `execution_request.py`, `backend.py`, `preflight.py`, `mesh_runner.py`, `adapter_openfoam.py`, `manifest.py`, `strict_json.py`, `safe_files.py`, `pipeline.py`, `diagnostics.py`, `__main__.py`, `__init__.py`	`python -m petwcfd` with `resolve`, `explain`, `plan`, `diagnose`, restricted `run`	`run` limited to `--until STATIC_OK` or `MESH_OK`; solver unreachable by construction	`IMPLEMENTED`
L02	Cylinder AutoCFD, `petwcfd_cylinder_autocfd_v1/petwcfd_cylinder/`	19 Python files	`__main__.py`, `runner.py`, `workflow.py`, `receipts.py`, `restart.py`, `replay.py`, `science_contract.py`, `science_analysis.py`, `campaign_lock.py`, `safe_io.py`, `task8_campaign.py`, `task6_candidate.py`, `task6_case_spec.py`, `task6_comparison.py`, `task6_generation_evidence.py`, `task6_execution_lock.py`, `task6_staged_dispatch.py`, `task6_mesh_runner.py`, `__init__.py`	`python -m petwcfd_cylinder` with six modes	`run`, `resume`, `campaign` reach `PACKET_SEALED`; Task 6 dispatch stops at `MESH_QUALIFIED`	`SOLVER_RAN`, `POSTPROCESSED`, `PACKET_SEALED`
L03	Case011Up building, `autocfd_case011up/`	16 Python files	`contract.py`, `render_case.py`, `preflight.py`, `route_b_plan.py`, `route_b_request.py`, `route_b_cli.py`, `route_b_evidence.py`, `route_b_smoke.py`, `route_b_postprocess.py`, `source_manifest.py`, `stampede3_targets.py`, `mesh_profiles.py`, `openfoam_backends.py`, `canonical.py`, `locations.py`, `__init__.py`	`route_b_cli.py` with 24 subcommands in three families	`fake-exec` only; real execution external under Condor and Slurm	`SOLVER_RAN` externally; `TERMINAL_CLASSIFIED`; `CLAIM_BLOCKED_FOR_ACCEPTANCE` for the medium case
L04	Case02Up layer-force interface	0 generation or execution modules in the snapshot	Postprocessing and review chain recorded in `CASE02UP_LAYERFORCE50_RESULT_REVIEW.md`	No product-line CLI in the snapshot	None; the run pre-exists	`POSTPROCESSED`
L05	D420 empty domain, `autocfd_d420_empty_domain/`	13 Python files	`contract.py`, `packet_reader.py`, `mask_reference.py`, `initial_field_mapper.py`, `boundarydata_exporter.py`, `boundary_dicts.py`, `mesh_generator.py`, `render_m0.py`, `render_c0.py`, `preflight.py`, `post_analysis.py`, `tacc_bundle.py`, `__init__.py`	Renderer and exporter entry points	None; `analyze_m0_raw` raises as a gated outline	`PACKAGE_READY`; historical runs `SOLVER_RAN`
L06	D420 M1 URANS, `autocfd_d420_m1_urans/`	5 Python files	`window.py`, `subwindow_extractor.py`, `mesh_generator.py`, `render_m1.py`, `__init__.py`	Renderer entry point	None	`PACKAGE_READY`; historical M1 `SOLVER_RAN`
L07	Figure generation	1 Python file	`generate_work_summary_figures.py`	Script	None	`IMPLEMENTED`
L08	Evidence and reporting artifacts	22 Markdown and JSON files	Round 5 disposition, Condor32 completion evidence, coarse and medium ABL reports, terminal receipts and classification, Case02Up review, D420 protocol, terminal and audit JSON, agent updates, frozen manifest	Not executable	Not applicable	Source of every numerical value in this study

6.3 Backend and evidence-service coverage

Table T12. Backend and evidence-service coverage

ID	Backend or service	Identity pinned	Where declared	Coverage in the snapshot	Status
B01	Local OpenCFD v2506, WSL, serial	Environment script `/usr/lib/openfoam/openfoam2506/etc/bashrc`, build `_d6aa836d-20260127`, patch `260127`, `WM_OPTIONS` `linux64GccDPInt32Opt`, `WM_COMPILER` `Gcc`, `WM_MPLIB` `SYSTEMOPENMPI`; fifteen `probe_environment` checks returning `ENV_OK`	`control_plane/petwcfd/backend.py`	Cylinder five-case campaign and Task 6 mesh dispatch	`SOLVER_RAN`
B02	Canonical executable fingerprints for B01	`blockMesh` `11bfeeb3...`, `checkMesh` `95efe750...`, `setExprBoundaryFields` `c7b6c9e4...`, `potentialFoam` `c253b638...`, `pimpleFoam` `76523e34...`	`petwcfd_cylinder/runner.py`	Enforced before every command; symlink and non-file rejected	`IMPLEMENTED` and exercised
B03	TACC Stampede3 OpenCFD v2412	Backend id `tacc_openfoam_com_v2412`, host `TACC_Stampede3`, `exact_version` `v2412`, activation `module purge; module load intel/24.0 impi/21.11 openfoam/v2412`, seven required executables, `qualification` restricted to `STATIC_ONLY` or `UNQUALIFIED`	`autocfd_case011up/openfoam_backends.py`	Jobs `3382272`, `3383251`, `3383356`, `3396039`, and the D420 historical and postprocess jobs	`SOLVER_RAN`; backend qualification remains `STATIC_ONLY` or `UNQUALIFIED`
B04	Stampede3 execution targets	`empty_abl` coarse, `skx-dev`, 1 node, 4 ranks, `00:30:00`, six gates; `empty_abl_medium` medium, 4 ranks, `00:45:00`; `building_rans` medium, 48 ranks, `02:00:00`, eight gates led by `empty_abl_acceptance`; `building_urans` blocked pending mesh-derived time controls, eleven gates	`stampede3_targets.py`	All authorization flags false, `allocation` null, `evidence_status` `DATED_COMPUTEOPS_ADVICE_REQUIRES_FRESH_RECHECK`	`IMPLEMENTED`; `building_rans` `BLOCKED` behind an ungranted gate
B05	HTCondor at CRC	Cluster `1542.0`, 32 CPUs, 16 GB requested memory, submit host , isolated run root	`CONDOR32_PRODUCTION_COMPLETION_EVIDENCE.md`	One building continuation to `11 s`	`SOLVER_RAN`
B06	ComputeOps request service	`authorizes_submission`, `remote_write_authorized`, `submission_authorized` all false; `fresh_presubmit_required` true; `run_root.create_only` true; `resource_binding.status` `REQUESTED_FRESH_COMPUTEOPS_ADVICE`; ten terminal classes	`route_b_request.py`	Case011Up requests; D420 three-arm candidate envelope `skx`, 4 nodes, 192 ranks, 720 minute walltime request, 600 minute estimate, benchmark-only and queue-heuristic	`IMPLEMENTED`; authorizes nothing
B07	Stage receipt service	Schema `petwcfd.stage-receipt.v2`, ten frozen stage names, non-overridable claim boundary, normalized absolute `run_root`, safe ASCII `case_id`	`receipts.py`	Cylinder only	`IMPLEMENTED` and exercised
B08	Terminal receipt and classification service	Schema `petw-case011up-v046-two-phase-terminal-v1`; six-category terminal classification	`terminal_receipt_3382272.json`, `TERMINAL_CLASSIFICATION_3396039.json`	Case011Up coarse and medium; adapted for D420 terminal JSON	`TERMINAL_CLASSIFIED`
B09	Source manifest and evidence-grade service	Six source roles with grades verified, not adopted, and partially verified	`source_manifest.py`	Case011Up	`IMPLEMENTED`
B10	Package identity service	D420 three-arm: 5,293 manifest-listed files matched on path and size; six critical files including three initial `U` fields and both package identity reports matched on hash	`20260811_D420_three_arm_production_package_ready.md`	D420 three-arm	`PACKAGE_READY`
B11	Packet sealing and final verification service	Exact 21-path enumeration, byte-for-byte `SHA256SUMS.txt`, semantic rebuild of three campaign documents, four uncertainty booleans hard-coded false	`task8_campaign.py`	Cylinder only	`VERIFIED_MODEL_OUTPUT_PACKET`, final SHA-256 `479ff703...` recorded as historical frozen-file evidence
B12	Display and provenance archive service	V12: 34 package files, 33 SHA entries, 8 figure source and derivative entries, ZIP 34 members `testzip=PASS`, SHA-256 `5d0ca61b...`, `1096344` bytes; F7 and F8 source script and raw-array manifest not independently located	`AU-20260812-005`, `AU-20260812-007`	D420 CM1-to-CFD display	`ACCEPTED_BOUNDED_DISPLAY_ARCHIVE` under `D-20260812-001`

6.4 Product-line maturity matrix

The maturity levels used below are defined as follows. `FULL` means the capability is owned by unified product-line code with execution evidence inside the snapshot. `PARTIAL` means the capability is implemented but is carried externally, is recovered after the fact, or is not unified into the same state machine. `PACKAGE` means static package level only, with no execution ownership. `NONE` means no implementation appears in the snapshot. These levels record program ownership, not scientific validation, and no cell of this matrix implies that any result has been accepted.

Table T13. Product-line maturity matrix

Capability	Cylinder five-case	Case011Up package	Case011Up HPC execution	Case02Up layer force	D420 M0/M1 historical	D420 three-arm
Contract validation	FULL	FULL	FULL	NONE	FULL	FULL
Deterministic rendering	FULL	FULL	FULL	NONE	FULL	FULL
Static preflight	FULL	FULL	FULL	NONE	FULL	FULL
Mesh build and qualification	FULL	PARTIAL	PARTIAL	NONE	PARTIAL	PACKAGE
Solver execution owned by the line	FULL	PARTIAL	PARTIAL	PARTIAL	PARTIAL	NONE
Checkpoint and resume semantics	FULL	NONE	NONE	NONE	NONE	NONE
Raw-data recomputation	FULL	PARTIAL	PARTIAL	FULL	PARTIAL	NONE
Terminal classification	FULL	FULL	FULL	PARTIAL	FULL	PACKAGE
Packet sealing and final verification	FULL	PARTIAL	PARTIAL	PARTIAL	PARTIAL	PACKAGE
Governed request and submission control	FULL	FULL	FULL	NONE	FULL	FULL
Claim gate enforcement	FULL	FULL	FULL	FULL	FULL	FULL

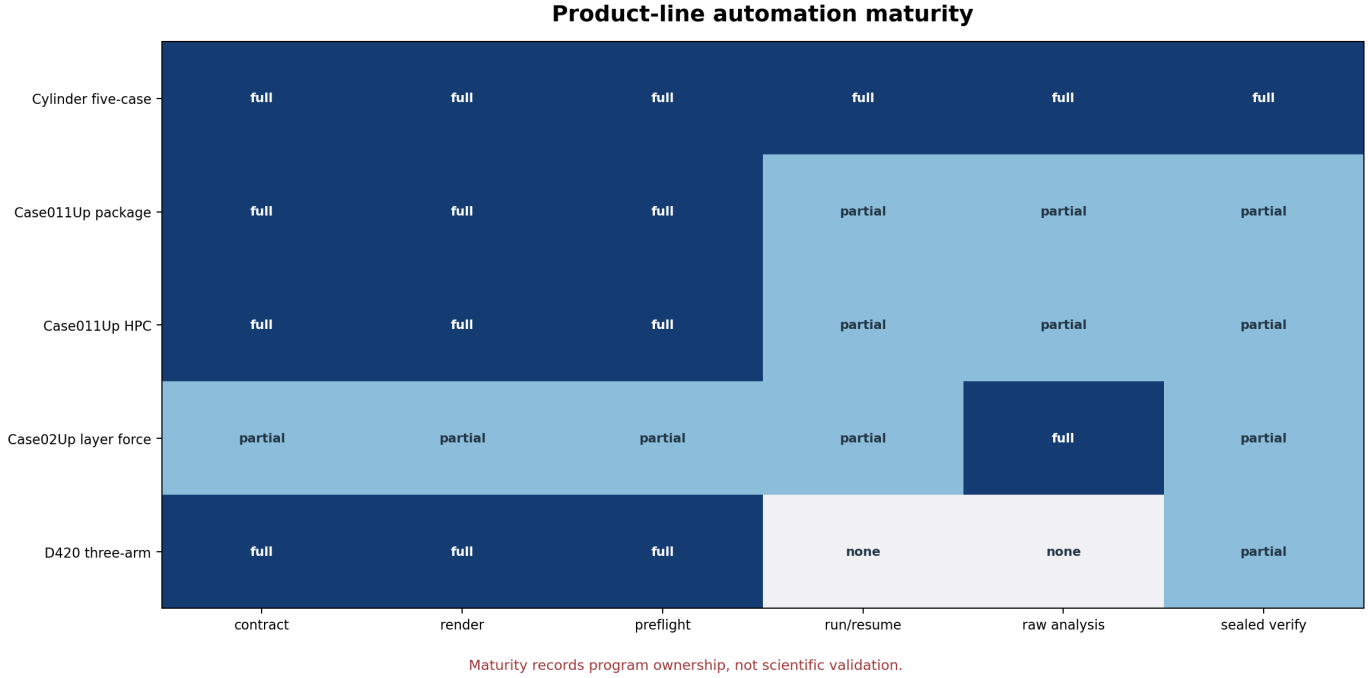


Figure F20. Program evidence. Automation maturity for five product lines across contract, render, preflight, run and resume, raw analysis and sealed verify, showing the cylinder line full across all six, both Case011Up views full through preflight and partial thereafter, Case02Up full only at raw analysis, and D420 three-arm full through preflight with none at run and raw analysis. As the figure states directly, maturity records program ownership, not scientific validation. This is program evidence, not a validation result.

6.5 Reading the matrix correctly

Three readings of Table T13 must be avoided, because each corresponds to a prohibited promotion in Section 1.2.

A `FULL` claim-gate row across every column does not mean the lines are equally mature. It means the opposite of a maturity claim: the claim gate is the one capability that is enforced identically everywhere, precisely so that differences in engineering maturity cannot leak into scientific status. The cylinder line and the D420 three-arm package have the same claim-gate strength and radically different execution maturity.

A `FULL` in the packet sealing row for the cylinder line does not mean the cylinder results are scientifically accepted. The sealed packet carries `MODEL_OUTPUT_REANALYSIS`, `MODEL_OUTPUT_DIAGNOSTICS_NOT_APPROVAL` and `NOT_RELEASED`, with four uncertainty booleans hard-coded false. Sealing proves that the packet is exactly what the raw data implies, and nothing further.

A `PACKAGE` in the D420 three-arm column is not a weaker form of `SOLVER_RAN`. It is a different terminal entirely: `submitted` is `false` and `solver_ran` is `false`, and the package claim boundary states that it proves input identity and static case completeness only, with no proof of mesh quality, solver completion, numerical validity, downburst mechanism or scientific acceptance.

The engineering position at this date is that one product line closes the loop completely, two carry mature static and interface automation with externally owned execution, one is postprocessing-only but deep, and one is prepared to the point where a single approved submission is the remaining step. The scientific position is unchanged across all five: no line has an instance of `SCIENCE_ACCEPTED`, and the bottleneck for the most prepared lines is not method but the decision between a prepared state and one approved execution.

Chapter 7 AutoCFD Model-Output Evidence

Every numerical value in this chapter is a model output or an execution observation. None of them has been approved as a physical validation result, and no instance of `SCIENCE_ACCEPTED` exists anywhere in the frozen snapshot. The figures in this chapter are labelled by class at the point of use, and a figure classified as solver/model output or postprocessing diagnostic is never a validation result.

7.1 The cylinder five-case campaign

The benchmark is the DFG 2D-2 time-periodic channel-cylinder case at `Re = 100` based on `Umean = 1.0 m/s` and `D = 0.1 m`, with a `Umax = 1.5 m/s` parabolic inlet and front and back patches of type `empty`, executed on OpenFOAM.com v2506 build `_d6aa836d-20260127` patch `260127`, local serial only. All five cases satisfied the same recorded execution gates: `blockMesh` return code 0, `checkMesh` return code 0, `Mesh OK=true`, solver final time 30 s, `pimpleFoam` return code 0, `End` marker true, `FOAM FATAL` marker false, and periodic lift detected. The science contract freezes exactly three analysis windows, `20_to_25_s`, `22_5_to_27_5_s` and `25_to_30_s`, with exactly one primary window, `25_to_30_s`, which is the window reported below. The analyzer fixes `D_REF_M = 0.1`, `U_REF_M_S = 1.0`, `RHO_REF_KG_M3 = 1.0` and `A_REF_M2 = 0.001`, with pressure probes at `(0.15, 0.2, 0.005)` and `(0.25, 0.2, 0.005)`.

The values in Table T14 are the corrected values. The first parser used a fixed old-format column index and read `Cd(r)` as `Cl` in the v2506 coefficient file; the corrected parser resolves the uniquely named `Time`, `Cd` and `Cl` columns from the actual header and recomputes every result from unchanged raw files, so the table is a recomputation from frozen raw data rather than a set of numbers copied forward from an earlier report.

Table T14. Cylinder five-case model outputs, analysis window 25 to 30 s

Case	Cells	dt (s)	Final time (s)	Cd_mid	Cl_pp	St	dp_mean	Max Courant	Terminal
coarse_dt005	5,028	0.0050	30	3.11875033	1.626863619	0.288362513	2.387296926	2.6019984	SOLVER_RAN
medium_dt005	9,800	0.0050	30	3.148927695	1.747781529	0.295670539	2.425753026	3.6578467	SOLVER_RAN
fine_dt005	19,285	0.0050	30	3.170944565	1.858084637	0.298507463	2.442709666	5.1420472	SOLVER_RAN
medium_dt010	9,800	0.0100	30	3.143860035	1.710200418	0.292275574	2.422215908	7.3114959	SOLVER_RAN
medium_dt0025	9,800	0.0025	30	3.15067269	1.763972837	0.296296296	2.42670796	1.8295271	SOLVER_RAN

All five cases completed 5/5 to 30 s under one public workflow, and each carries the stage terminal `SOLVER_RAN` followed by `POSTPROCESSED` metrics extraction. The campaign-level terminal is [label], which is a claim terminal and not an acceptance.

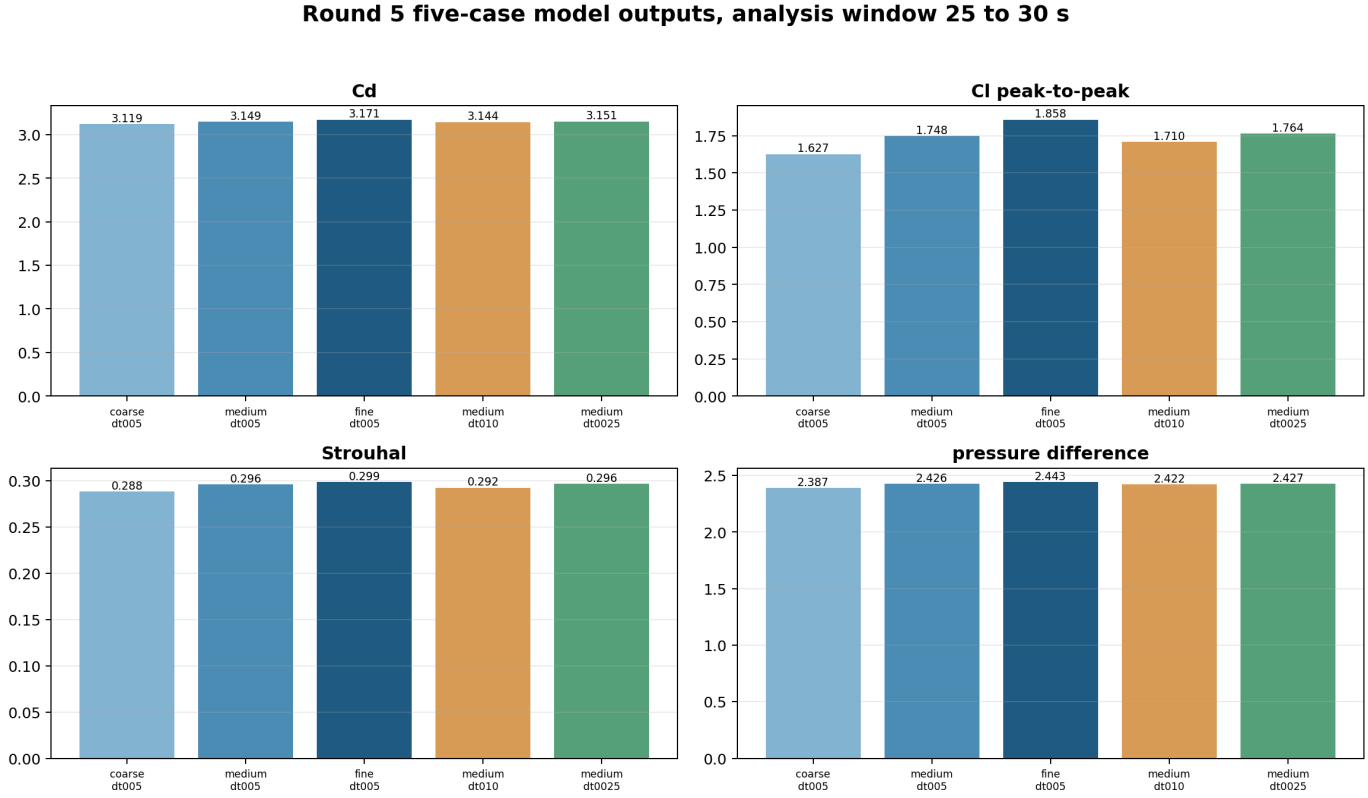


Figure F09. [Classification: solver/model output] `Cd_mid`, `Cl_pp`, `St` and `dp_mean` for the five cases over the 25 to 30 s window. This is a comparison of model outputs produced by the same frozen contract and is not a validation result.

Table T15. Cylinder sensitivity and authority differences

Comparison axis	Cd	Cl_pp	St	dp_mean	Interpretation boundary
Locked authority value	3.19584 37857	2.00740 583536	2.46411554 83424013	0.301886 792453	The authority curve is a bound reference and is not an automatic pass criterion
`fine_dt005` difference from authority (%)	-0.7791 125715	-7.43851 57067	-0.8687044 836	-1.11940 29851	Arithmetic difference between a model output and a reference, not a validation conclusion
Medium-to-fine change (%)	0.69918 6267	6.31103 52278	0.69902581 04	0.959488 2729	The `Cl_pp` change is approximately `6.31%`, so grid convergence is not approved
Medium mesh, `dt` from 0.005 to 0.0025 (%)	0.05541 55309	0.92639 1985	0.03936869 17	0.211640 2116	Lift amplitude approximately `0.93%` and Strouhal number approximately `0.21%`; no automatic time-step-convergence claim is made

The code-level honesty statement sits in `NUMERICAL_UNCERTAINTY.json`, where `formal_order_of_accuracy_estimated`, `richardson_extrapolation_performed`, `gci_performed` and `uncertainty_bound_approved` are permanently `false`. The five-case contrasts expose trends only. They establish no order of accuracy, no mesh independence and no approved uncertainty interval. The approximately `6.31%` medium-to-fine lift-amplitude change is the one substantive open numerical question in this product line, and it is carried forward as Q3 in Table T20 and as R5 in Table T27.

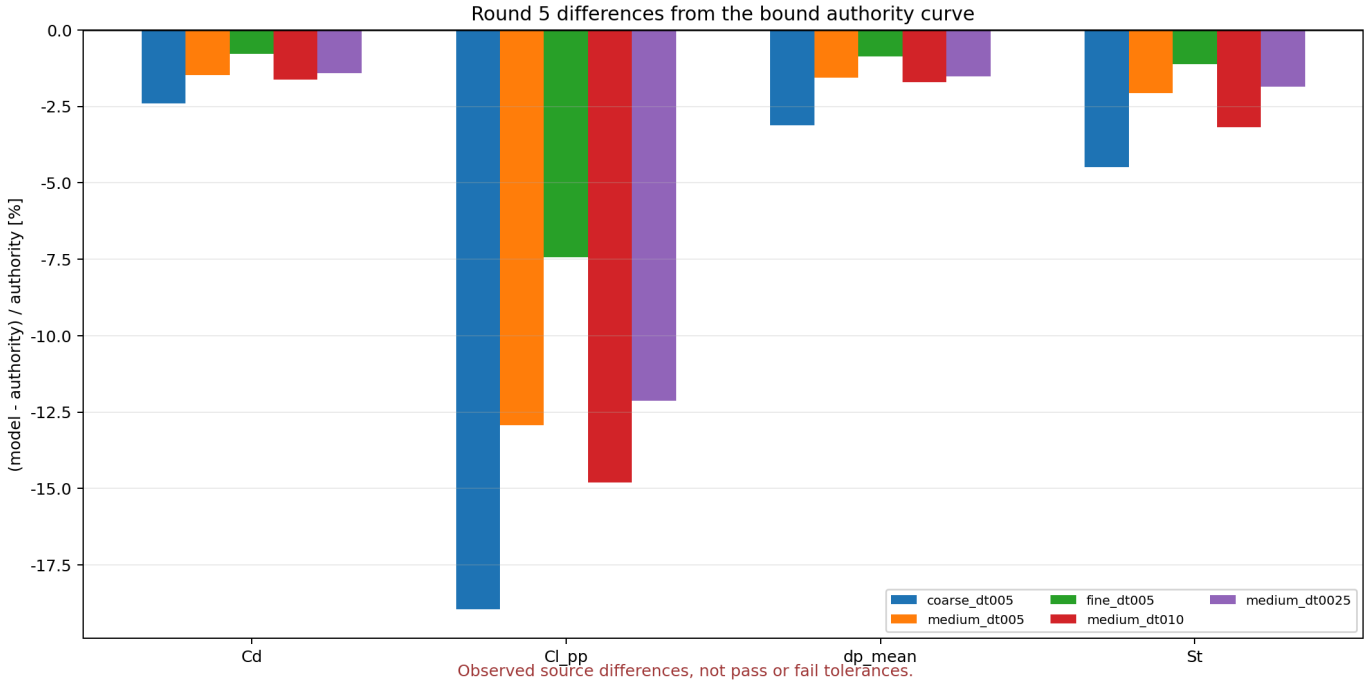


Figure F10. [Classification: solver/model output] Percentage differences of the `fine_dt005` model output from the locked authority values. This is an arithmetic comparison between a model output and a bound reference curve and is not a validation result.

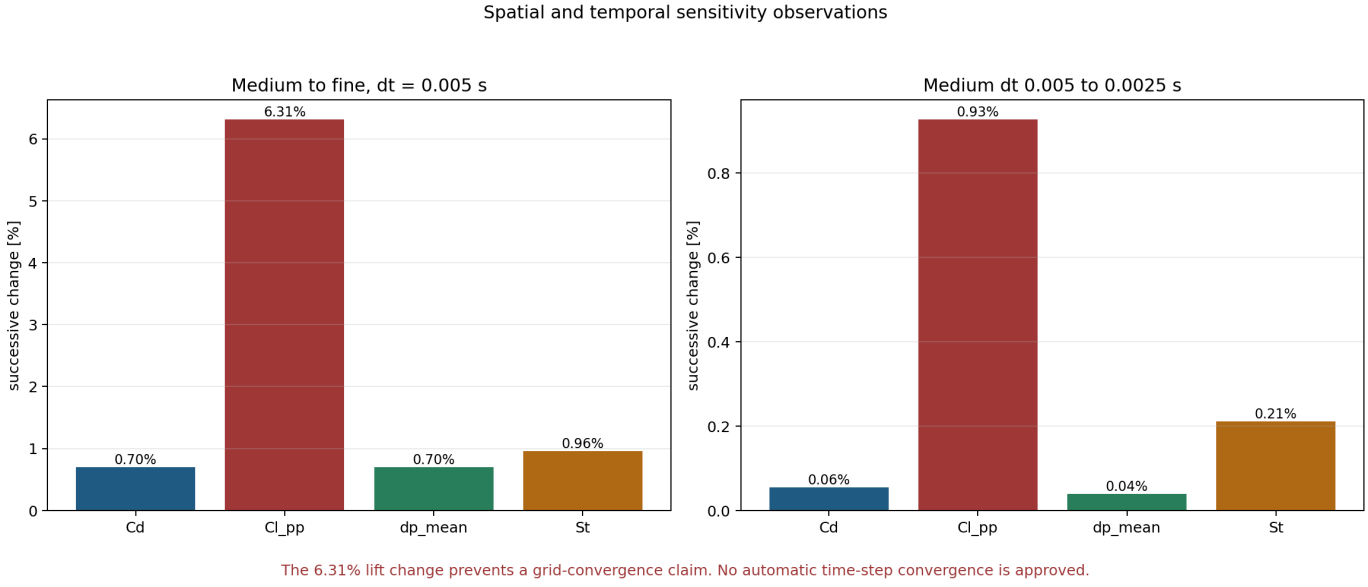


Figure F11. [Classification: solver/model output] Medium-to-fine change and medium-mesh time-step change across the four reported quantities. This figure is not evidence of mesh independence or time-step independence.

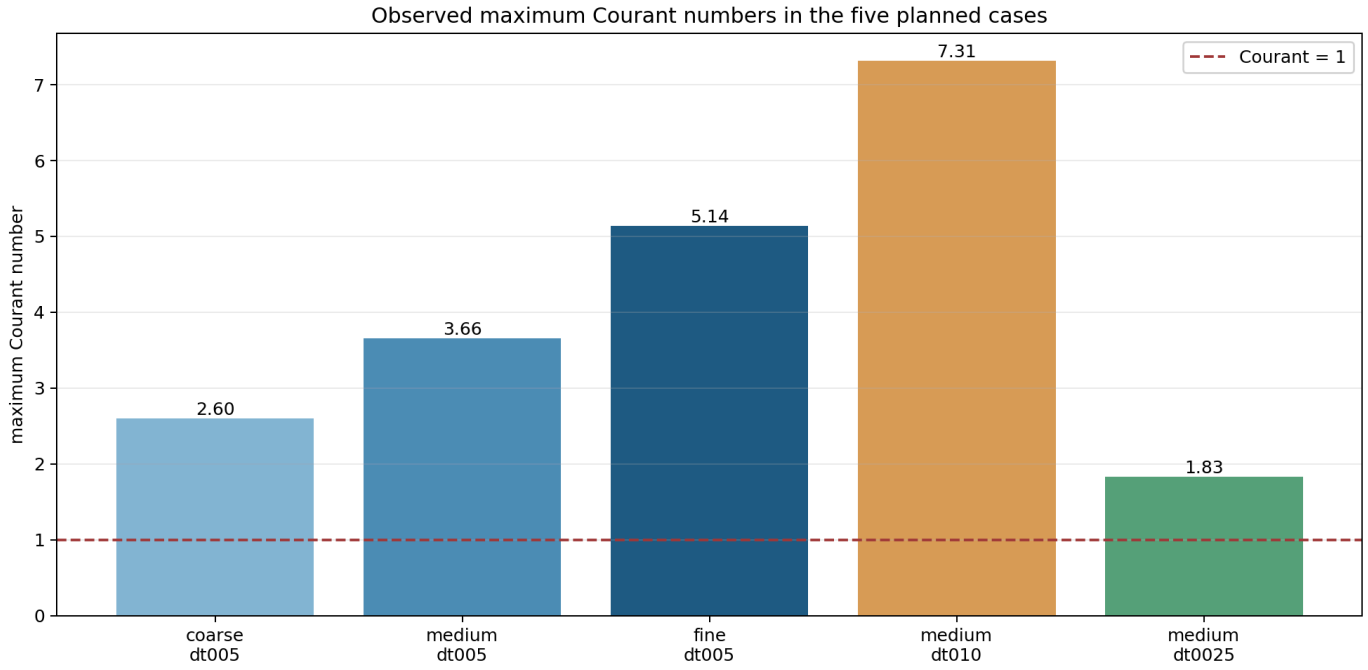


Figure F12. [Classification: postprocessing diagnostic] Maximum Courant number per case, recomputed from the raw solver logs, retained to explain the time-step error risk at `dt = 0.005 s` that motivates the open question in T15.

7.2 Case011Up Condor32 production

`CONDOR32_PRODUCTION_COMPLETION_EVIDENCE.md`, recorded 2026-07-11 13:01 EDT, is the first real building time history in this project.

Table T16. Case011Up Condor32 terminal record

Item	Value	Interpretation boundary
HTCondor cluster	`1542.0`	Execution identity
Allocation	32 CPUs, 16 GB requested memory	Recorded at submission
Remote host		Recorded at submission
Isolated run root	`/building1/Case011Up_PETW_condor32_production_1542_0`	No existing root was reused or overwritten
Continuation source state	Reconstructed written time `0.00909599` from the preserved Condor64 source run	Continuation baseline
Target end time	`11`	Contract target
Final logged physical time before OpenFOAM termination	`10.999827824854936`	Followed by the terminal step and completion markers
Continuation interval relative to the restart state	Approximately `10.99090401` simulation-time units	Arithmetic difference
Cumulative timing	`ExecutionTime = 59874.78 s` (`16:37:54.78`), `ClockTime = 61095 s` (`16:58:15`)	Resource observation
Local read-only monitor process	`proc_19a9abfa94bd` , left the queue normally, exit code `0`	The monitor fetched bounded log tails only
Remote log tail markers	Contains both `End` and `Finalising parallel run`	Application terminated normally
Terminal-block cumulative continuity error	`-1.5812877e-08`	Solver diagnostic
Terminal total base force	`(0.02886224 -0.0094251721 0.0016325056)`	Unit-density solver force
Terminal total base moment	`(-0.0014989393 -7.6898194e-05 -4.4764249e-05)`	Unit-density solver moment
Terminal coefficients	`Cl = 0.043533482` , `Cs = -0.25133792` , `CmPitch = -0.06835395` , `CmRoll = -1.3323905` , `CmYaw = -0.039790443`	The configured `forceCoeffs` block explicitly describes its coefficients as reference-only for this nonstationary case
Terminal field extrema	`max(mag(U)) = 4.2731876` , `min(p) = -5.9094764` , `max(p) = 1.835067`	Model output
Function objects in the terminal block	`baseForces` , `storyForces` , `forceCoeffs` , `fieldMinMax` , `fieldAverage`	Five executed
`Cd_ref`	`0.76965972` , using `rhoInf=1` , `magUInf=1` , `Aref=0.075`	Arithmetically consistent with the reported streamwise force, but not an approved physical coefficient under nonstationary inflow
Terminal	`SOLVER_RAN`	Reference quantities, units, a scientifically selected time-dependent or fixed reference velocity, complete time histories, restart continuity, probe caveats and physical validation all remain unresolved

The evidence file separately records that the monitor captured bounded tails of the scheduler, reconstruction, decomposition and solver logs, and that this is not a full-log warning audit. The Condor32 terminal therefore proves that a building case can run to its target time on a real scheduler and leave a readable terminal record. It proves nothing about the physical admissibility of the recorded forces.

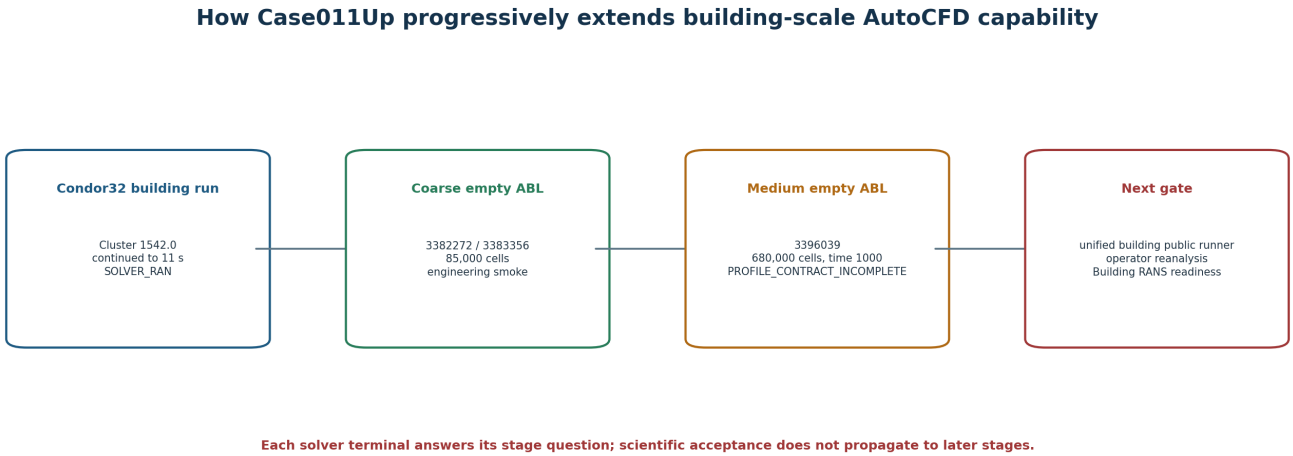


Figure F13. [Classification: method diagram] The four-step Case011Up progression, from the Condor32 building run through the coarse empty-ABL jobs and the medium empty-ABL job to the next gate. Each node carries only its own recorded terminal, and the diagram states the governing rule that scientific acceptance does not propagate from one stage to the next.

7.3 Coarse and medium empty ABL

The coarse line consists of three separate jobs, `3382272`, `3383251` and `3383356`, and must not be merged with the single medium job `3396039`. The coarse line closes a sampling-operator question on an 85,000-cell mesh and is authorized for engineering pipeline smoke only. The medium line runs a 680,000-cell production mesh to time 1000 and returns an incomplete profile contract. Table T17 keeps the two columns strictly separate.

Table T17. Coarse and medium empty-ABL diagnostics

Item	Coarse empty ABL	Medium empty ABL
Jobs	Source job `3382272` ; serial same-operator diagnostic `3383251` ; patch-aware job `3383356`	`3396039`
Scheduler terminal	`3382272 status PASS` , `completion_class END_TIME_1000_REACHED` ; `3383251 COMPLETED 0:0` ; `3383356 COMPLETED 0:0` , elapsed 35 s	`COMPLETED` , exit code `0:0` , elapsed `00:12:08` , node `c455-011` , no queue record after completion
Host and account	, account `TG-CIV260040` , partition `skx-dev` , `ntasks 4`	Same account family and partition family, OpenCFD v2412
Application execution	Pressure smoke of 5 iterations to time 5, return code 0; production of 1000 iterations to time 1000, return code 0; `profile_postprocess_used false`	`simpleFoam` reached time `1000` , ended normally, finalized the parallel run; 6000 residual records, 1000 continuity records, no nonfinite solver-log lines
Mesh	Background mesh `100x34x25` ; the formal empty-domain mesh of 85,000 cells is identical to the pure background mesh; domain height `2.5 m` , `dz = 0.1 m` , first cell centre `z = 0.05 m` , first cell top `z = 0.1 m`	Production mesh of 680,000 cells reporting `Mesh OK`
Near-wall record	Of the 41 ABL points the first 2 lie below the first cell centre and the first 4 below the first cell top, and the fifth point at `z = 0.105002 m` lies just above the first cell top; a candidate `y+` at the first cell centre with `u* = 0.5739 m/s` and `nu = 1.5e-5 m2/s` is approximately 1913; the Wiki [A04] range `30 <= y+ <= 500` is a smooth-wall wall-function guideline and not the frozen gate for this rough-wall case	`Ks/z0 = 12.66` while the wiki software-specific fully-rough approximation would give approximately `Ks = 30 z0 = 0.00237 m` , recorded as a compatibility item and not an automatic failure; the recovered runtime packet contains no `y+` , wall distance, wall shear or actual first-cell centre, so no exact `y+` is claimed; the first available heights are `0.0096002 m` for `upstream_profile` and `0.0100 m` for `abl_middle`
Sampling counts	The original boundary-plane volume sampler returned `22/41/24` in parallel; the serial same-operator diagnostic still returned `22/41/24` , which excludes parallel decomposition as the cause; the patch-aware job returned `41/41/41` with all values finite, terminal class `PATCH_EXTRACTION_EXACT_41_COMPLETE` ; maximum returned coordinate errors were inlet `3.388e-18 m` , middle `8.217e-32 m` and outlet `3.388e-18 m` , far below the `1e-6 m` gate; execution, retrieval and source-immutability checks passed 35/35	The contract declares 6 flow profiles, 10 output times from 100 to 1000 and 41 points per profile. Observed at every time: `inlet_profile` returned **39/41**; the additional ABL diagnostic profiles returned **33/41** for `abl_inlet` , **41/41** for `abl_middle` and **31/41** for `abl_outlet` ; the five interior or wake profiles returned 41 rows at every time; all observed rows had nine finite numeric columns
Numerical consistency	patchCloud agrees with the original volume sampler at every height the original operator could return: inlet overlaps at 22 heights, outlet at 23 frozen grid heights (the extra non-frozen `z = 0.2 m` point in the original file is excluded) and middle at 41 heights, with a maximum absolute difference of `1e-6` corresponding to the last digit of output precision; the v3 and v2 serial outputs are value-identical at overlapping points	Column mapping is set by the runtime serialized suffix `epsilon_k_nut_U` , parsed as `x y z epsilon k nut Ux Uy Uz` ; the `controlDict` field declaration `(U k epsilon nut)` is retained as provenance and does not determine the mapping
Preservation diagnostics	Outlet relative to inlet over all 41 points: `Ux 3.358%` , `k 5.863%` , `epsilon 8.836%` , `nut 4.404%` . Middle resolved heights relative to target: `Ux 0.2157%` , `k 3.446%` , `epsilon 20.42%` , `nut 2.391%` . Upstream resolved heights relative to target: `Ux 1.811%` , `k 4.953%` , `epsilon 41.88%` , `nut 11.24%`	At time 1000: inlet maximum relative deviation approximately `2.47%` from the neutral log-law velocity reference and `39.4%` from the neutral epsilon reference; middle approximately `73.4%` and `66.5%` ; outlet approximately `1.38%` and `11.1%` . Maximum absolute deviation of `k` from the contract reference was `5.09e-05 m2/s2` at inlet, `0.0779 m2/s2` at middle and `0.0276 m2/s2` at outlet; maximum relative deviation of `nut` was approximately `0.23%` , `23.7%` and `5.84%`
Drift comparison	Labels are fixed: inlet and outlet are `PATCH_EXTRACTION` , middle and upstream are `VOLUME_EXTRACTION` ; patch results must not be renamed volume-sampler evidence	Because row counts differ (`33/41` , `41/31` , `33/31`), the inlet-to-middle, middle-to-outlet and inlet-to-outlet drift comparisons are all recorded as `UNUSABLE`
Candidate interpretation	The candidate turbulence-intensity diagnostic `Iu = sqrt(2k/3)/|Ux|` is forced to the label `CANDIDATE_INTERPRETATION_NOT_CONTRACT` : outlet relative to inlet gives `6.128%` over the full 41 points, middle relative to target gives `1.895%` at `z >= 0.2237545 m` , upstream relative to target gives approximately `3.0%` at `z >= 0.1 m` . The Wiki mean `U` and turbulence-intensity 10% clause is a `Normative draft` and not a Case011Up gate; even though a counterfactual calculation under the two hmax window assumptions listed by Fable5 shows no 10% failure at middle or upstream, that result may not be presented as prestandard compliance	accepts the wall-function RANS treatment for this medium empty-ABL baseline, so the absence of wall-resolved prism layers and of a runtime `y+` report is not a blocker for this baseline. The decision does not waive the remaining evidence limits, and the next scientific issue is profile completeness and ABL interpretation rather than conversion of this baseline to a wall-resolved mesh
Missing-coordinate localization	Not applicable	`READ_ONLY_PROFILE_GRID_REANALYSIS_3396039.md` locates the gaps exactly at time 1000: `upstream_profile` and `abl_middle` are complete at 41 rows; `inlet_profile` has 39 rows and is missing `z = 0.0096002 m` and `0.569051855 m` ; `abl_inlet` has 33 rows and is missing `0.0100000` , `0.0337505` , `0.6750140` , `0.6987645` , `0.7225150` , `0.7462655` , `0.7700160` and `0.7937665 m` ; `abl_outlet` has 31 rows and is missing `0.0100000` , `0.0337505` , `0.0575010` , `0.0812515` , `0.3187565` , `0.4612595` , `0.6037625` , `0.6275130` , `0.8650180` and `0.8887685 m` . The row-count pattern is invariant across times 100 through 1000, and each retained profile is finite and monotone in z. The reanalysis applied no interpolation, padding or reconstruction and ran no solver, postProcess, scheduler or remote write

Item	Coarse empty ABL	Medium empty ABL
Identity digests	Archive SHA-256 `7adb6c1263bf21a4261beecf5a2435163f0e8df57d9e4508b44c9ca52d73a074`, package SHA-256 `dd9fc73b0307e30456b5b6c8337bb3573c59734d4db7db2ebc7027c5c063ff6f`, terminal time `2026-08-07T19:36:12.093067+00:00`	Package SHA-256 `09b175d52bb45469ac5e8abd6c76b2e431f0c4329b161c82df4856d444abef12`, input identity SHA-256 `921e263f8f9d6ed9bf78507f077f6ee7195a335e74256783f93c53329ba2c044`, remote run root `l_09b175d5_20260811_05`
Terminal and status	Verified status `FABLE5_FINAL_SCIENCE_VERIFIED_WITH_PARENT_CORRECTIONS`, scientific status , recommended path `B_ENGINEERING_SMOKE_ONLY_PENDING_RESOLVED_GATE`, that is, coarse building-RANS engineering and pipeline smoke are allowed while scientific production is deferred	Artifact terminal `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE`, analysis status `PROFILE_CONTRACT_INCOMPLETE`, claim terminal `CLAIM_BLOCKED_FOR_ACCEPTANCE`. The single approved claim is that a real medium empty-ABL model output was generated and analyzed with an incomplete profile lattice. Claims that are `BLOCKED` include that medium empty ABL is accepted, that ABL is preserved or validated, that the result is wind-tunnel validated and that the result supports Building RANS. The stop condition states that one scheduler submission reached terminal state and that no retry, resubmission, cancel, cleanup, root reuse or successor execution is authorized from this iteration

The medium case gives the most instructive terminal separation in the snapshot. Scheduler `COMPLETED`, normal application termination, a 680,000-cell mesh reporting `Mesh OK` and 6000 residual records are all true at the same time as a blocked scientific output. Terminals must therefore be recorded separately across scheduler, application, wrapper, artifact, usability and claim, and success in one category cannot substitute for another. The job is `POSTPROCESSED` and `TERMINAL_CLASSIFIED`, and it is not accepted.

One parent-level factual correction is recorded. The original Fable5 decision package later described resolved-height `epsilon` as preserved within a few percent, which is inconsistent with its own tables and is not adopted by this study. This is the second recorded instance of an external review correcting an internal narrative, matching the pattern of the Round 5 parser event.

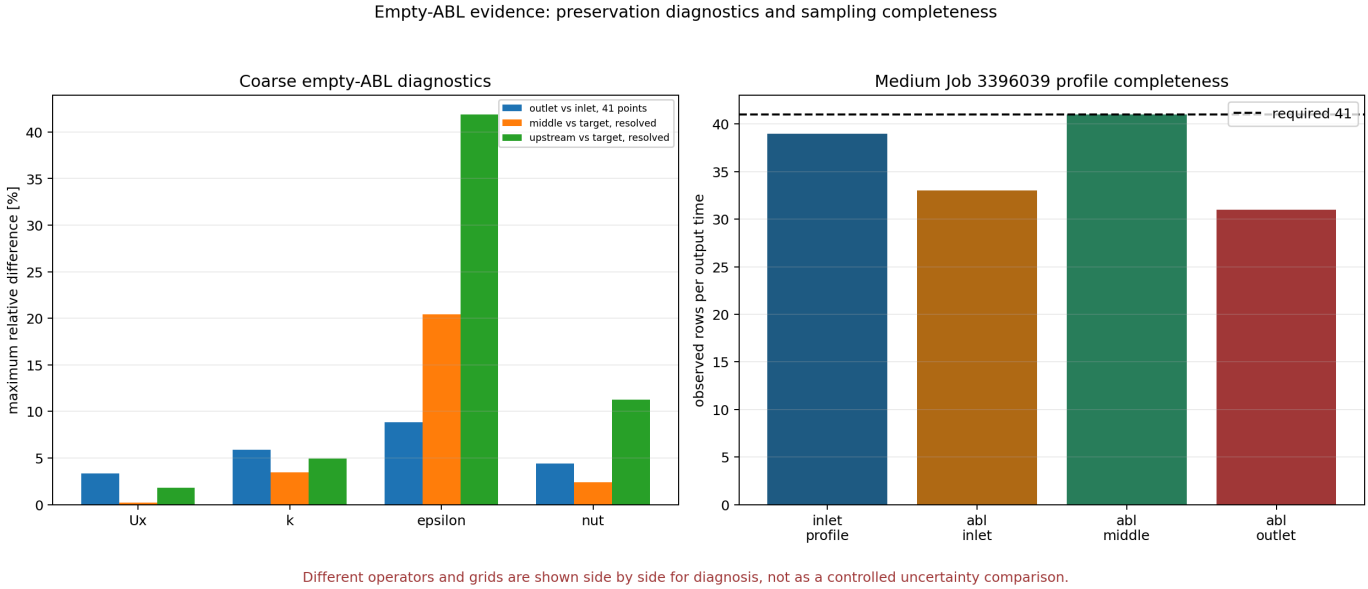


Figure F14. [Classification: postprocessing diagnostic] Profile deviations and sampling-count diagnostics for the coarse and medium empty-ABL cases. This is a postprocessing diagnostic of model output and is not evidence that the ABL is preserved or validated.

7.4 Case02Up LayerForce50

Case02Up represents a different product line. Real solver output already exists, and the automation is concentrated in postprocessing and the structural handoff. The analysis reads all base-name and OpenFOAM rollover `forceBin*.dat` files from the completed attempt14 run and obtains `50,877` unique, strictly increasing samples spanning `0.0000600024 s` through `11.0 s`, with 601 columns per row, that is, time plus 50 non-cumulative layers with 12 values per layer.

One semantic correction underlies the entire analysis. OpenFOAM v2506 source code shows that the header `z co-ords` are bin end points rather than centers. The corrected 50 layer centers span `z = 0.0265350` to `2.4737147`, the printed end points span `0.0515062` to `2.4986859`, and the nominal spacing is `0.0499424`. This is the third semantic correction obtained by reading source code or through independent review, matching the parser and `epsilon` cases.

Table T18. Case02Up LayerForce50 response

Item	Value	Interpretation boundary
Samples	`50,877` unique strictly increasing samples, `0.0000600024 s` to `11.0 s`, 601 columns per row	50 non-cumulative layers with 12 values per layer
Layer geometry	50 corrected layer centers from `z = 0.0265350` to `2.4737147`; printed end points `0.0515062` to `2.4986859`; nominal spacing `0.0499424`	Header `z co-ords` are bin end points, not centers
Initialization transient	First four samples: `t=0.0000600024 s: Fx=-455.7531`; `t=0.0001320090 s: Fx=284.3468`; `t=0.0002184160 s: Fx=-10.2555`; `t=0.0003221050 s: Fx=-1.7519`. After `0.001 s` and before 1 s the maximum observed `|Fx|` is only `0.1748`	The impulse contaminates full-record extrema and the `0-1 s` RMS but does not affect the main `t >= 1 s` results unless the initialization procedure itself is under study
Along-wind peak	`Fx,max = 17.4044453` at `t = 6.82791 s`	Unit-density solver force
Cross-wind extrema	`Fy,max = 13.7320110` at `t = 6.76312 s`; `Fy,min = -14.1442953` at `t = 6.83168 s`	The maximum absolute cross-wind force is about `81.27%` of the maximum along-wind force; at the negative cross-wind peak `|Fy|/Fx = 85.60%`
Vertical component	`Fz,min = -0.00521332` at `t = 5.03907 s`; `Fz,max = 0.00436915` at `t = 6.15746 s`	Much smaller than the horizontal components
Selected total along-wind values	`1 s: 0.09028`; `2 s: 0.20295`; `3 s: 4.88001`; `4 s: 15.21119`; `5 s: 15.37405`; `6 s: 16.14888`; `7 s: 14.93421`; `8 s: 4.65684`; `9 s: 0.19020`; `10 s: 0.14700`; `11 s: 0.15380`	Between 6 and 8 s, `Fx` decreases by `71.16%`; between 8 and 9 s it decreases by another `95.92%`
Interval statistics	`1-3 s`: mean `Fx = 2.01824`, RMS `Fx = 2.53582`, RMS `Fy = 1.56595`; `3-6 s`: `14.30790`, `14.60946`, `4.91703`; `6-8 s`: `13.32081`, `13.76842`, `6.63507`; `8-11 s`: `1.57006`, `2.18533`, `0.70161`	Because the inlet is prescribed and nonstationary, these are explicitly descriptive statistics and not stationary turbulence estimates. Cross-wind RMS is about `34.37%` of mean `Fx` during `3-6 s` and `49.81%` during `6-8 s`; the increase during the decay interval is an observed feature of this run and not an approved physical interpretation
Height-resolved response at the peak instant	The 50 layers sum exactly to `17.4044453`; layer-force centroid `z = 1.2493315`; layer mean `Fx = 0.3480889`; layer coefficient of variation `7.94%`; minimum layer `Fx = 0.2711711`, maximum `0.4120568`; the strongest layer at that instant is centered at `z = 1.5248083`; the five strongest layers are centered near `z = 1.18` to `1.77`, each contributing only about `2.18%` to `2.37%` of the total	Distribution of unit-density solver forces
Non-simultaneous extremum	Across all times after 1 s the largest individual layer `|Fx|` is bin 43 at `z = 2.17406003` with `Fx = 0.483829959` at `t = 5.38056 s`, which is not simultaneous with the total-force maximum	Structural use must retain the synchronized layer time histories and must not combine independent layer maxima into one artificial profile
Force decomposition at the peak	Normal `Fx = 17.4151476` (`100.0615%` of total); tangential `Fx = -0.0107023` (`-0.0615%`)	Arithmetic decomposition
Scientific boundary	The active output uses `rhoInf=1`	Every value in this table is a unit-density solver force and not an approved physical design load. No wind-tunnel comparison, grid-sensitivity study, time-step-sensitivity study, density conversion, reference-area normalization or structural-response transfer has been approved
Contract-side consistency observation	`autocfd_case011up/contract.py` fixes `force_outputs.n_bins` to 20 while this case uses 50 layers	Confirms that layer-force postprocessing belongs to a separate product line and is not governed by the Case011Up contract
Terminal	`POSTPROCESSED`	Recomputation from raw data is complete; scientific acceptance is not granted

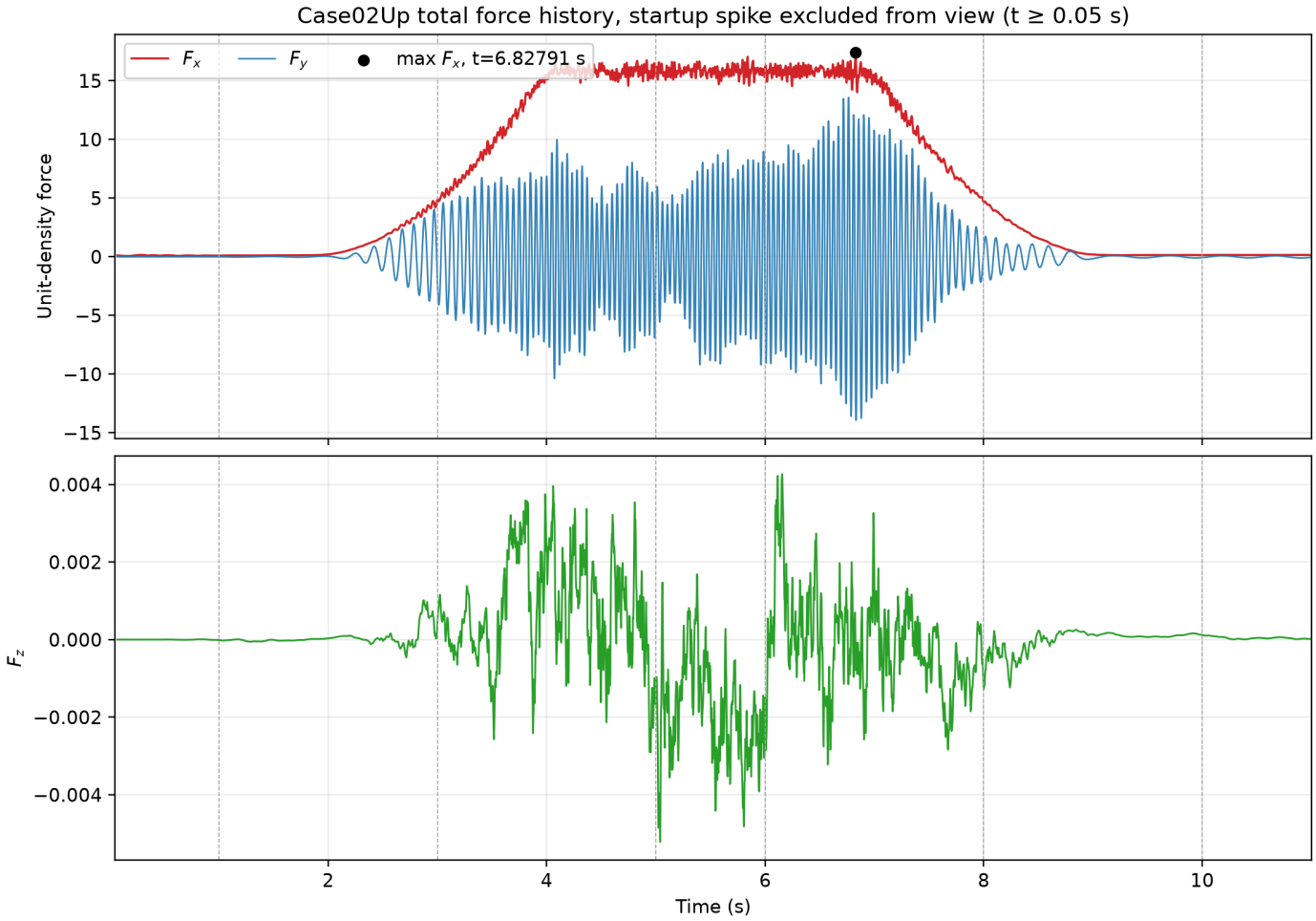


Figure F15. [Classification: postprocessing diagnostic] Along-wind, cross-wind and vertical total-force histories recomputed from the `50,877` raw samples. The values are unit-density solver forces at `rhoInf=1` and are not design loads.

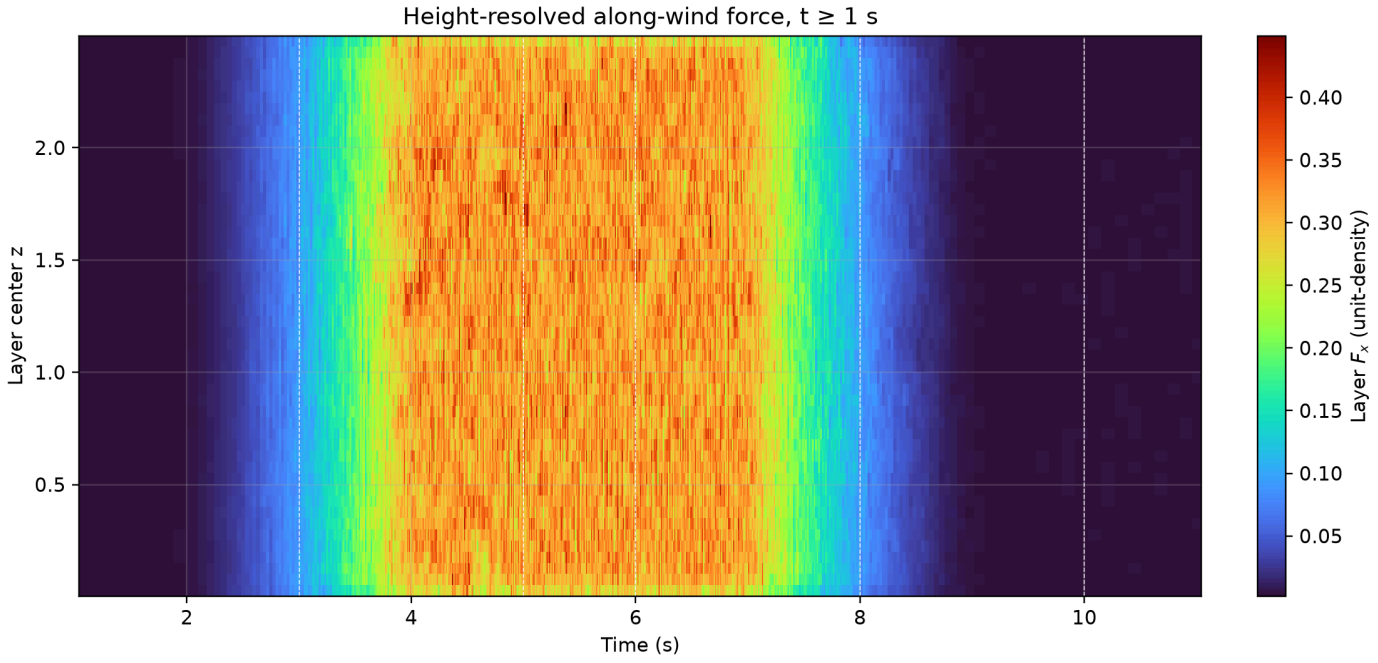


Figure F16. [Classification: postprocessing diagnostic] Time and height distribution of the 50 synchronized layer forces, using the corrected layer centers `0.0265350` to `2.4737147`. This is a postprocessing diagnostic and is not a structural load.

Chapter 8 Evidence Levels, Scientific Limitations and Next Data Actions

Chapter 7 reported what was produced. This chapter states what that production is worth as evidence, what it cannot support, and which next actions would generate new data rather than new prose.

Table T19. Engineering terminal classes versus the scientific terminal class

Class	Criterion	Instances in this snapshot	What cannot be inferred
Engineering implementation terminal	The program object exists and is statically verifiable, that is, `IMPLEMENTED`	70 source files; six cylinder CLI modes; 24 `route_b_cli` subcommands for Case011Up; the D420 interface chain	Cannot infer that any case has been solved or that any result is usable
Static package terminal	Package identity and static completeness verified, that is, `PACKAGE_READY`	Three-arm package with `5,293` files and `STATIC_THREE_ARM_PACKAGE_OK`; the Task 6 `fine_dt0025` deterministic candidate	Cannot infer `SUBMITTED` and cannot infer that a solver ran
Scheduler terminal	The job state recorded by the scheduler	`3382272` `PASS`; `3396039` `COMPLETED 0:0`; `3391286` `FAILED 2:0` after being `SUBMITTED` exactly once; `3383251` and `3383356` `COMPLETED 0:0`; Condor32 `1542.0` exit code 0	Cannot infer scientific acceptance
Application terminal	Normal solver termination markers, that is, `SOLVER_RAN`	`End` and `Finalising parallel run`; `simpleFoam` to time 1000; `pimpleFoam` for five cases to 30 s; `FOAM FATAL ERROR` appears in `3391286`	Cannot infer physical validation
Wrapper evidence terminal	Completeness of the wrapper-layer evidence	`3396039` wrapper `TERMINAL_EVIDENCE_COMPLETE` with `PASS`	Cannot infer that the artifacts are complete
Artifact terminal	Whether the artifacts required by the contract are present	`3396039` `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE`; `3391286` scientific output count 0	Cannot infer usability
Usability terminal	The scope over which the artifacts may be used	`3396039` `PARTIAL_MODEL_OUTPUT_USABLE`; coarse ABL `engineering_pipeline_smoke_only`	Cannot infer scientific acceptance
Postprocessing terminal	Whether raw output has been recomputed, that is, `POSTPROCESSED`	Cylinder `METRICS_EXTRACTED`; Case02Up `50,877` samples; `3393320` four-time reconstruction, which is successful postprocessing and not a new solve	Cannot infer a controlled comparison
Claim terminal	The verdict of the claim gate, recorded through `TERMINAL_CLASSIFIED` artifacts	`CLAIM_BLOCKED_FOR_ACCEPTANCE`; `NOT_CONSISTENT_CONTROLLED_COMPARISON`; `ASED`	This is the only class that can connect to a scientific conclusion, and every instance in this snapshot is `BLOCKED` or not released
Scientific acceptance terminal	Explicit approval, that is, `SCIENCE_ACCEPTED`	No instance exists	Cannot be inferred automatically from any of the nine classes above

Table T20. Unresolved scientific questions and the minimum next check for each

ID	Unresolved scientific question	Current evidence position	Minimum next check	Substitutions that are not permitted
Q 1	Why the 41-point sampling contract of the medium empty-ABL case fails at 20 specific heights (`inlet_profile` 2, `abl_inlet` 8, `abl_outlet` 10), and whether the cause is boundary-plane and cell-intersection behavior, the production postprocessing condition, or their interaction	`READ_ONLY_PROFILE_GRID_REANALYSIS_3396039.md` has localized the blocker to coordinate level and the pattern is invariant across all ten output times	Rerun the same serial volume sampler in a create-only reanalysis root, with any patchCloud supplement labelled separately	Interpolation, row padding, renaming patch extraction as volume-sampler evidence, or a blind solver rerun
Q 2	Whether `empty_abl_acceptance` can be granted	`stampede3_targets.py` lists it as the first required gate for `building_rans` ; the coarse side stands at EPTANCE	Close Q1 first, then a determination	Treating coarse patch completeness as closure of the medium sampling contract
Q 3	Whether the `6.31%` medium-to-fine change in `Cl_pp` is materially affected by the Courant number or time-step error at `dt = 0.005 s`	Table T14, Table T15 and the frozen research-question text in `task6_candidate.py`	Grant solver-stage execution to `fine_dt0025` , producing a fourth spatial point and a new temporal point	Declaring mesh independence or an order of accuracy from the existing three spatial points
Q 4	How the Case02Up unit-density layer forces convert into a structurally usable load	`50,877` samples, corrected layer centers, the peak-instant layer distribution, centroid `z = 1.2493315` , and the evidence that the largest single layer is not simultaneous with the total peak	Define physical density, reference area and coordinate system, and deliver the 50-layer synchronized time histories	Combining independent layer maxima into one profile; producing a design load
Q 5	What the resolved-height `epsilon` deviations mean (middle `20.42%` , upstream `41.88%`)	The tables in the coarse full-evidence file	Re-evaluate after the profiles are completed under the same operator	Adopting the incorrect description that resolved-height `epsilon` is preserved within a few percent
Q 6	How reference velocity and reference coefficients should be defined under nonstationary inflow	Condor32 `Cd_ref = 0.76965972` and the reference-only statement in the `forceCoeffs` block	A time-varying or fixed reference velocity definition	Treating an arithmetic coefficient as a physical coefficient
Q 7	How large the M0 versus M1 difference is under matched-drive conditions	The three-arm package is `PACKAGE_READY` with no solver result	One three-arm submission	Substituting historical M0/M1 metrics for a matched-drive conclusion
Q 8	Whether the near-wall compatibility items (coarse candidate `y+` approximately 1913, medium `Ks/z0 = 12.66`) affect interpretation	The two ABL evidence files	Evaluate after wall distance and wall shear stress are added to the runtime output	Using a smooth-wall wall-function guideline as a frozen gate for a rough-wall case

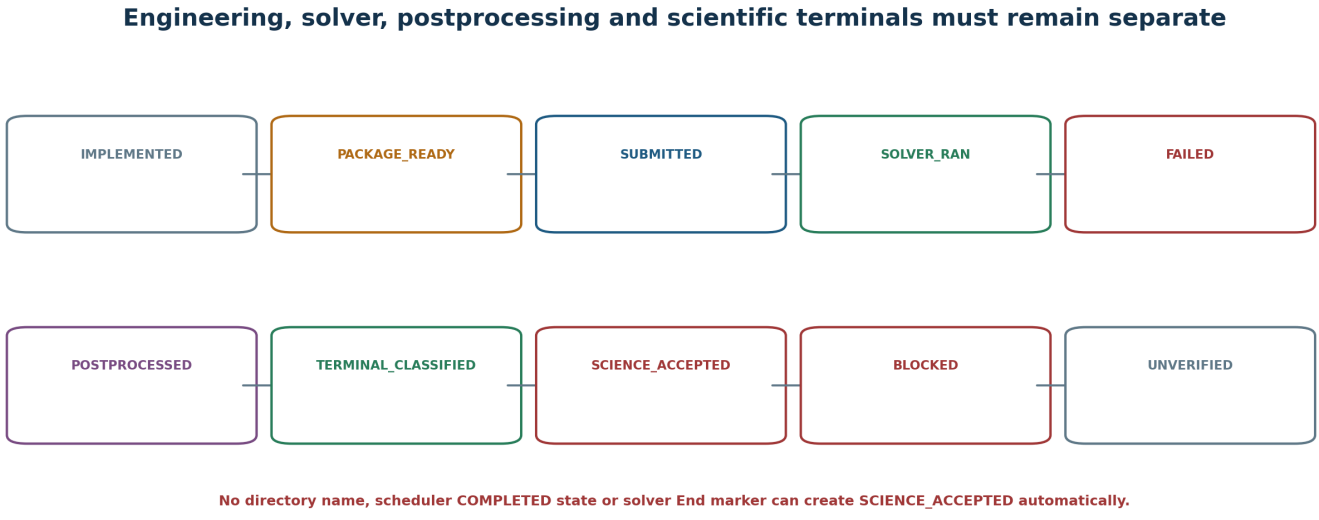


Figure F08 (repeated reference in Chapter 8). [Classification: method diagram] Relative position of the evidence levels and the claim gate. The diagram carries no numerical results and is used here to show why an engineering terminal cannot be promoted into the scientific terminal class.

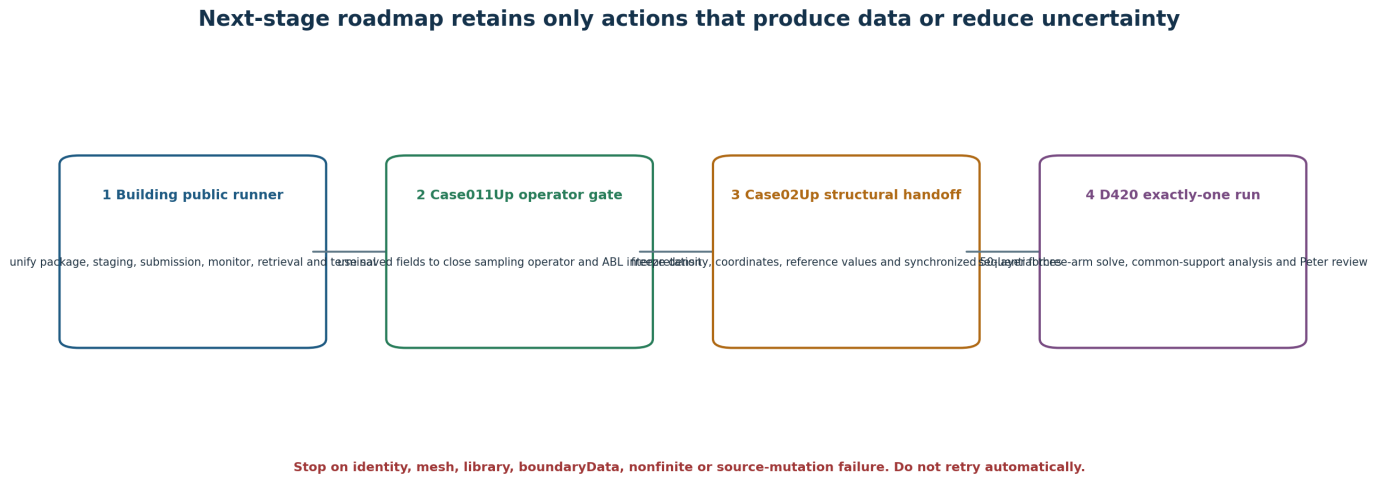


Figure F22. [Classification: method diagram] The five data-producing actions R1 to R5 with their prerequisites and stopping conditions. The figure contains no result values.

The ordering rationale is as follows. R1, the Job `3396039` operator reanalysis, ranks first because it is the only scientific issue whose blocker is already localized to coordinate level, with 2, 8 and 10 specific missing heights invariant across ten output times, and because it requires no new solving. It also directly determines whether `empty_abl_acceptance` can be granted, and that gate is the first required gate for `building_rans` in `stampede3_targets.py`. R2, unifying Case011Up under the public runner, ports the verified cylinder mechanism to the building line; its value is not governance but the removal of manual recovery, since the Condor32 and TACC terminals are currently assembled after the fact from receipts, with no checkpoint semantics and no per-stage receipts, so the question of whether a continuation resumed from the correct state cannot be answered. R3, the Case02Up structural handoff conversion, has explicit scientific content, namely `rhoInf=1` to physical density, reference area, coordinate system and 50-layer synchronized histories, and the inputs for that conversion are already complete. R4, one matched-drive three-arm submission, is the only action that can produce matched-drive data. R5, solver authorization for Task 6 `fine_dt0025`, targets the one substantive numerical gap in the cylinder line, the approximately `6.31%` medium-to-fine lift-amplitude change.

Chapter 9 CM1-CFD Purpose and Coupled-Workflow Method

CM1-CFD pushes the AutoCFD method into time-varying, multiscale, conservative input mapping and matched-drive comparison. It is the only line that simultaneously exposed an input-contract defect, coupled confounders and the gap between package readiness and execution. Its purpose is to carry a downburst wind field produced by CM1 into an OpenFOAM case in a way that is byte-reproducible, non-corrective and auditable, so that any later difference between two CFD arms can be attributed to a designed variable rather than to the interface.

9.1 Source identity and interface qualification

`2026-08-05_CM1_D420_G2_readback_and_CFD_pilot_contract_review.md` records a read-only readback of three write indices (`36`, `245`, `455`) by 48 ranks, with parent verification passing 25/25. The same file records two engineering failures and their handling. The first module attempt failed before NetCDF open because NetCDF required an explicit compiler module, and that failure was preserved rather than deleted. A complete first read then found that a 0.1 mm threshold on `zhval` versus `1000*zh` was too strict for independent float32 unit representations, with an observed maximum difference of `0.954 mm`; the comparison tolerance was changed to a documented 2 mm without changing scientific or remote scope, and the final full rerun passed. This is a clear case of a tool tolerance that is not a scientific tolerance.

`2026-08-05_CM1_Opus5_D420_to_CFD_review.md` gives the independent verdict `PROCEED_TO_LOCAL_EXTRACTION_PREFLIGHT_ONLY`. D420 passes the scheduler, application, actual-time, rank completeness, finite-field and SHA256 gates but does not pass coupling Gate 0, because the frozen evidence lacks coordinate arrays, grid extents, per-rank ownership, halo rules, actual per-variable stagger, AGL semantics and five source-identity hashes. The authoritative CFD time origin is `2041.5858154296875 s` and the usable target interval is `0` to `418.4141845703125 s`; a synthetic 2040 s state and the substitution `t_CFD = t_CM1 - 2040` are forbidden. The current six-field package can support only a neutral velocity-first route after Gate 0, and a buoyant or moist continuation remains `BLOCKED` because pressure, density, vapor, total thermodynamic state and base-state closure are absent.

Table T21. D420 source, coordinates and window

Item	Value	Boundary
Ranks and topology	48 ranks, `6 x 8`	Per-rank ownership is not given in the frozen evidence and is one reason Gate 0 does not pass
Global scalar grid	`800 x 800 x 84`	Source grid
Shared interfaces	120 u shared interfaces, 126 v shared interfaces	Byte-exact continuity checks passed
Levels from 0 to 3 km	29, that is `zh` indices 0 through 28 inclusive	Top scalar height `2973.0770587921143 m`
Readback write indices	`36`, `245`, `455`; real times `2041.5858154296875`, `2250.0`, `2460.0 s`	Zero rank spread; 144 files and `1,539,204,481` represented source bytes; parent verification 25/25
Shared control volume	`x = [7800, 11800] m`, `y = [7700, 11700] m`, `z = [0, 1000] m`, downburst center `(9800, 9700) m`	Structured order `cell_id = (k*ny + j)*nx + i`
Matched window	Source time `2120` to `2380 s`; CFD relative time `0` to `260 s`; 261 records; cadence `1.0 s`	Write index `[115, 375]`; `no_extrapolation` is `true`
Crop anchor	The rounded historical anchor `(62.75, 28.25) km` maps to `i=527:726`, `j=182:381` with an actual center of about `(62.70000267, 28.20000076) km`; the evidence anchor `(62.775, 28.2) km` maps to `i=528:727`, `j=182:381` with an actual center of about `(62.80000114, 28.20000076) km`; the previously conjectured `j=183:382` mapping is not supported by the nearest real coordinates	No anchor has been selected, and the crop evidence is deliberately left unresolved
Frozen one-frame volume	Native dimensions `time=1`, `xh=200`, `xf=201`, `yh=200`, `yf=201`, `zh=29`, `zf=30`; retains native `u`, `v`, `w` and `zhval` plus coordinate-aware centered velocity products	Temperature, humidity, microphysics and CM1 absolute pressure remain excluded
CFD-side review identity	IAC `IAC-20260805-72b9a99eca`, kanban task `t_93348741`, route `petw.cfd.task`, final state `DONE`, run `868`, artifact SHA-256 `e828c2824bdd06a0ae1c1f1ef2c35cc304d1f84c21c2fe23a9879942d40c0e25`; CFD semantic validation 16/16 PASS; CM1 parent validation 16/16 PASS	The review recommends the `evidence_anchor_candidate` (write index `36`, exact source time `2041.5858154296875 s`) only for format, rank assembly, destaggering and same-frame initialization-interface verification, and explicitly does not treat that frame as a primary-peak or physically representative frame
Gate state	`G2_TACC_PREFLIGHT=PASS`; `CFD_G2_CONTRACT_REVIEW=PASS`	`cfd_handoff_allowed=false` remains unchanged

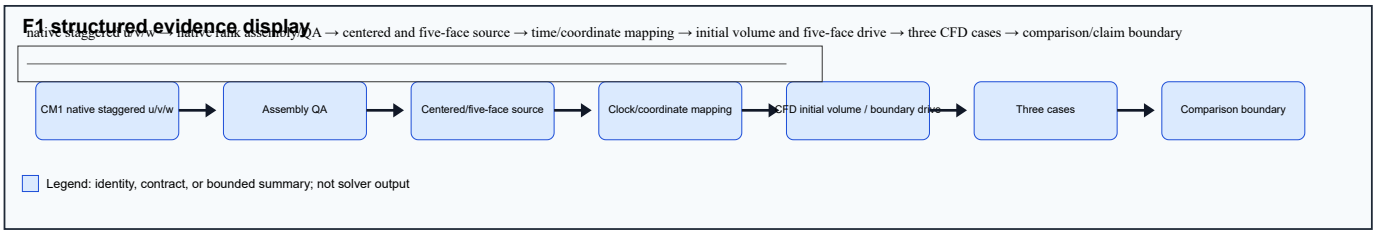


Figure F23. [Classification: method diagram] Data flow from rank-tiled staggered NetCDF through five-face `boundaryData`, the initial field and the OpenFOAM case tree. The diagram contains no solver results.

9.2 The interface program chain and its conservation semantics

`packet_reader.py` freezes the production schema `petw-d420-d4b-production-packet-v1` and the production dimensions: `time 261`, `west/east/south/north_point` 5800 each, `top_point 40000`, `bottom_point 40000`, `zh 29`, `yh 200` and `xh 200`. Face order is `west, east, south, north, top`; face normals, face point orders (`k(zh)_outer_j(yh)_inner` for west and east, `k(zh)_outer_i(xh)_inner` for south and north, `j(yh)_outer_i(xh)_inner` for top) and the bottom normal `(0,0,-1)` are fixed; the flux replay relative tolerance is `1.0e-9`; and the cadence is `1.0 s`. Production mode requires a caller-supplied mandatory byte count and SHA-256 identity and rejects finite implicit NetCDF fill sentinels.

The D420 `contract.py` freezes a fuller identity: `case_id D420EmptyDomain`; production packet `178929854` bytes with SHA-256 `b0dc479d5a48d649fe29a485c681992e227c3ed500dda1459f46c39a83471069`, scientific-content SHA-256 `7cc2aa62f475cf239fb4bdc582dbc52e5f9020acc37fd380bfd22b630758a50a` and fixture SHA-256 `b2140991a3f7ce452a4f787846dde6959a69c77d1fea1b720c2e8c1932cf8481`; a clock contract of 261 records; M0 topology `[200,`

`200, 29]` with `1,160,000` cells and `1,212,030` points; C0 topology `[100, 100, 29]` with `290,000` cells and scientific use `NONE_ENGINEERING_MECHANICS_ONLY`; boundary condition types `d420TimeVaryingMappedInletOutletVelocity` and `d420ComplementaryPressure`; and a velocity policy of `preserve_raw_source_vectors_exactly_no_correction`.

Table T22. Destaggering, boundaryData, initial mapping and flux correction

Element	Rule	What is explicitly not done	Bound implementation
Native stagger to centered	Native dimensions `u(time,zh,yh,xf)`, `v(time,zh,yf,xh)`, `w(time,zf,yh,xh)`; coordinate-aware centered velocity products are retained together with the frozen one-frame volume	No conservative correction of any kind	G2 readback record; `packet_reader.py`
Inflow and outflow switching	The per-element inflow mask is determined by the dot product of the raw mapped source vector and the outward face normal: `dot < 0` is inflow with `U` fixed to the full mapped vector, `dot >= 0` is outflow with zero normal gradient, and the exact-zero case is frozen to outflow	Solver-modified fields never redefine the mask	`mask_reference.py`, the Python golden reference for the C++ boundary library and the C0 runtime gate
Time interpolation	Permitted only between two adjacent 1 s records with a weight inside `[0, 1]`	Anything else raises rather than silently clamping or extrapolating	`mask_reference.py`
Flux correction	No flux correction of any kind is computed there or anywhere downstream	No flux correction	Module declaration in `mask_reference.py`
Initial field mapping	Maps `initialization/velocity_center[0, k, j, i, :]` to internal-field cell `(k*ny + j)*nx + i` and validates the formula string and dimensions in `CELL_ORDER_MAP.json` under the case root	A pure reorder with no smoothing, scaling, correction or projection; refuses to run unless the science contract freezes the projection policy to `PROJECTION_FORBIDDEN_WITHOUT_SEPARATE_APPROVAL`; refuses to overwrite an existing `0/U`	`initial_field_mapper.py`
Discrete diagnostics	Discrete divergence, the ground wall-normal component and the `t=0` boundary versus adjacent-cell mismatch are reported as observations	They never trigger any action	`initial_field_mapper.py`
boundaryData export	Writes five-face `constant/boundaryData` with one `points` file per face and one `U` file per packet clock record; points use float64 `%.17g` and velocities float32 `%.9g`, both of which round-trip exactly, so replaying the same packet into a fresh root yields byte-identical files and an identical manifest	The output root must be an owned root (under the workspace `04_Outputs` or below a `.cfd_automation_owned` marker); an existing boundaryData tree or manifest is never overwritten; all data is written into a temporary directory that is atomically renamed on success and removed on failure; no bottom directory and no time directory outside the packet clock is ever produced	`boundarydata_exporter.py`, which also records that it deliberately does not plumb the production identity parameters because production boundaryData generation is a separately action with default `DO_NOT_GENERATE`
M1 sub-window	Frozen to `x=[7800, 11800)`, `y=[7700, 11700)`, `z=[0, 1000]` centered on `(9800, 9700)`, resolving on the production grid (100 m centres from 50 to 19950 m) to index slices `x=[78, 118)` and `y=[77, 117)`, that is 40 by 40 source cells of 100 m	Not a nested-boundary extraction	`window.py`
M1 mesh	`400 x 400 x 100` uniform hexahedral cells of exactly 10 m; any non-uniform combination raises; west, east, south, north and top are `patch`, ground is `wall`	Non-uniform combinations are not accepted	`mesh_generator.py`
M1 URANS parameters	`U_REF_M_S=15.0`, `TURBULENCE_INTENSITY=0.10`, `TURBULENCE_LENGTH_M=500.0`, `CMU=0.09`, `NU_M2_S=1.5e-5`, `END_TIME_S=260.0`, `INITIAL_DELTAT_S=0.05`, `MAX_CO=0.5`, `MAX_DELTAT_S=0.5`, `WRITE_INTERVAL_S=10.0`, binary write format, 48 ranks; `k = 1.5 (U_ref I)^2` and `omega = sqrt(k) / (Cmu^0.25 L)`	No unfrozen turbulence parameter is introduced	`render_m1.py`
Independent cross-check	Those formulas give `k = 1.5 x (15.0 x 0.10)^2 = 3.375` and `omega = sqrt(3.375) / (0.09^0.25 x 500) = 0.00670820393`, matching the M1 `initial_k` of `uniform 3.375` and `initial_omega` of `uniform 0.00670820393` recorded in `D420_MESHFIX_V7_BOUNDARY_CONSISTENCY_AUDIT_20260811.json`	This is one of the few places in the snapshot where source code and a remote case dictionary agree independently	Cross-reference between the two files
Recorded engineering approximation	The sub-window side planes are interior to the 20 km CM1 domain while the packet carries only the 20 km outer faces, so M1 applies the outer-face forcing at the sub-window planes as directed	This is an engineering approximation and not a nested-boundary extraction	Module documentation in `subwindow_extractor.py`; the approximation later becomes one of the central confounders in the M0 versus M1 comparison

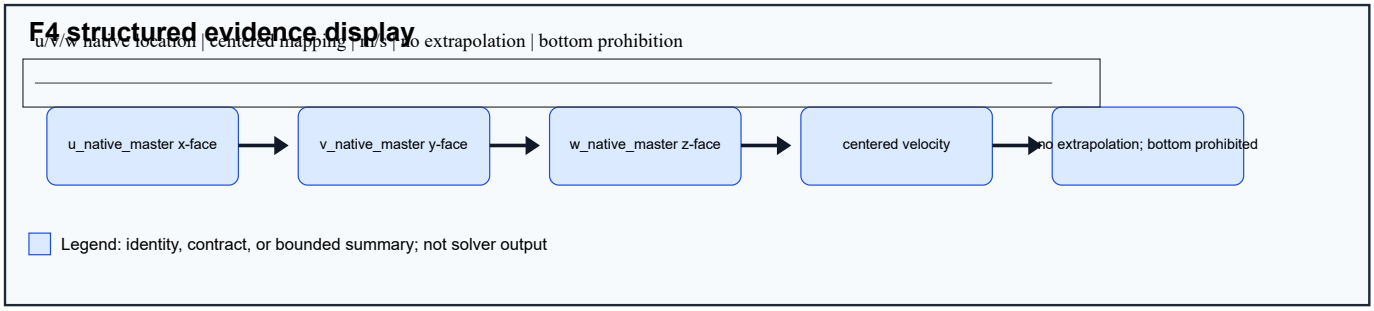


Figure F24. [Classification: method diagram] Correspondence between the native staggered velocity dimensions and the coordinate-aware centered velocity products. The diagram contains no numerical results.

The interface chain is therefore `IMPLEMENTED` and byte-reproducible. That status describes the mapping software only. It does not describe any CFD result obtained through the mapping.

Chapter 10 CM1-CFD Execution History and Evidence Terminals

10.1 Historical M0 and M1

Table T23. Historical M0 and M1

Item	M0	M1
Job	`3384063`	`3386652`
Scheduler and application	Application-complete OpenCFD v2412 run to 260 s	Scheduler `COMPLETED 0:0`, solver return code 0, final time 80 s, postflight PASS
Mesh	`[200, 200, 29]`, `1,160,000` cells, `1,212,030` points	`400 x 400 x 100`, `cell_count` `16,000,000`, uniform 10 m
Ranks	48	48
Writes	1 s writes	`writeInterval 10`; nonzero `U`, `p`, `k`, `omega` and `nut` on all 48 ranks at 50, 60, 70 and 80 s
Numerical diagnostics	Maximum reported Courant number `0.228`; cumulative continuity error approximately `-6.0e-8`; no fatal or nonfinite marker	No equivalent Courant or continuity history is given in this snapshot
Domain	20 km x 20 km x 3.038 km	4 km x 4 km x 1 km
Ground treatment	`slip`	`noSlip`
Turbulence closure	`laminar`	URANS `kOmegaSST` with `k`, `omega` and `nut` wall functions
Outer velocity fallback	`uniform (0 0 0)`	`$internalField`
Outer pressure fallback	`uniform 0`	`$internalField`
Initial field	Nonuniform `U` field, `1,160,000` points, component means `[8.657224357301942, -4.6082312361829185, 0.35855460887963203]`, speed mean `10.66461219660349`, minimum `0.1900968295684176`, maximum `31.828220595488165 m/s`	The original `0/U` is uniform zero; the selected run carries restart evidence that must be independently aligned
`controlDict`	`startFrom startTime`, `startTime 0`, `endTime 260`, `deltaT 0.05`, `writeInterval 1`	`startFrom latestTime`, `startTime 0`, `endTime 80`, `deltaT 0.05`, `writeInterval 10`
Outer-face forcing	Outer faces belong to the full-domain case	Outer faces apply the interior sub-window forcing approximation
Terminal	`SOLVER_RAN`	`SOLVER_RAN`

`D420_M1_VS_M0_SCIENTIFIC_ACCEPTANCE_PROTOCOL.md` states plainly that these facts establish execution and output availability only, and that no matched-domain M1 versus M0 scientific evidence currently exists. Treating the historical D420 M0 and M1 outputs as a controlled comparison is prohibited.

The comparison method is the methodological contribution of this line. The packet-derived numeric M0 first center is `50.00000074505806 m` and is snapped to the intended 50 m height within a frozen `1e-4 m` coordinate identity check; M1 has cell centers at 45 and 55 m, so the M1 50 m field is their linear interpolation before horizontal coarsening. Coarsening reshapes the `400 by 400` M1 plane into `40 by 10 by 40 by 10` blocks, and each 100 m M0 cell receives the arithmetic mean of its 100 equal-area M1 cells; the averaged M1 coordinates must replay the selected M0 coordinates within `1e-9 m`, which is a numerical identity check and not a scientific acceptance tolerance. A 500 m boundary buffer is frozen as a diagnostic partition leaving a 3 km by 3 km central region, and every principal metric is reported for the full shared plane, the central region and the boundary buffer. Pressure comparison removes each plane mean, because pressure datum and domain differences would otherwise dominate through an arbitrary offset. A shared-control-volume net-flux estimate is computed from cell-centered boundary-adjacent velocities and exact cell-face areas and is explicitly a diagnostic closure estimate rather than a replacement for face-flux `phi` integration, with missing `phi` or mapped-face output recorded as unavailable rather than synthesized. M1 at 5 m is reported separately as a model diagnostic and must not be called 10 m wind, `s10`, `Vhat_3s` or an engineering gust. The protocol introduces no numeric scientific

acceptance tolerance and leaves four interpretations to PI accept M1 for bounded mechanism comparison, conditionally accept while requiring a named follow-up, reject M1 because boundary or model artifacts dominate, or declare the result inconclusive and request one named additional diagnostic.

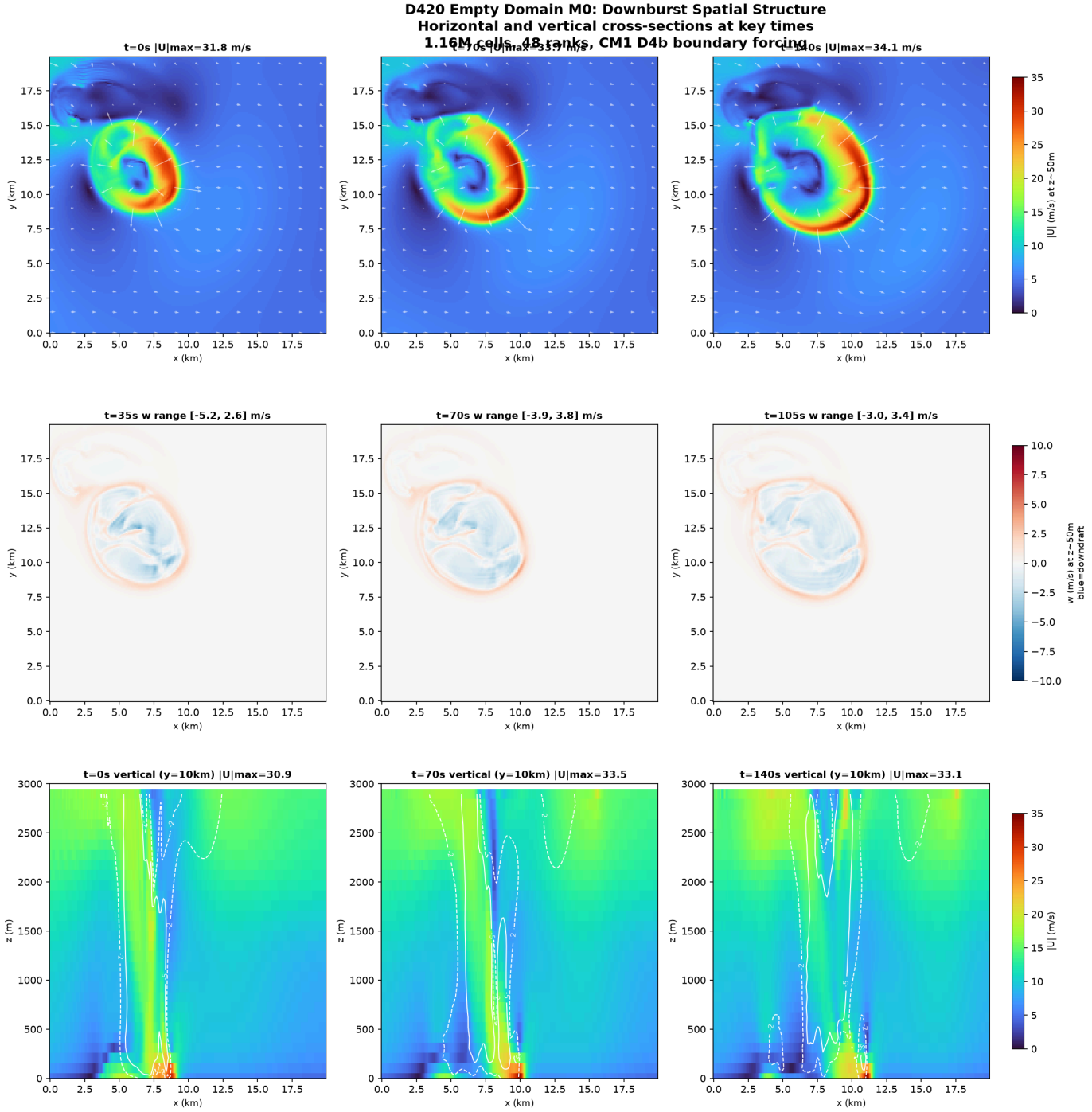


Figure F17. [Classification: historical diagnostic display] Spatial diagnostic display of the historical M0 and M1 outputs on the selected height plane. The two runs were executed under different case setups, and this figure is not a controlled comparison result.

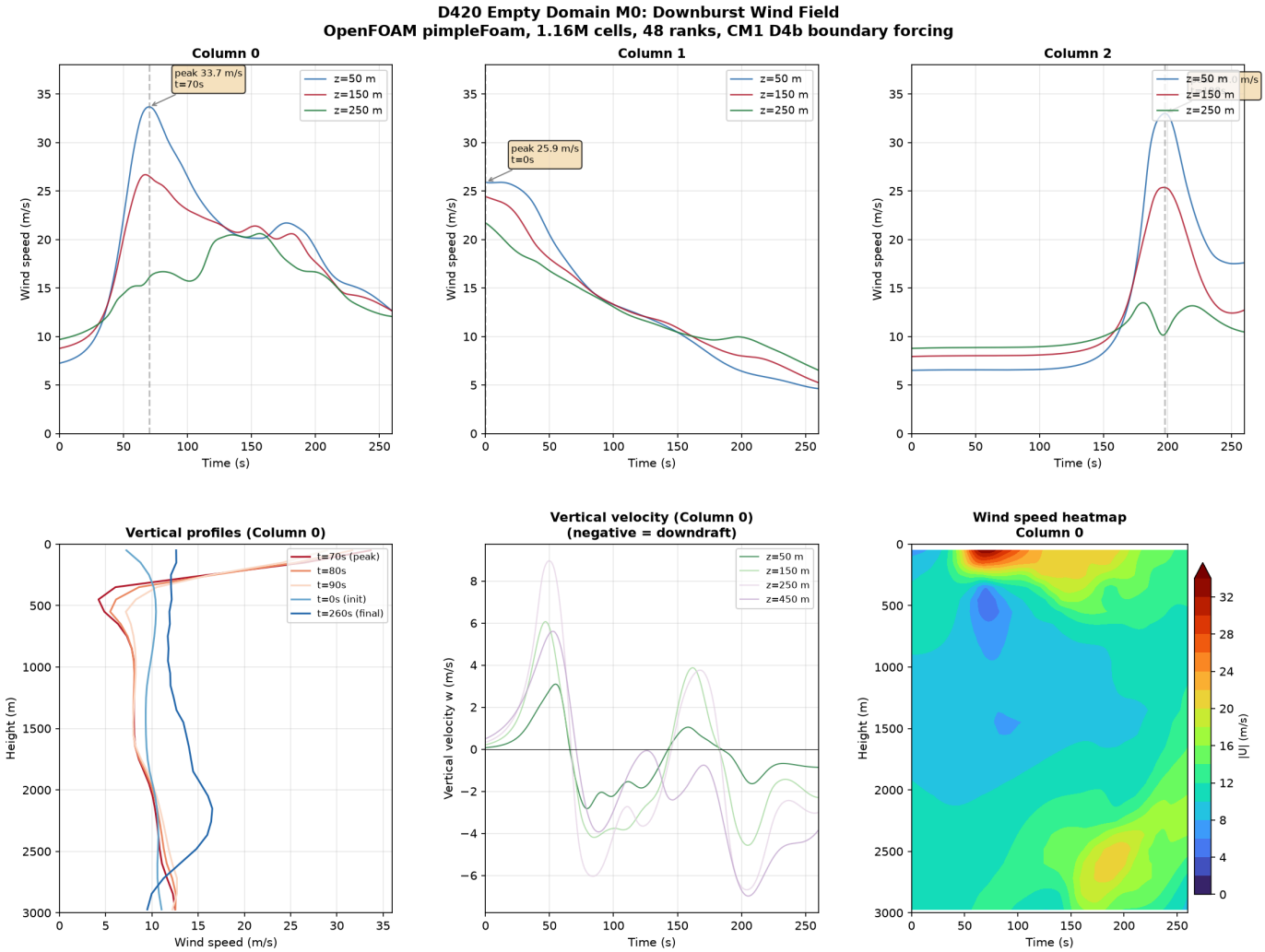


Figure F18. [Classification: historical diagnostic display] Time-series diagnostic display for the historical M0 and M1 outputs. M0 runs to 260 s and M1 to 80 s with different clocks and write cadences, so this figure is not a matched-drive comparison.

10.2 Jobs 3391286 and 3393320

Table T24. Jobs 3391286 and 3393320

Item	Job 3391286	Job 3393320
Status	`FAILED_BEFORE_RECONSTRUCTION`, that is, `FAILED`	, that is, `POSTPROCESSED`
Scheduler terminal	`FAILED`, exit code `2:0`, elapsed `00:00:57`, start `2026-08-10T13:00:07-05:00`, end `13:01:04-05:00`, node `c454-101`, partition `skx-dev`, account `TG-CIV260040`; allocation of 1 node and scheduler allocation of 48 CPUs against a requested application shape of 1 task and 1 CPU	Output root `0_v7`
Submission semantics	`SUBMITTED` exactly once after live account, partition, source and output checks and a successful `sbatch --test-only`	Executed under the postprocessing contract
Application failure	`FOAM FATAL ERROR: (openfoam-2412) Cannot find file "points" in directory "polyMesh" in times "0" down to constant`, raised during the first M0 `reconstructPar` call	None
Root cause	The root-level clean M0 source view has a readable `constant/polyMesh/points` at `62,279,881` bytes, but each of its 48 processor directories contains only the selected time directories `50,60,70,80`, and both M0 and M1 have `processorN/constant/polyMesh/points` present at `0/48`. The failure is therefore not caused by the root symlink target or by file permissions: `reconstructPar` requires the processor-level decomposed mesh payload, and the clean-view QA checked selected processor time fields without checking that inventory	Not applicable
Scientific output	Count 0. The job did not reach M1 reconstruction, compact extraction, comparison, solver execution, `Vhat_3s` or Figure 5-style production	M0 reconstruction of `U,p` at 50, 60, 70 and 80 s passed; M1 reconstruction of `U,p,k,omega,nut` at the same four times passed; four metrics JSON files and four figure-input NPZ files were produced with `automatic_acceptance false`
Key qualification	The supported conclusion is that the current clean source-view contract is insufficient for `reconstructPar`. The unsupported conclusions are CM1 mapping error, solver divergence, TACC resource failure and M1 versus M0 scientific disagreement	The `interpretation_boundary` states directly that no `pimpleFoam` or URANS simulation was executed by this postprocessing contract, that continuity, Courant and mapped-face histories are unavailable and not synthesized, and that `Vhat_3s` is not available and not inferable from s1diag or these snapshots. Job `3393320` is successful postprocessing of existing outputs and is not a new solve; `new_solver_run` is `false`
Repair boundary	Copying the root global mesh into every processor directory is not an acceptable repair; a safe repair requires either the original processor-level mesh payloads from the source of truth or a separately reviewed exact decomposition procedure whose generated processor mesh and field ordering are independently proven compatible; running `decomposePar`, regenerating processor meshes, modifying source views or submitting a repaired successor requires a new package, new output and control roots and a new explicit approval. The one approved successor submission was consumed by this job, no retry was performed, and the failed job remains preserved	An independent audit classifies the same job as `NOT_CONSISTENT_CONTROLLED_COMPARISON`
Exploratory metric ranges over the four times	Not applicable	M1 to M0 peak horizontal speed ratio `0.44025700464974993` to `0.45701284756890426`; central horizontal speed bias `-9.701145014615593` to `-7.288756870256819 m/s`; central horizontal speed correlation `0.5606162092008797` to `0.6220281334400477`; central horizontal vector RMSE `13.21909373385293` to `14.343349979427432 m/s`; full-plane horizontal speed correlation `0.3179760263930257` to `0.4020569123259768`; boundary horizontal speed correlation `-0.027374855570347016` to `-0.009068803588600575`; pressure spatial correlation `0.24828318970760035` to `0.31412250539305153`; M0 minimum `w` `-3.8013528172164293` to `-2.9224795626958007 m/s` while M1 minimum `w` is `-0.9911255959993416` to `-0.9403806612929588 m/s`; vertical-section horizontal speed correlation `0.16949527754847446` to `0.28885167372915715`
Eight direct conflicts	Not applicable	Ground velocity M0 `slip` versus M1 `noSlip`; turbulence closure M0 `laminar` versus M1 URANS `kOmegaSST`; M1 has `k`, `omega` and `nut` wall functions while M0 has no turbulence fields; outer velocity fallback `uniform (0 0 0)` versus `$internalField`; outer pressure fallback `uniform 0` versus `$internalField`; domain 20 km x 20 km x 3.038 km versus 4 km x 4 km x 1 km; M0 starts from a nonuniform `U` field while the M1 original `0/U` is uniform zero and the selected run carries restart evidence requiring independent alignment; M1 outer faces apply an interior sub-window forcing approximation while M0 outer faces belong to the full-domain case
Byte-level evidence	Not applicable	M0 and M1 `boundaryData` both have 5 patches, 261 time directories and 1305 files, yet M0 is `606,314,744` bytes and M1 is `60,276,966` bytes. For `east/50/U`, M0 declares 5800 points at `214157` bytes with `u_mean 8.279567809813793`, `v_mean -5.255522186532414` and `w_mean 0.36474442184454653`, while M1 declares 1160 points at `42755` bytes with `u_mean 8.330903653465517`, `v_mean -5.24861428045431` and `w_mean 0.47129194130517243`; `same_path_hashes_differ` is `true`. Identical directory labels are not identity
Valid and	Valid: the current clean source view does not satisfy the input requirements of `reconstructPar`	Valid: the existing M1 and M0 outputs differ substantially under their respective case setups while both retain a negative- `w` downburst

Item	Job 3391286	Job 3393320
invalid claims		diagnostic. Invalid: that these metrics prove the differences were caused by mesh refinement, that M1 preserves M0, or that M1 is scientifically accepted. The near-zero boundary correlation supports prioritizing investigation of interior-plane forcing and domain truncation but is not causal proof

The failure of Job `3391286` changed the method by promoting the execution-input contract from an implicit assumption to an explicit inventory item. The meshfix successor supplied `points`, `faces`, `owner`, `neighbour` and `boundary` for every processor view and passed multiple fail-closed intermediate gates.

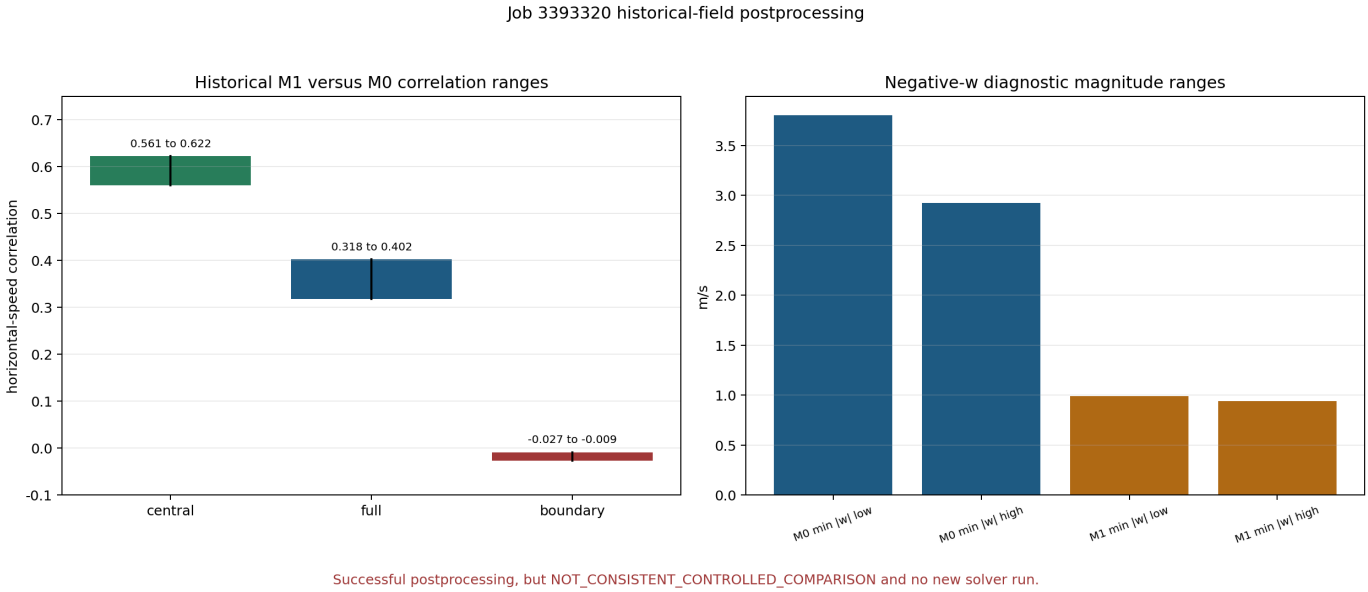


Figure F19. [Classification: postprocessing diagnostic] Exploratory metric ranges across the four reconstructed times. The corresponding job executed no solver, and it is independently classified as `NOT_CONSISTENT_CONTROLLED_COMPARISON`.

10.3 The matched-drive three-arm static package

Table T25. Matched-drive three-arm package

Item	Value	Boundary
Source	The CM1 production packet from job `3379790` was read back locally with an identity matching the caller-verified production bytes and SHA-256; the three-arm package was rendered from that production packet and not from the fixture	Input identity verified
Formal case identifiers	`D420_MC100_LAMINAR` (100 m, laminar); `D420_MF010_LAMINAR` (10 m, laminar); `D420_MF010_KOMEGASST_URANS` (10 m, kOmegaSST URANS)	The three-arm design separates the confounders exposed by the historical comparison into discriminable variables
Common constraints	All three cases bind the same five-face time-varying `boundaryData` manifest, the same `0-260 s` clock, the same `noSlip` ground treatment, no restart and the same CM1-derived initial velocity mapping; only the URANS case contains `k`, `omega` and `nut`	Confounders are designed out rather than argued away
Local package identity verification	All `5,293` manifest-listed files matched their recorded paths and sizes; six critical files, including all three initial `U` fields and both package identity reports, matched their manifest hashes	Static status `STATIC_THREE_ARM_PACKAGE_OK`
Execution status	No solver, scheduler or remote write has occurred	Status is `PACKAGE_READY`; the package has not been submitted and there is no solver result
ComputeOps live consultation	Candidate execution envelope `TACC_Stampede3`, account `TG-CIV260040`, `skx`, 4 nodes, 192 ranks, a 720 minute walltime request and an estimated 600 minute runtime	The candidate is marked benchmark-only and queue-heuristic with `authorizes_submission=false`; the exact package and a new remote root still require direct approval
Package claim boundary	Proves input identity and static case completeness only	Does not prove mesh quality, solver completion, numerical validity, downburst mechanism or scientific acceptance

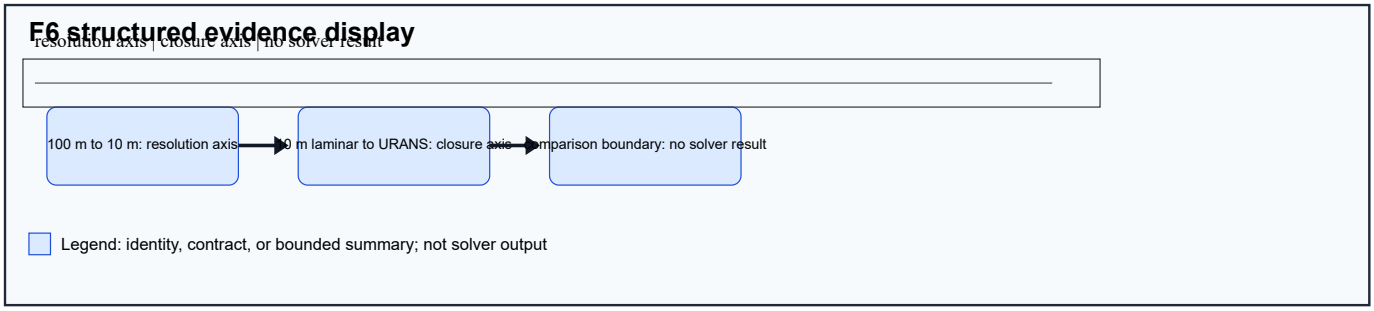


Figure F25. [Classification: method diagram] The common constraints held fixed across the three arms and the two variables that are separated, namely mesh scale and turbulence closure. The package has not been submitted and carries no solver result.

10.4 The V12 bounded display archive

`AU-20260812-005` and `AU-20260812-007` record the formation and acceptance of the V12 display archive. V12 performs a read-only reconstruction over CM1 job `3379790`, the one-frame pilot, M0 job `3384063`, M1 job `3386652`, job `3393320`, failed jobs `3389204` and `3391286`, two existing M0 PNG images, the V8 SVG and the three-arm package. Verification reports 34 package files, 33 SHA entries and 8 figure source and derivative entries; JSON, CSV, SVG and PNG parsing pass; F7 and F8 pixel semantic QA passes with no visible probe annotation; source and derivative replay passes; SHA replay passes; and the ZIP has 34 members with `testzip=PASS`, SHA-256 `5d0ca61b329626f4aadf199aa6784978b9ceb378685464b56407af8cdc5955e7` and `1096344` bytes.

Two items deserve separate record. First, V12 actively removed a factual statement about "87 probes" that could not be bound to the D420 M0 source, and noted that the local combined terminal fragment directly shows the launcher smoke `writeInterval 1000` rather than the M0 production cadence. Second, V12 records a new blocker: the F7 and F8 source-side generation script and raw-array manifest were not independently located, so those images remain source-traceable existing display derivatives with file-level and pixel-level verification rather than raw solver arrays.

The PI instruction to archive was converted into decision record `D-20260812-001`, with current result `ACCEPTED_BOUNDED_DISPLAY_ARCHIVE`. V12 is accepted for project reference, result display and future controlled-CFD planning, and it remains a display and provenance archive rather than a controlled scientific CFD result.

One inconsistency must be recorded. Section 8.1 of the existing work summary states that the V12 display shows M0 domain maximum speeds on the `z` about 50 m plane of approximately `31.8 m/s` at `t = 0 s`, `33.7 m/s` at `t = 70 s` and `34.1 m/s` at `t = 140 s`. The `31.8 m/s` value at `t = 0 s` agrees with the M0 initial `U` speed maximum of `31.828220595488165 m/s` recorded in the consistency audit. The `33.7 m/s` and `34.1 m/s` values have no corresponding source among the 92 files of this snapshot, and neither V12 Agent Update contains them. This study therefore marks those two values as `UNVERIFIED`, meaning that they appear only in the existing report and are not rebound within this snapshot, and does not use them as verified values.

Chapter 11 Integrated Conclusions, Stopping Conditions and PI Decisions

11.1 Four transferable methodological conclusions

First, terminals must be layered. The single most valuable artifact in this snapshot is `TERMINAL_CLASSIFICATION_3396039.json`, in which the scheduler is `TERMINAL_COMPLETED`, the application is `TERMINAL_COMPLETED`, the wrapper is `TERMINAL_EVIDENCE_COMPLETE`, the artifact is `PARTIAL_SCIENTIFIC_ARTIFACT_AVAILABLE`, usability is `PARTIAL_MODEL_OUTPUT_USABLE` and the claim is `CLAIM_BLOCKED_FOR_ACCEPTANCE`. Six terminals coexist and none implies another. Any CFD automation system that reports only the first two will systematically overstate its own maturity.

Second, corrections came from outside rather than from self-checking. The three substantive corrections, namely the Round 5 parser, the coarse ABL `epsilon` description and the Case02Up `z co-ords` semantics, were found respectively by an independent CFD review, by parent verification and by reading solver source code. None was found by the producing side. The deterministic checkers of AutoCFD are good at intercepting identity and structural errors and poor at intercepting semantic misreadings, so semantic misreadings must be carried by independent review.

Third, the input contract matters as much as the output contract. The entire failure of Job `3391286` follows from `reconstructPar` requiring the processor-level mesh payload while the clean-view QA checked only processor time fields. The correct repair was not to relax the check but to add `points`, `faces`, `owner`, `neighbour` and `boundary` to the required inventory and to forbid explicitly the plausible-looking shortcut of copying the root global mesh into every processor directory.

Fourth, the gap between package readiness and execution is a real cost. The D420 three-arm package has `5,293` frozen files, has designed out eight confounders and has a candidate 4-node 192-rank envelope, yet `authorizes_submission` is `false` and there is still zero matched-drive data. Similarly, Task 6 `fine_dt0025` has a deterministic candidate and mesh authorization but no solver authorization. The current AutoCFD bottleneck is not method but the decision between a prepared state and one approved execution.

11.2 Current position of the two tracks

AutoCFD has advanced from a static case generator into three distinguishable things: a complete program loop on a fixed two-dimensional benchmark covering generation, meshing, initialization, solving, continuation, raw-data recomputation, diagnostics, packet sealing and final verification; a building vertical slice that has produced real building time histories and real ABL profiles while execution and recovery are not unified inside the same state machine; and a highly mature but postprocessing-only structural-interface line. It supports the engineering conclusion of controlled automation under a fixed scientific contract. It does not support a general AutoCFD for arbitrary geometry, arbitrary model and arbitrary HPC, and it cannot approve physical validation automatically. The maturity of the cylinder line is a property of that line and is not transferred to the building, HPC or D420 lines.

CM1-CFD has solved the principal engineering interface problems from rank-tiled staggered NetCDF to CFD-centered velocity, five-face `boundaryData`, initial fields and an OpenFOAM case tree, and it has restructured the eight confounders exposed by the historical comparison into a discriminable three-arm design. It supports the conclusion that the interface is `IMPLEMENTED`, that historical solves exist as `SOLVER_RAN`, that one postprocessing job is `POSTPROCESSED` while classified `NOT_CONSISTENT_CONTROLLED_COMPARISON`, and that controlled three-arm inputs are `PACKAGE_READY`. It does not support the conclusion that a matched-drive comparison has been completed or that CM1 to building loads has been validated.

Table T26. Supported and unsupported claims

Category	Content
Supported (model output and execution observation)	Five cylinder cases completed 5/5 under one public workflow, covering generation, meshing, initialization, solving to 30 s, raw-data recomputation and sealing; the Condor32 `1542.0` building case continued to `10.999827824854936` and wrote terminal forces and moments; Case02Up produced `50,877` strictly increasing 601-column samples and 50 synchronized layer forces at `rhoInf=1`; coarse empty-ABL job `3382272` completed 1000 iterations to time 1000, job `3383251` excluded parallel decomposition as the cause of the sampling deficit, and job `3383356` returned `41/41/41` by patch extraction; medium empty-ABL job `3396039` completed to time 1000 with a 680,000-cell mesh reporting `Mesh OK`; the D420 interface chain generates five-face `boundaryData` and initial fields; the three-arm package passed identity verification of `5,293` files; job `3393320` completed four-time reconstruction and metric computation as postprocessing
Candidate interpretations (consistent with model output, not specifically tested)	The Case02Up cross-wind response is consistent with wake-driven instability; the upper-middle bias of the Case02Up layer response must be compared with the prescribed height-dependent inlet field; the coarse ABL candidate `Iu` diagnostic, labelled `CANDIDATE_INTERPRETATION_NOT_CONTRACT`; the medium ABL middle deviation is consistent with wake or downstream flow effects; the near-zero D420 boundary correlation supports prioritizing investigation of interior-plane forcing and domain truncation
Prohibited	Automatic benchmark pass, mesh independence, time-step independence, time-step convergence, ABL preservation or validation, Building RANS readiness, wind-tunnel validation, attribution of M1 differences to resolution, scientific acceptance of M1, `Vhat_3s`, a validated 10 m 3-second gust, design wind speed, design pressure, structural design load, design guidance, code recommendation and Path C promotion; treating a `final`, `accepted` or `ready` directory name or a scheduler `COMPLETED` as scientific acceptance; renaming patch extraction as volume-sampler evidence; treating coarse patch completeness as closure of the medium sampling contract; treating the historical D420 M0 and M1 as a matched-drive controlled comparison; calling V12, the package manifest, the comparison design, F7 and F8, job completion or historical M0/M1 metrics a controlled solver result; treating unit-density forces as a structural design load
Values not rebound	The V12 display values `33.7 m/s` and `34.1 m/s` appear only in the existing report, have no corresponding source among the 92 files of this snapshot and are recorded as `UNVERIFIED`; the caller-supplied external anchor `f278ed027895a8e1d3eb76e36b9acd9559c3fbdbd7acd3bbc8f73f49a0b49419` was not recomputed in this session
Historical evidence, not replayed here	The Round 5 frozen files record 195 control-plane tests passed, 12 benchmark tests passed, 15 analysis and verifier tests passed, Ruff checks passed, a security pattern scan with 0 matches and final verification SHA-256 `479ff703...`. Those are historical entries inside that frozen file. No test suite was replayed during this report packaging session, and this study makes no claim that the full suite passes at present

Table T27. Staged next actions and stopping conditions

ID	Action	New data produced	Prerequisite	Completion condition	Stopping condition	[redacted] decision
R1	Job `3396039` operator reanalysis	Full profiles from the same serial volume sampler in a create-only reanalysis root, with any patchCloud supplement labelled separately	Retained processor volume fields, `polyMesh` and sampling dictionaries are sufficient	Volume-sampler and patch extraction are distinguished, common coordinates are replayed, and the cause of the 20 missing heights (`inlet_profile` 2, `abl_inlet` 8, `abl_outlet` 10) is explained	No interpolation, no row padding, no modification of the original root, no blind solver rerun; if the retained fields are insufficient, record a missing-data terminal	Whether to approve one minimum new solver action if the fields prove insufficient
R2	Unify Case011Up under the public runner	Per-stage receipts, checkpoint semantics and replayable terminals for one building case	Port the cylinder state machine; adapt to Condor and Slurm	Package, staging, scheduler, monitor and retrieve, raw reanalysis and terminal classification are owned by one CLI	No change to the existing science contract; no retroactive rerun of historical jobs	Whether to prioritize the building line over R3
R3	Case02Up structural handoff conversion	A structural load packet with physical density, reference area, coordinate system and 50-layer synchronized histories	The existing `50,877` samples and the corrected layer centers `0.0265350` to `2.4737147`	The handoff contains synchronized layer histories rather than combined independent layer maxima	No design load is produced; no unapproved normalization	Whether to prioritize this over R2
R4	One matched-drive three-arm submission	Mesh, application terminal, raw fields, common-support comparison and full, central and buffer metrics for `D420_MC100_LAMINAR`, `D420_MF010_LAMINAR` and `D420_MF010_KOMEGASST_URANS`	[redacted] direct approval of the exact package, platform, account, partition, resources, new run root, command family and one submission scope	All three arms complete and all metrics are reproduced by independent local replay	Any identity, mesh, library, `boundaryData`, nonfinite, source-mutation or case-ordering failure reports a terminal with no automatic retry	Whether to authorize this one submission
R5	Task 6 `fine_dt0025` solver authorization	A fourth spatial point and a new temporal point, testing whether the `Cl_pp` trend is materially affected by the Courant number or time-step error at `dt = 0.005 s`	Deterministic candidate, lock `2cc9b6c5...`, authorization `229a08f7...`, mesh dispatch authorization currently stopping at `MESH_QUALIFIED`	All four comparison axes are produced and pass final campaign verification	No command outside the allowlist; no change to the physics contract	Whether to extend execution authorization from `MESH_QUALIFIED` to `PACKET_SEALED`

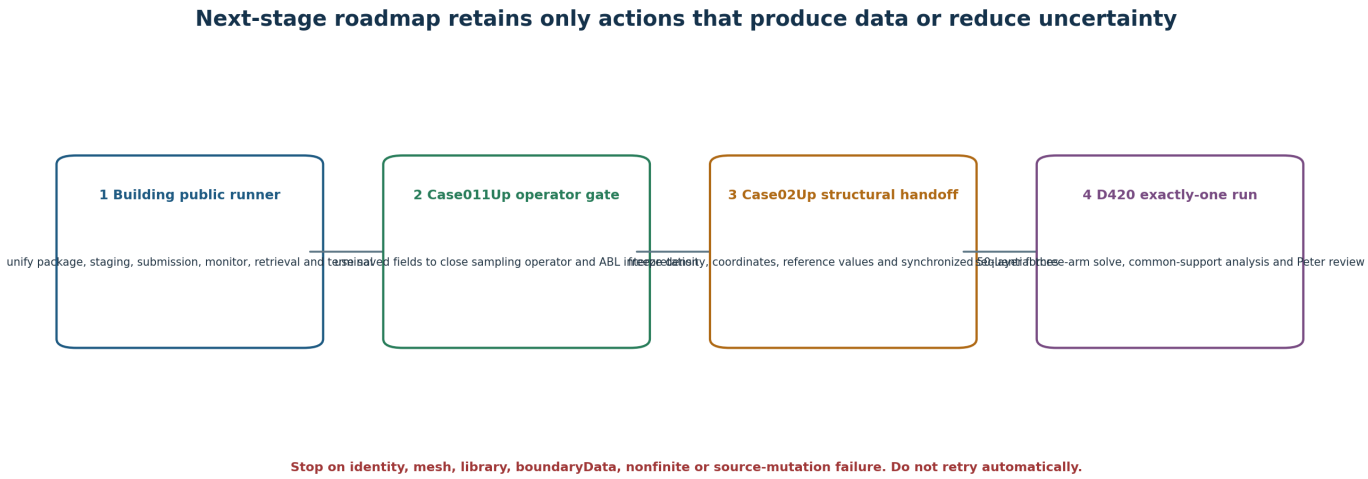


Figure F22 (repeated reference in Chapter 11). [Classification: method diagram] The dependency relations among R1 to R5 and the PI decision point attached to each. R1 removes the named blocker on `empty_abl_acceptance` without new solving; R2 and R3 are the competing building-line priorities; R4 is the only action that can produce matched-drive data; R5 closes the one substantive numerical gap in the cylinder line. Each node carries its stopping condition, so a failure produces a recorded terminal rather than an automatic retry.

11.3 PI decision points

1. Whether R1, the minimum operator reanalysis for job `3396039`, executes immediately, and whether one minimum new solver action is approved if the retained fields prove insufficient.
2. Building-line priority: R2, unifying Case011Up under the public runner, or R3, the Case02Up structural handoff conversion.
3. Whether to authorize one D420 matched-drive three-arm submission with its exact resource envelope (`TACC_Stampede3`, `TG-CIV260040`, `skx`, 4 nodes, 192 ranks, 720 minute walltime request).
4. Whether to extend Task 6 `fine_dt0025` execution authorization from `MESH_QUALIFIED` to `PACKET_SEALED` in order to address the approximately `6.31%` spatial sensitivity gap in the cylinder line.
5. Whether to grant `empty_abl_acceptance`, which is the first required gate for `building_rans` in `stampede3_targets.py`, once R1 reports.
6. Promotion of any model output to a scientific conclusion, which remains a separate PI approval and has no automatic path in code.

11.4 Final boundary and stopping condition of this delivery

No result to date has been approved as a thunderstorm design wind speed, `Vhat_3s`, a design pressure, a structural design load, a code recommendation or formal design guidance. Engineering loop closure, static package identity, scheduler completion, solver completion, postprocessing completion and display-archive acceptance cannot cross the scientific claim gate automatically, and no instance of `SCIENCE_ACCEPTED` exists anywhere in this snapshot.

Three boundaries are restated explicitly because they are the most likely to be misread. The historical D420 M0 and M1 outputs must not be treated as a controlled comparison; the eight recorded direct conflicts and the byte-level `boundaryData` differences prevent attribution of any difference to any single cause. Job `3393320` is successful postprocessing of existing outputs and is not a new solve; `new_solver_run` is `false`, and the job is independently classified `NOT_CONSISTENT_CONTROLLED_COMPARISON`. The matched-drive three-arm package `D420_MC100_LAMINAR`, `D420_MF010_LAMINAR` and `D420_MF010_KOMEGASST_URANS` is `PACKAGE_READY` with `5,293` verified files, has not been submitted and has no solver result. Job `3391286` is `FAILED` with zero scientific output, and its one approved successor submission was consumed.

This study is advisory text requiring parent verification. It neither requested, simulated nor performed any write, remote access, scheduler action or solver execution; it replayed no test suite during report packaging; and wherever the evidence was insufficient, the deficiency is recorded as found together with its source path or evidence identity.

DELIVERY_COMPLETE_WITH_RECORDED_SCIENTIFIC_LIMITATIONS